

Face selection for slowly switching Markov chains and the planar persistent walk

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Abstract

A ball bounces down a Galton board with N rows, and each bounce repeats the previous one with probability p . Kagey asked when the middle bin becomes as likely as an end bin, which only the ball that never switches can reach. We ask the same question for a ball with any finite number of directions. Its direction is a Markov chain that rarely switches, and its bin records how many steps it made in each direction. The most likely bin uses only some set of directions, and we find which set. Two effects compete. The ball must never switch to a direction outside the set, which is likely when the set is hard to leave. But a larger set has more bins to share its probability, and each single bin loses a factor N for each extra direction. When the expected number of switches is a fixed multiple of $\log N$, both effects are powers of N , and the winning set minimizes the sum of the two exponents. A local limit theorem inside a set then gives the constant in front, and with it the number of switches at which two sets are equally likely, up to an error that tends to zero, together with the next term. This needs that the ball can pass between any two directions of a set without leaving it, and a ball that cycles through three directions shows that the answer changes otherwise. Our main application is Kagey's board in the plane, a ball on the square lattice that keeps its direction, turns left or right, or reverses. Its far corner and the origin are equally likely at exactly one switching rate, which we find to first order. When the ball reverses more often than it turns, this crossing agrees with the one on Kagey's board with $N/2$ rows, up to a negative power of N . When it turns more often, the crossing comes from a local limit theorem in two dimensions. On one curve of rates the planar walk is exactly two independent copies of Kagey's board, and with turns only its crossing is Kagey's up to a small error. We also apply the rule to sticky priors for hidden Markov models.

Keywords: persistent random walk; correlated random walk; Galton board; occupation times of Markov chains; mode; principal Dirichlet eigenvalue; quasi-stationary distribution; local limit theorem; hidden Markov model.

MSC2020: 60J10; 60F05; 60F10; 60G50; 62M05.

1 Introduction

A ball falls through a Galton board with N rows of pegs. At the first peg it goes left or right with probability $\frac{1}{2}$. At every later peg it repeats its previous direction with probability p and switches with probability $q = 1 - p$. Problem 131 of Kagey's Open Problem Collection [1] asks for the law of the bin where the ball lands, and in particular when the middle bin is exactly as likely as an endpoint. An endpoint is reached by a single path, the one that never switches, so

it has probability $\frac{1}{2}p^{N-1}$. When p is close to 1 this path wins, and when p is smaller the middle wins. Paper A [2] answers the question at every N . The two are equally likely when the expected number of switches $T = Nq$ is close to $\log N$. More precisely, the crossing value z_N of T satisfies $z_N + \frac{1}{2} \log z_N = \log N + \frac{1}{2} \log \frac{\pi}{8} + o(1)$.

Kagey also asks what changes on other boards. This paper answers his main question, when the middle is as likely as the ends, for boards on which the ball can move in more than two directions. Our main example is Kagey's board in two dimensions. A ball moves on the square lattice $\mathbb{Z}^2$, one step east, north, west or south at a time. It starts in a uniform direction, and at each step it keeps its direction, turns left or right, or reverses. Turns and reversals are rare. A turn to each side has probability rT/N and a reversal has probability sT/N , where $r, s \geq 0$ are fixed rates with $2r + s > 0$. After N steps, is the ball more likely to be at the corner $(N, 0)$, which only the path that never switches can reach, or back at the origin? We show that for every even $N \geq 4$ there is exactly one value T_N^* of T at which the two are equally likely (Proposition 6.2). The corner is more likely below it and the origin above it, and as $N \rightarrow \infty$ through even integers

$$\frac{T_N^*}{\log N} \rightarrow \min\left(\frac{2}{2r+s}, \frac{1}{s}\right) \quad (1)$$

(Theorems 6.3 and 6.4). The two values in the minimum belong to two kinds of paths to the origin, and to see why we first describe the general rule behind this limit.

1.1 Face selection

The planar result is an instance of a general rule, which is the main subject of this paper. Let the direction be a Markov chain on d states that switches from state i to state j with probability $q_{ij}T/N$ at each step, and count the visits to each state. This count vector, the *composition*, plays the role of the bin. Its support is a set F of states, which we call a *face* (it indexes a face of the simplex of visit fractions). Which face carries the most likely composition, and at which T does the mode move to another face? On Kagey's board there are two states. The two faces of one state are the endpoints, and the face of two states is the set of interior bins.

Put $T = \tau \log N$ and let λ_F be the rate at which the chain leaves F , that is, the principal Dirichlet eigenvalue on F of $-Q$, where Q is the matrix of the rates q_{ij} . How likely the best composition with support F is depends on two effects.

- *The cost of staying.* The chain must stay inside F for all N steps. This has probability about $e^{-T\lambda_F} = N^{-\tau\lambda_F}$. A larger face is harder to leave, so the chain stays on it more easily.
- *The cost of spreading.* The mass of F is shared among the compositions with support F , and there are about $N^{|F|-1}$ of them, one factor N for each free coordinate. Up to powers of $\log N$, the best single composition gets a share $N^{-(|F|-1)}$.

So the most likely composition with support F has probability $N^{-\Phi_\tau(F)+o(1)}$, where

$$\Phi_\tau(F) = \tau\lambda_F + |F| - 1,$$

and every mode sits on a face that minimizes Φ_τ (Theorem 3.3). In words, a sticky walk prefers a well-separated community, a set of states that is hard to leave for its size. As τ grows staying counts for more, and the mode moves to larger faces. On Kagey's board an endpoint has $\Phi_\tau = \tau$ and the interior has $\Phi_\tau = 1$, so the middle takes over at $\tau = 1$, which is the first-order form of the

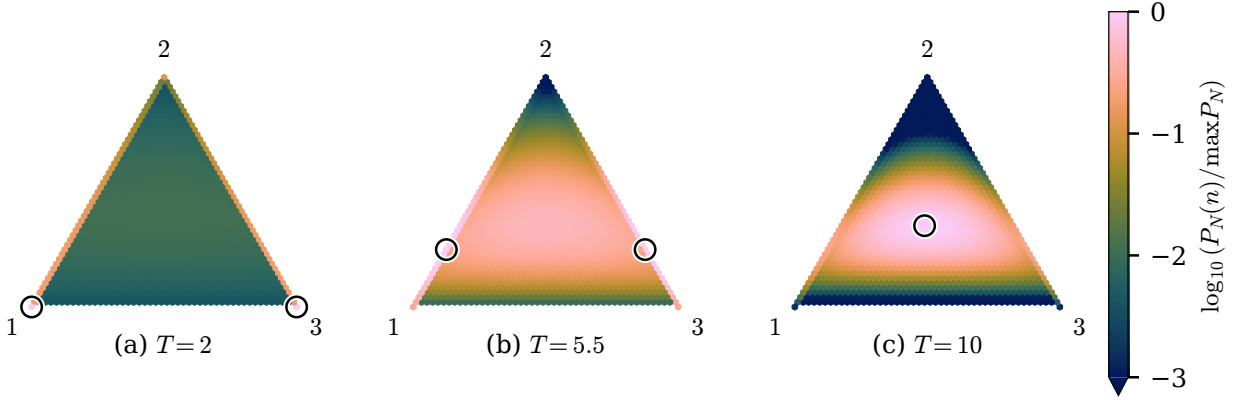


Figure 1: The exact law of the composition for the path 1 – 2 – 3 (rate 1 on each edge, uniform start, $N = 48$), on the simplex of visit fractions. Each cell is one composition, colored by its probability relative to the mode, and the circles mark the modes. (a) For small T the mode is at an end of the path. (b) Then it sits on an edge, that is, on a pair of neighbors. (c) Then it is inside. The bottom edge is the pair $\{1, 3\}$, which the chain cannot use without visiting 2, and it never carries the mode. Here λ_F is 1 for an end, $(3 - \sqrt{5})/2$ for a pair of neighbors and 0 for all three states, so to first order the mode moves at $\tau = (1 + \sqrt{5})/2$ and at $\tau = (3 + \sqrt{5})/2$, that is, at $T = 6.3$ and $T = 10.1$. At $N = 48$ it moves near $T = 4.0$ and $T = 7.1$. Corrections of this size are the subject of Section 4.

answer of Paper A. Section 3.1 works out four states on a cycle by hand, and Theorem 3.4 reads the winners off the convex hull of the points $(\lambda_F, |F| - 1)$.

Figure 1 shows the rule at work on three states in a row.

The planar walk fits this picture. The corner is the face of the single direction east, with exit rate $2r + s$, so it has probability about $N^{-(2r+s)\tau}$. A path to the origin either stays on one axis, a face of two states with exit rate $2r$, or uses all four directions, a face with exit rate 0. On an axis the origin is a single composition, of probability about $N^{-(2r\tau+1)}$. With all four directions a single composition has probability about N^{-3} , but about $N/2$ of them end at the origin, which gives N^{-2} . This suggests that the origin beats the corner when $(2r + s)\tau > \min(2r\tau + 1, 2)$, which is (1). It is only a heuristic, since the origin is not a single composition, and Section 6 proves the limit directly.

Second order. First order fixes a crossing only up to a factor $1 + o(1)$. To locate it up to $o(1)$ we need the constant in front. Say that a face F satisfies (H) if the chain can move between any two states of F without leaving F , and the start gives F positive mass. For such faces we prove a local limit theorem inside the face (Theorem 4.4). It shows (Theorem 4.6) that the best composition with support F , with $k = |F|$ states, has probability

$$M_F = N^{-(k-1)} T^{(k-1)/2} K_F e^{-T\lambda_F} (1 + O(1/T + T^2/N))$$

uniformly for $1 \leq T \leq N^{1/3}$, with an explicit constant K_F . The factor $T^{(k-1)/2}$ appears because the visit fractions fluctuate on the scale $T^{-1/2}$ in each of the $k - 1$ free directions. Now let F and G satisfy (H), and suppose that F has $\Delta \geq 1$ more states than G and an exit rate smaller by $\delta > 0$. Then for large N every $T \in [1, N^{1/3}]$ with $M_F = M_G$ satisfies

$$\delta T + \frac{\Delta}{2} \log T = \Delta \log N - \log \frac{K_F}{K_G} + O\left(\frac{1}{\log N}\right)$$

(Theorem 4.7). So the coefficient $\frac{1}{2}$ of $\log \log N$ that Papers A and B found holds for every such pair, whatever the graph, the rates, the sizes or the start, and without reversibility. When several faces tie at first order, Theorem 4.8 describes the window of T in which they compete. For reversible faces K_F involves the square root of a weighted count of spanning trees (Proposition 4.2), and the constants of Papers A and B are short corollaries (Corollaries 4.10 and 4.11). Proposition 4.14 adds the next term, of relative order $1/T$, to the local law, to M_F and to the crossing equation. Without (H) the coefficient can change. On a directed 3-cycle it is 1 at one crossing and 0 at the next (Proposition 4.13). Section 5 computes all the constants on the 4-cycle, shows on the complete graph K_3 that the start can decide which face wins, and lists the winners on other graphs.

The planar walk at second order. The origin is not a single composition on the full face, so the planar crossing needs its own argument. The mass at the origin splits exactly into the paths that stay on one axis and the paths that use all four directions (Proposition 6.2). When reversals dominate ($s > 2r$) the axis paths carry the origin mass at the crossing. Then $sT_N^* = z_{N/2} + O(N^{-\gamma})$ for some $\gamma > 0$, where $z_{N/2}$ is the crossing of Kagey's board with $N/2$ steps (Theorem 6.4). When turns dominate ($s < 2r$) the paths with all four directions carry it. Their mass is given by a local limit theorem in two dimensions (Theorem 6.5), and the crossing then satisfies

$$(2r + s)T_N^* + \log T_N^* = 2 \log N + \log \frac{\pi}{8(r + s)} + o(1)$$

(Corollary 6.6). Here $\log T_N^*$ has the coefficient 1 and not $\frac{1}{2}$, because the origin is a point of a local limit in two dimensions. There is also one curve of switching probabilities on which everything is exact. If the walk turns to each side with probability $\rho(1 - \rho)$ and reverses with probability ρ^2 , its two diagonal coordinates are independent copies of Kagey's board with switching probability ρ . So on this curve the planar crossing is Kagey's crossing at every N (Proposition 6.7). With turns only ($s = 0$) we show that $rT_N^* = z_N + O((\log N)^2/N)$ (Theorem 6.9).

The paper closes with an application, and nothing before it depends on it. Section 7 applies face selection to sticky priors for hidden Markov models. There the exit rate of a set of hidden states is one minus the Perron root of the kernel of the switches restricted to the set, and as the stickiness decreases the most likely count vector under the prior moves from one state to larger sets (Corollary 7.1 and Proposition 7.2). Among symmetric kernels only the uniform one jumps from one state straight to all of them (Corollary 7.3). Section 1.4 says what each section does.

1.2 Papers A, B and D

This paper is one of four on Problem 131. We call them Paper A [2], Paper B [3], Paper C (this paper) and Paper D [4]. They share the following notation.

- Kagey's board has N steps (his row n has $N = n - 1$). The first step is left or right with probability $\frac{1}{2}$, and each later step repeats the previous one with probability p . We write $q = 1 - p$ and $u = q/p$, the odds of a switch.
- $P_N(k)$ is the probability of k right steps, that is, that the ball lands in bin k . The endpoints have $P_N(0) = P_N(N) = p^{N-1}/2$.
- $L = \log N$, and $T = Nq$ is the expected number of switches (up to the first step).
- The crossing is the value of p (or T) at which a bin is exactly as likely as an endpoint. For the middle bin it is p_N , and $z_N = N(1 - p_N)$.

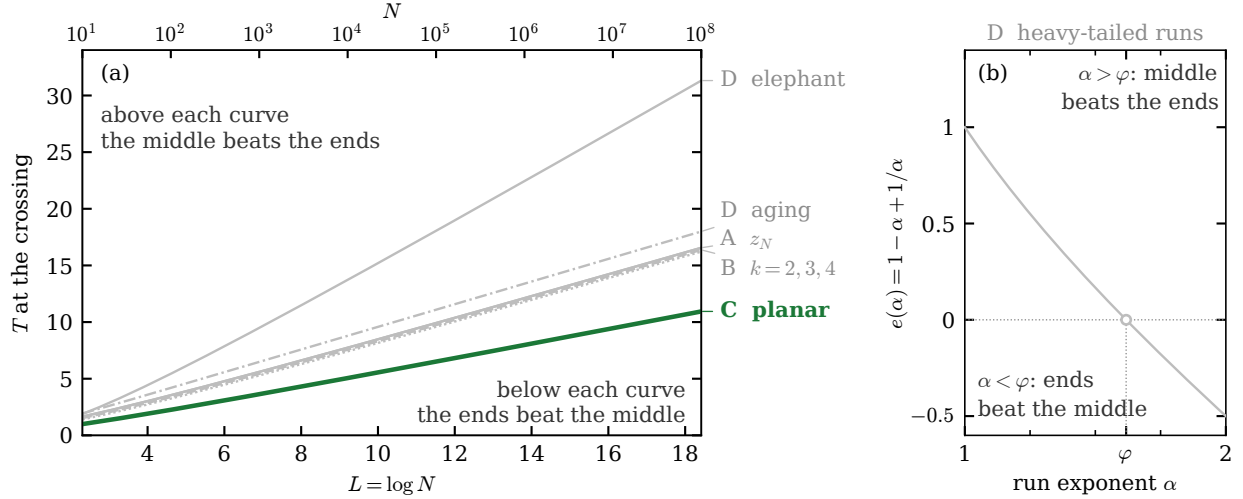


Figure 2: (a) The crossing on each model's switching scale against $L = \log N$ for Papers A to D. The scale is T for Papers A to C, $x = N\varepsilon$ for the elephant walk and the expected number of switches for the aging walk of Paper D. The bold curve solves the first crossing equation of Corollary 6.6 for the planar walk with $r = s = 1$, without its $o(1)$ term. The curve A is z_N , and the curves B are the face crossings of Paper B. (b) The exponent $e(\alpha) = 1 - \alpha + 1/\alpha$ of Paper D for runs with heavy tails, which changes sign at the golden ratio φ . (A version of this figure also appears in Paper D.)

- With d states, a face F is a nonempty set of states, and the vertices are the faces of size 1. The exit rate λ_F is the principal Dirichlet eigenvalue of $-Q$ on F , and K_F is the second-order constant.

Paper A solves Kagey's board. Its uniform Bessel bounds bracket z_N for every $N \geq 2$ and give $z_N + \frac{1}{2} \log z_N = L + c_0 + 1/(8z_N) + R_N$ with $c_0 = \frac{1}{2} \log(\pi/8)$ and an explicit R_N . Paper B lets the ball switch to each of $d - 1$ other directions at the same rate. There every face ties at first order, and constants a_k break the tie at second order. Paper D asks what changes when the ball remembers more than its last step. The present paper treats every rate matrix Q . It explains which face carries the mode and why, computes the constants K_F , recovers c_0 and a_k as special cases, and applies all this to the planar walk.

We quote a few results from the earlier papers. From Paper A we use the run count, the fact that the bin polynomials grow toward the middle, the uniform Bessel bound (recalled in (12)) and the expansion of z_N . From Paper B we use only its expansion of the balanced bins, for one constant in Proposition 4.14(iv). Figure 2 compares the crossings of the four papers.

1.3 Related work

The rate I_F of Section 3 is the Donsker–Varadhan rate of the occupation measure of the chain killed on leaving F . Dupuis and Liu give this rate explicitly for reversible jump processes [5, Eq. (3.2)], and on a finite chain their formula agrees with Proposition 3.1(c). Chen and Deng [7] study large deviations for absorbing Markov chains, and according to their abstract they obtain a variational representation of the Dirichlet principal eigenvalue from the large deviation rate. Guillin, Nectoux and Wu [6] prove a large deviation principle for the occupation measure of a process conditioned to stay in a set. For reversible processes they show that the exit rate of the

set is the least value of the Donsker–Varadhan rate over laws on the set (their Cor. 2). In general they prove an inequality between their rate, the Donsker–Varadhan rate and the exit rate, and they conjecture equality (their (2.15)). For a finite chain with Q_F irreducible, the fact that λ_F is the least value of I_F is the variational formula of Donsker and Varadhan [8], which needs no reversibility. Proposition 3.1(a) gives a short proof in our notation. This does not address the conjecture of Guillin, Nectoux and Wu, which concerns their conditional rate. The quasi-ergodic law goes back to Darroch and Seneta [9, 10], the book of Collet, Martínez and San Martín [11] is the standard reference for quasi-stationary and quasi-ergodic laws, and Colonus and Rasmussen [12] treat the reducible case. Dietz and Sethuraman [13] study chains that switch from i to j at time n with probability $G(i, j)/n$, where every $G(i, j) > 0$. Their chains also make about $\log n$ switches, but spread log-uniformly in time, and their occupation fractions have a limit law with full support on the simplex. In [14] they prove large deviations for chains whose transition matrices converge to a reducible matrix.

Exact laws of transition counts go back to Whittle [15], Dawson and Good [16] and Goodman [17]. Polettini [18] and Carugno, Vivo and Coghi [19] derive them from the BEST theorem and the matrix-tree theorem when every count grows linearly. We group the words by runs instead (Lemma 2.3), which suits a regime with $O(\log N)$ switches. In continuous time, Brydges, van der Hofstad and König [33] give the joint density of the local times of a chain on the event that its range is a given set, and Huang, Kious, Sidoravicius and Tarrès [34] write it with Bessel functions and spanning trees. Brydges, van der Hofstad and König remark that their formula is hard to evaluate asymptotically, and they prove upper bounds. Theorem 4.4 gives the leading term with its constant in our regime. Bertini, Gabrielli and Landim [20] and Landim [21] expand the level-two rate functions of finite chains whose rates depend on a parameter. They describe the time scales of such chains but do not ask where the mode of the composition sits. The sticky HDP-HMM of Fox, Sudderth, Jordan and Willsky [22] adds a self-transition bias to the hierarchical Dirichlet process hidden Markov model.

On the lattice the correlated random walk goes back to Gillis [23]. Chen and Renshaw [24] treat the Gillis–Domb–Fisher walk, and in [25] they show that the characteristic function of any correlated random walk satisfies a recurrence. So in principle the law of our planar walk is known. Marris and Giuggioli [26] give lattice propagators for walks with forward, backward and lateral steps. In the continuum, Santra, Basu and Sabhapandit [27] find the marginals of a run-and-tumble particle with four directions, and Angelani [28] treats switching among four orthogonal directions. With turns only, the continuous motion is a rotation of two independent telegraph processes (Cinque and Orsingher [29, Theorem 2.2], who credit Orsingher [30] for constant rates, and Smith, Le Doussal, Majumdar and Schehr [32]). With reversals the two processes are no longer independent [29, Section 3]. Proposition 6.7 is a lattice form of this, and on the lattice the independent case does have reversals. In the continuum the corners carry atoms and the origin does not (it has a density in the plane, and with reversals also along the axes [31]), so the comparison of a corner with the origin is a lattice question. We did not find the face-selection question or the planar crossing of the corner and the origin in the literature.

1.4 Plan of the paper

Section 2 fixes the notation and gives the exact law of the composition as a sum over runs. Section 3 works out four states on a cycle by hand and then proves first-order face selection (Theorems 3.3 and 3.4). Section 4 proves the local limit theorem inside a face, the face maxima, the crossing equation and the tie windows. It then recovers the constants of Papers A and B, shows on a

directed 3-cycle what can change without (H), and adds the next term. Section 5 computes the 4-cycle with all its constants and lists the winners on other graphs. Section 6 treats the planar walk, first for general rates and then on the product curve and with turns only. Section 7 applies face selection to sticky priors for hidden Markov models, and Section 8 describes the computer checks and the open questions. Appendix A proves the local law, the uniform upper bound of Lemma 3.2 and the first-order bounds for reducible faces. Appendix B proves the Laplace lemma that Appendix A uses, and the next term. Appendix C has the proofs for the planar walk, Appendix D the proofs for the table of outcomes, and Appendix E the limit of large concentration for the direct jump under the sticky prior.

2 Setting, notation and the exact law

2.1 Setting and notation

The chain. Fix $d \geq 2$ and a generator Q on $[d] = \{1, \dots, d\}$. Its rates are $q_{ij} \geq 0$ for $i \neq j$, the exit rates are $q_i = \sum_{j \neq i} q_{ij}$, $Q_{ii} = -q_i$, and $q_{\max} = \max_i q_i$. When we write q for the vector (q_i) this is clear from the context. (The scalar $q = 1 - p$ of Kagey's board appears only where we recall Paper A, and the scalar q of Proposition 4.9 is a common exit rate.) The start μ is a probability law on $[d]$, and $\mu(F) = \sum_{i \in F} \mu_i$. For $N \geq 2$ and $T > 0$ put

$$h = \frac{T}{N}, \quad L = \log N, \quad \tau = \frac{T}{L}.$$

The chain $X_1, \dots, X_N$ has $X_1 \sim \mu$ and transition matrix $I + hQ$. So at each step it jumps from i to j with probability hq_{ij} , and N steps look like the continuous-time chain with generator Q run for time T . We assume $hq_{\max} \leq \frac{1}{2}$ unless we say otherwise. The *composition* is the vector of visit counts $n = (n_1, \dots, n_d)$, with $n_i = \#\{t \leq N : X_t = i\}$, so that $n_1 + \dots + n_d = N$. We write $P_N(n)$ for the probability that the chain has composition n , $\text{supp } n = \{i : n_i > 0\}$ for its support, and $\alpha = n/N$ for the visit fractions.

Faces. A face is a nonempty set $F \subseteq [d]$, of size $k = k_F = |F|$. Let Δ_F be the set of probability vectors on $[d]$ that vanish off F , and Δ_F° the set of those that are positive on F . We number the states of F as $1, \dots, k$ and use the coordinates $(\alpha_2, \dots, \alpha_k)$ on Δ_F and $(t_2, \dots, t_k)$ on $\{t \in \mathbb{R}^F : t_1 + \dots + t_k = T\}$. We write $x \cdot y = \langle x, y \rangle = \sum_{i \in F} x_i y_i$ and $x \circ y = (x_i y_i)_{i \in F}$. Let $A_F = (q_{ij})_{i, j \in F}$ with zero diagonal, and $Q_F = A_F - \text{diag}(q_i)_{i \in F}$. Since q_i includes the rates of leaving F , Q_F generates the chain killed when it leaves F . The switching graph G_F has an edge $i \rightarrow j$ whenever $i, j \in F$, $i \neq j$ and $q_{ij} > 0$.

Perron roots. A square matrix B is Metzler if its off-diagonal entries are nonnegative, and then its Perron root $\varrho(B)$ is its largest real eigenvalue. If B is irreducible, $\varrho(B)$ is simple and has positive left and right eigenvectors [36].

Definition 2.1 (faces). Let F be a face. We call F *irreducible* if G_F is strongly connected, so every singleton is irreducible. We call F *admissible* if for some N some path $x_1 \cdots x_N$ of positive probability has support exactly F (for $0 < hq_{\max} < 1$ this does not depend on h). We say that F satisfies (H) if F is irreducible and $\mu(F) > 0$. The *exit rate* of F is $\lambda_F = -\varrho(Q_F)$, so $\lambda_{\{i\}} = q_i$. If F is irreducible, r_F and l_F are right and left eigenvectors of Q_F for $-\lambda_F$ with positive entries and $\langle l_F, r_F \rangle = 1$, and $\alpha_F^* = l_F \circ r_F$ is the *quasi-ergodic law* of F [9, 10]. The *face maximum* is $M_F = \max\{P_N(n) : \text{supp } n = F\}$, with $\max \emptyset = 0$.

We call λ_F the exit rate for the following reason. If F is irreducible, then $e^{TQ_F} r_F = e^{-T\lambda_F} r_F$ and r_F is positive. So the chance that the continuous-time chain started at a state of F stays in F up to

time T , which is a row sum of e^{TQ_F} , lies between $e^{-T\lambda_F} \min r_F / \max r_F$ and $e^{-T\lambda_F} \max r_F / \min r_F$. If F is reducible, we order the strongly connected components $C_1, \dots, C_r$ of G_F so that every edge between components goes forward. Then Q_F is block triangular with diagonal blocks Q_{C_j} , and $\lambda_F = \min_j \lambda_{C_j}$. In all cases $0 \leq \lambda_F \leq \min_{i \in F} q_i$, since the rows of Q_F have sums at most 0 and a Perron root is at least every diagonal entry. For irreducible F the quasi-ergodic law α_F^* is the long-run law of the visit fractions of the chain conditioned to stay in F , and the best composition on F sits near $N\alpha_F^*$ (Theorem 4.6).

Later sections add their own symbols where they are needed, namely the rate I_F (Section 3), the tilted chain and the constant K_F (Definition 4.1), and the planar notation of Section 6.

Kagey's board. For $a, b \geq 1$ let $S_{a,b}(u) = \sum_w u^{s(w)}$, the sum over the words w with a right steps and b left steps, where $s(w)$ is the number of switches of w . Each such word has probability $\frac{1}{2}p^{N-1}(q/p)^{s(w)}$ on Kagey's board with $N = a + b$ steps. So $S_{a,b}(q/p)$ is the probability of bin a divided by the probability $\frac{1}{2}p^{N-1}$ of an endpoint. For $N \geq 2$ let $S_N = S_{\lfloor N/2 \rfloor, \lfloor N/2 \rfloor}$, which belongs to a middle bin. It is a nonzero polynomial with nonnegative coefficients and no constant term, so $S_N(u) = 1$ has exactly one positive root u_N , and $p_N = 1/(1 + u_N)$ is the crossing of the middle and the endpoints. Put $z_N = N(1 - p_N)$ and $c_0 = \frac{1}{2} \log(\pi/8)$. Paper A proves the crossing equation $z_N + \frac{1}{2} \log z_N = L + c_0 + 1/(8z_N) + R_N$ with an explicit remainder $R_N = O(1/L^2)$.

Conventions. Unless we say otherwise, Q , μ and the faces under discussion are fixed, and every constant, every implied constant in $O(\cdot)$ and every threshold N_0 depends only on them and on the compact set $\mathcal{K}$ when one appears. A bound that holds “uniformly for $1 \leq T \leq N^{1/3}$ ” holds for all $N \geq N_0$ and all T in that range with one implied constant. Subscripts as in $O_{\mathcal{K}}$ stress a dependence.

2.2 Admissible faces

Lemma 2.2 (admissible faces). *Let F be a face and $C_1, \dots, C_r$ the strongly connected components of G_F . (a) F is admissible if and only if the components can be numbered so that $\mu(C_1) > 0$ and, for each $j < r$, $q_{xy} > 0$ for some $x \in C_j$ and $y \in C_{j+1}$. (b) An irreducible face is admissible if and only if $\mu(F) > 0$. So (H) holds exactly for the irreducible admissible faces. (c) If Q has symmetric support, that is $q_{ij} > 0$ exactly when $q_{ji} > 0$, then every admissible face is irreducible.*

Proof. The states of the maximal runs of a path of positive probability with support F form a walk in G_F that starts in $\text{supp } \mu$ and visits every state of F , and every such walk arises in this way for large N . (a) Along such a walk a component C cannot be visited again after the walk has left it for a state $y \notin C$, since then y and C would be reachable from each other. Hence a path of positive probability visits each strongly connected component of its support in one block of consecutive times, and it passes from one block to the next by a single switch. Numbering the components in the order of these blocks gives the condition. Conversely, given the numbering and edges $x_j \rightarrow y_{j+1}$, take $y_1 \in C_1$ with $\mu_{y_1} > 0$ and any $x_r \in C_r$, and join walks from y_j to x_j through all of C_j by these edges. (b) is (a) with $r = 1$. (c) Here an edge between two components would give an edge back, so no edge joins two components, and $r = 1$ by (a). $\square$

So $M_F > 0$ for large N if F is admissible, and $M_F = 0$ otherwise.

2.3 The exact law

For $m \in \mathbb{Z}_{\geq 1}^F$ put $M = \sum_{i \in F} m_i$ and $W(m) = \sum_c \mu_{c_1} \prod_{j < M} q_{c_j c_{j+1}}$, the sum over the words $c = c_1 \cdots c_M$ in F with $c_j \neq c_{j+1}$ in which each $i \in F$ occurs m_i times. Such a word lists the states of the runs of a path, and m_i is the number of runs in state i .

Lemma 2.3 (run expansion). *Let $hq_{\max} \leq 1$ and let n have support F . Then, with $\binom{n_i-1}{m_i-1} = 0$ for $m_i > n_i$ and $0^0 = 1$,*

$$P_N(n) = \sum_{m \in \mathbb{Z}_{\geq 1}^F} W(m) h^{M-1} \prod_{i \in F} \binom{n_i-1}{m_i-1} (1 - hq_i)^{n_i-m_i}.$$

Proof. The states $c_1 \cdots c_M$ of the maximal runs of a path form a word as in $W(m)$. The path has $M-1$ switches, of probabilities $hq_{c_j c_{j+1}}$, and $n_i - m_i$ stays in state i , of probability $1 - hq_i$ each. For a fixed c the lengths of the runs of state i form one of the $\binom{n_i-1}{m_i-1}$ compositions of n_i into m_i positive parts. Summing over c gives the claim. $\square$

Lemma 2.3 is checked in Lean (Section 8). For $F = \{i\}$ the only path is i^N , so uniformly for $1 \leq T \leq N^{1/2}$

$$M_{\{i\}} = \mu_i (1 - hq_i)^{N-1} = \mu_i e^{-q_i T} (1 + O(T^2/N)). \quad (2)$$

For $t \in (0, \infty)^F$ let $H(t) = \sum_m W(m) \prod_{i \in F} t_i^{m_i-1} / (m_i-1)!$, which is finite since $W(m) \leq k^M q_{\max}^{M-1}$. The next lemma says that $e^{-\langle q, t \rangle} H(t)$ is the density of the occupation times of the continuous-time chain on the event that it stays in F . It is the continuum form of Lemma 2.3, in which the binomial coefficients become the powers $t_i^{m_i-1} / (m_i-1)!$. Lemma 4.5 below shows that the discrete law is close to it.

Lemma 2.4 (the continuum density). *Let $(Y_s)_{s \geq 0}$ be the continuous-time chain with generator Q and $Y_0 \sim \mu$, let $\mathcal{O}_i = \int_0^T \mathbf{1}\{Y_s = i\} ds$, and let E_F be the event that Y stays in F during $[0, T]$ and visits every state of F . Then $\mathbb{P}(E_F, \mathcal{O} \in \cdot)$ has the density $e^{-\langle q, t \rangle} H(t)$ in $(t_2, \dots, t_k)$ on $\{t \in (0, \infty)^F : \sum_i t_i = T\}$, and $\mathbb{P}(E_F, \mathcal{O}/T \in \cdot)$ has the density*

$$f_T(\alpha) = T^{k-1} e^{-T\langle q, \alpha \rangle} H(T\alpha)$$

in $(\alpha_2, \dots, \alpha_k)$. For n with support F and $t = hn$, $N^{-(k-1)} f_T(n/N) = h^{k-1} e^{-\langle q, t \rangle} H(t)$. For $F = \{i\}$, $f_T = \mathbb{P}(E_F) = \mu_i e^{-q_i T}$.

Proof. We split E_F by the sequence of states that Y visits and integrate out the holding times. Up to a null set, E_F is the disjoint union of the events E_c that Y visits exactly $c_1, \dots, c_M$ in this order during $[0, T]$, over the words c in the sums $W(m)$. Fix c , let $s_1, \dots, s_{M-1}$ be the first $M-1$ holding times and $s_M = T - \sum_{j < M} s_j$. The last holding time only has to exceed s_M . So on $\{s_j > 0 \text{ for } j \leq M\}$ the event E_c has density $\mu_{c_1} \prod_{j < M} q_{c_j c_{j+1}} e^{-\langle q, \mathcal{O} \rangle}$ in $(s_1, \dots, s_{M-1})$, where $\mathcal{O}_i$ is the sum of the s_j with $c_j = i$. For each state $i \neq c_M$ we replace the last holding time in state i by $\mathcal{O}_i$, a triangular change of variables with unit diagonal. For fixed $\mathcal{O} = t$ the other $m_i - 1$ holding times in each state i (for $i = c_M$, those before the last one) range over a simplex of volume $t_i^{m_i-1} / (m_i-1)!$. Integrating them out and summing over c gives $e^{-\langle q, t \rangle} H(t)$, and any $k-1$ of the coordinates t_i give the same Lebesgue measure. The rest follows from $t = T\alpha$ and $T/N = h$. $\square$

Example 2.5 (the four models). (i) For $d = 2$, $q_{12} = q_{21} = 1$ and uniform μ the chain is Kagey's board, the persistent walk of [2], with $p = 1 - h$. So $T = N(1 - p)$, the faces $\{1\}$ and $\{2\}$ are the endpoints, and Lemma 2.3 is the run count of [2]. (ii) For $Q = J - dI$, with J the all-ones matrix, and uniform μ it is the model of [3], which switches to each other state with probability h . (iii) For $Q = \Pi - I$, with Π stochastic with zero diagonal, $\lambda_F = 1 - \varrho(\Pi_F)$. This is the sticky chain of Section 7. (iv) Let $d = 4$, number the directions east, north, west and south as $0, 1, 2, 3$ modulo 4, and put $q_{j, j \pm 1} = r$, $q_{j, j+2} = s$ and μ uniform. Then X_t is the direction of step t of a planar walk

that turns left or right with probability rh each and reverses with probability sh (Section 6). For $s = 0$ and $r = 1$ this is the 4-cycle of Section 3.1. In (i), (ii) and (iv), Q has symmetric support and μ full support, so all admissible faces satisfy (H) by Lemma 2.2.

3 First-order face selection

3.1 A worked example with four states on a cycle

Before the general theorem we work out one example by hand. Take $d = 4$ states on a cycle, with rate 1 between neighbors (indices modulo 4), and the uniform start. So at each step the chain moves to each of its two neighbors with probability $h = T/N$ and stays with probability $1 - 2h$. This is the direction of the planar walk of Section 6 when it turns but never reverses (Example 2.5(iv) with $r = 1$, $s = 0$). Up to rotation and reflection there are four kinds of faces, namely a vertex, an adjacent pair, an arc of three states and the whole cycle. (A face such as $\{1, 3\}$ cannot be the support of a path, since the chain cannot pass from 1 to 3 without visiting 2 or 4.) Which kind carries the most likely composition?

For each kind we ask two questions. With what probability does the chain stay in the face for all N steps? Among how many compositions is this probability shared? The best composition of the face gets about the first answer divided by the second. In this subsection T is large and T^2/N is small, the sign $\approx$ means equality up to a bounded factor, and we do not give proofs. The proofs are in Theorems 3.3 and 4.6, and Corollary 5.1 has the constants.

A vertex, say $\{1\}$. The chain stays at state 1 only if it never switches, which has probability $(1 - 2h)^{N-1} \approx e^{-2T}$. So the exit rate is $\lambda_{\{1\}} = 2$. The only composition is $(N, 0, 0, 0)$, so nothing is shared, and $M_{\{1\}} = \frac{1}{4}(1 - 2h)^{N-1} \approx e^{-2T}$.

An adjacent pair, say $\{1, 2\}$. The chain leaves the pair by the switches $1 \rightarrow 4$ and $2 \rightarrow 3$, while the switches between 1 and 2 keep it inside. So at each step it leaves with probability h , whichever of the two states it is in, and it stays in the pair with probability $(1 - h)^{N-1} \approx e^{-T}$. The exit rate is $\lambda_{\{1,2\}} = 1$, half that of a vertex. Given that the chain stays, it switches between 1 and 2 about T times, so it makes about T runs of length about N/T each. The number a of visits to state 1 is the total length of half of these runs. So a is close to $N/2$, with fluctuations of order $\sqrt{T} \cdot N/T = N/\sqrt{T}$. Hence the mass e^{-T} is shared by about $N/\sqrt{T}$ compositions $(a, N - a, 0, 0)$, and the best one gets

$$M_{\{1,2\}} \approx \frac{e^{-T}}{N/\sqrt{T}} = N^{-1} T^{1/2} e^{-T}.$$

For the pair we can check this against an exact formula. A path with support $\{1, 2\}$ is a word in the letters 1 and 2. Each stay has probability $1 - 2h$ and each switch probability h , so a word with j switches has probability $\frac{1}{4}(1 - 2h)^{N-1} u^j$ with $u = h/(1 - 2h)$. Summing over the words with a letters 1 gives $P_N(a, N - a, 0, 0) = M_{\{1\}} S_{a, N-a}(u)$, the bin polynomial of Kagey's board (Section 2.1). It is largest for the balanced compositions, and the uniform Bessel bound of Paper A, recalled in (12) below, shows that $S_N(u)$ is $\frac{2}{N} \sqrt{2T/\pi} e^T$ times a factor that tends to 1. So $M_{\{1,2\}}$ is $\frac{1}{\sqrt{2\pi}} N^{-1} T^{1/2} e^{-T}$ times such a factor.

An arc of three states, say $\{1, 2, 3\}$. Only the end states 1 and 3 can leave (both to state 4), each at rate 1, and the middle state 2 cannot. So the chain stays longer if it prefers the middle state, and we need an eigenvalue to find the exit rate. The chain killed on leaving the arc has the matrix $I + hQ_F$ with $-Q_F = 2I - A$, where A is the adjacency matrix of the path $1 - 2 - 3$. Its largest eigenvalue is $1 - (2 - \sqrt{2})h$, with eigenvector $(1, \sqrt{2}, 1)$. So the chain stays

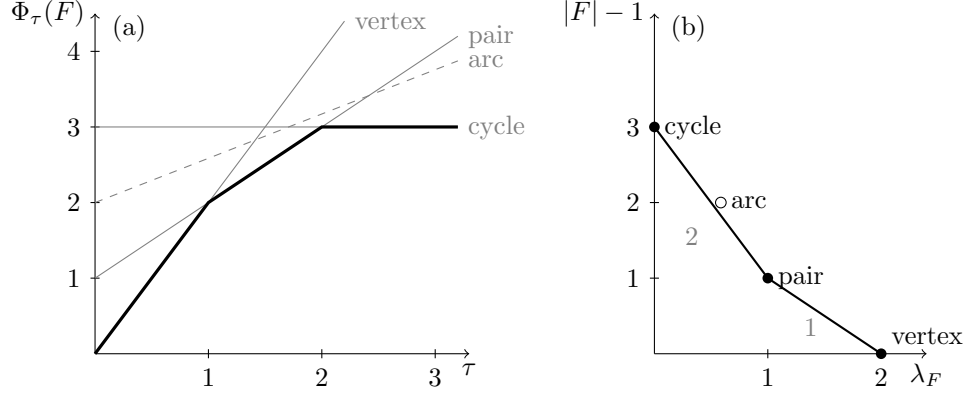


Figure 3: The 4-cycle. (a) The exponents $\Phi_\tau(F) = \tau\lambda_F + |F| - 1$ of the four kinds of faces against τ , and their lower envelope (bold). The mode sits on the face whose line is lowest, namely a vertex for $\tau < 1$, an adjacent pair for $1 < \tau < 2$ and the whole cycle for $\tau > 2$. The dashed line of the arc of three states is never lowest. (b) The same faces as points $(\lambda_F, |F| - 1)$, with the lower-left boundary of their convex hull. The gray numbers are minus the slopes of its edges, the values of τ where the winner changes (Theorem 3.4). The arc (open) lies above the boundary.

in the arc with probability about $e^{-(2-\sqrt{2})T}$, and $\lambda_{\{1,2,3\}} = 2 - \sqrt{2} = 0.586$. The visit fractions now fluctuate in two directions, so the mass is shared by about $(N/\sqrt{T})^2$ compositions, and $M_{\{1,2,3\}} \approx N^{-2} T e^{-(2-\sqrt{2})T}$.

The whole cycle. The chain never has to leave, so $\lambda_{[4]} = 0$ and the mass is 1. It is shared by about $(N/\sqrt{T})^3$ compositions near $(\frac{N}{4}, \frac{N}{4}, \frac{N}{4}, \frac{N}{4})$. So $M_{[4]} \approx N^{-3} T^{3/2}$.

face F	λ_F	it stays with probability	number of compositions	best composition M_F	exponent $\Phi_\tau(F)$
vertex	2	e^{-2T}	1	e^{-2T}	2τ
adjacent pair	1	e^{-T}	$N/\sqrt{T}$	$N^{-1} T^{1/2} e^{-T}$	$\tau + 1$
arc of three	$2 - \sqrt{2}$	$e^{-(2-\sqrt{2})T}$	$(N/\sqrt{T})^2$	$N^{-2} T e^{-(2-\sqrt{2})T}$	$(2 - \sqrt{2})\tau + 2$
whole cycle	0	1	$(N/\sqrt{T})^3$	$N^{-3} T^{3/2}$	3

Comparison. The table above collects the four cases, up to bounded factors. Now put $T = \tau \log N$. Then $e^{-\lambda T} = N^{-\tau\lambda}$, while the powers of T are only powers of $\log N$. So, up to such powers, $M_F \approx N^{-\Phi_\tau(F)}$ with the exponent $\Phi_\tau(F) = \tau\lambda_F + |F| - 1$ in the last column. The mode sits on the face with the smallest exponent. Figure 3(a) plots the four exponents against τ . The vertices win for $\tau < 1$, the adjacent pairs for $1 < \tau < 2$ and the whole cycle for $\tau > 2$. The arcs of three states never win. They beat the pairs only for $\tau > 1 + \sqrt{2}$, and there the whole cycle beats them, since that needs only $(2 - \sqrt{2})\tau > 1$. Note also that the whole cycle carries almost all the mass for every $\tau > 0$. It carries the mode only for $\tau > 2$, because it spreads its mass over about N^3 compositions.

The general rule. For any face F the same two effects give $M_F \approx N^{-(|F|-1)} e^{-T\lambda_F} = N^{-\Phi_\tau(F)}$ up to powers of $\log N$, as announced in Section 1.1. There is a second way to see the factor $N^{-(|F|-1)}$. A path with support F must enter each of the $|F| - 1$ states of F where it does not start, and each first entry is a switch of probability about T/N . Every later switch also has probability about T/N , but it multiplies the number of ways to place the runs by a factor of order

N , so it loses no power of N . Within F the best composition sits near the quasi-ergodic law α_F^* . Here it is $(\frac{1}{2}, \frac{1}{2})$ on a pair, $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ on an arc (the normalized squared Perron vector) and the uniform law on the cycle. Other visit fractions α lose an extra factor $e^{-T(I_F(\alpha) - \lambda_F)}$, with a rate I_F that we now introduce.

The rest of this section proves the rule. Proposition 3.1 studies the rate inside a face, Theorem 3.3 proves the rule for every Q and every start, and Theorem 3.4 reads the winners off a convex hull as in Figure 3(b). Section 5 returns to the 4-cycle with all constants (Corollary 5.1).

3.2 The rate inside a face

For $\alpha \in \Delta_F$ let

$$I_F(\alpha) = \sup_{u \in (0, \infty)^F} \sum_{i \in F} \alpha_i \left(-\frac{(Q_F u)_i}{u_i} \right) = \sup_{u \in (0, \infty)^F} \sum_{i \in F} \alpha_i \left(q_i - \sum_{j \in F \setminus \{i\}} q_{ij} \frac{u_j}{u_i} \right),$$

and let $q_i^{\text{out}}(F) = \sum_{j \notin F} q_{ij}$, the rate of leaving F from i . The function I_F is the Donsker–Varadhan rate [8] of the occupation times of the chain killed on leaving F (Section 1.3). The chance that the chain stays in F with visit fractions near α decays like $e^{-T I_F(\alpha)}$ (Theorem 3.3(i)). Taking $u = \mathbf{1}$ gives $I_F(\alpha) \geq \sum_i \alpha_i q_i^{\text{out}}(F)$, and dropping the terms with q_{ij} gives $I_F(\alpha) \leq \sum_i \alpha_i q_i$. So I_F lies between the rate of leaving F and the rate of leaving the current state, both averaged over α . On the pair $\{1, 2\}$ of Section 3.1, part (c) below gives $I_F(\alpha) = 2 - 2\sqrt{\alpha_1 \alpha_2}$, which is 2 at a vertex and has its least value 1 at $(\frac{1}{2}, \frac{1}{2})$. We need the properties of I_F without reversibility, and the proofs are short.

Proposition 3.1 (the rate on a face). *Let F be irreducible with $k = |F| \geq 2$.*

- (a) $I_F \geq \lambda_F$ on Δ_F , with equality exactly at $\alpha_F^* = l_F \circ r_F$.
- (b) I_F is convex on Δ_F and strictly convex on Δ_F° . For $\alpha \in \Delta_F^\circ$ the supremum defining $I_F(\alpha)$ is attained at a vector $u = r_\alpha$, which is unique up to a positive factor, and α is the stationary law of the generator on F with off-diagonal entries $q_{ij} r_{\alpha, j} / r_{\alpha, i}$.
- (c) If there is $\nu \in (0, \infty)^F$ with $\nu_i q_{ij} = \nu_j q_{ji}$ for all $i, j \in F$, then for every $\alpha \in \Delta_F$

$$I_F(\alpha) = \sum_{\{i, j\} \subseteq F, i \neq j} (\sqrt{\alpha_i q_{ij}} - \sqrt{\alpha_j q_{ji}})^2 + \sum_{i \in F} \alpha_i q_i^{\text{out}}(F).$$

- (d) For every nonempty face $F' \subsetneq F$ and every $\alpha \in \Delta_{F'}$, $I_F(\alpha) = I_{F'}(\alpha) \geq \lambda_{F'} > \lambda_F$.

Idea of the proof. We write $u = e^\psi$. Then the function inside the supremum is concave in ψ and affine in α , and ψ is a maximizer exactly when α is stationary for the chain on F with rates $q_{ij} e^{\psi_j - \psi_i}$. The lower bound $I_F \geq \lambda_F$ comes from one test vector u (a resolvent applied to $\mathbf{1}$), and the value at α_F^* comes from the test vector $u = r_F$. Strict convexity then shows that α_F^* is the only minimizer in the open face, and (d) handles the boundary.

Proof. Write $u = e^\psi$, so that $I_F(\alpha) = \sup_\psi \Psi_\alpha(\psi)$ with $\Psi_\alpha(\psi) = \sum_i \alpha_i q_i - \sum_{i \neq j} \alpha_i q_{ij} e^{\psi_j - \psi_i}$, where i, j run over F . It is concave in ψ and affine in α . Let G^ψ be the generator on F with off-diagonal entries $q_{ij} e^{\psi_j - \psi_i}$. Then $\nabla_\psi \Psi_\alpha = -\alpha^\top G^\psi$, so ψ is a maximizer exactly when α is stationary for G^ψ , which is irreducible.

The lower bound in (a), for every face F' . Write $Q_{F'} = B - cI$ with $B \geq 0$ and let $s > \varrho(Q_{F'})$. Then $u = (sI - Q_{F'})^{-1}\mathbf{1} = \sum_{n \geq 0} (s + c)^{-n-1} B^n \mathbf{1}$ is positive and $-(Q_{F'}u)_i/u_i = -s + 1/u_i > -s$. So $I_{F'} > -s$, and $s \downarrow \varrho(Q_{F'})$ gives the claim.

The value at α_F^* . With $\psi = \log r_F$ and $R = \text{diag}(r_F)$ we get $G^\psi = R^{-1}(Q_F + \lambda_F I)R$ and $(\alpha_F^*)^\top G^\psi = l_F^\top (Q_F + \lambda_F I)R = 0$. So $I_F(\alpha_F^*) = \Psi_{\alpha_F^*}(\log r_F) = \lambda_F$.

(b) I_F is convex as a supremum of affine functions. Let $\alpha \in \Delta_F^\circ$, let a be the least positive rate q_{ij} in F , normalize $\min_i \psi_i = 0$ and let $M = \max_i \psi_i$. Some edge $i \rightarrow j$ of a shortest path in G_F from a minimum of ψ to a maximum has $\psi_j - \psi_i \geq M/(k-1)$. So $\Psi_\alpha(\psi) \leq \sum_i \alpha_i q_i - a(\min_i \alpha_i) e^{M/(k-1)}$, and the supremum is attained. If ψ and $\psi + \delta$ both attain it, concavity makes each term $\alpha_i q_{ij} e^{\psi_j - \psi_i + t(\delta_j - \delta_i)}$ affine in $t \in [0, 1]$. So $\delta_j = \delta_i$ on every edge, and δ is constant. For strict convexity let $\alpha^0 \neq \alpha^1$ lie in Δ_F° , let $\alpha^t = (1-t)\alpha^0 + t\alpha^1$ with $0 < t < 1$, and let ψ attain $I_F(\alpha^t)$. Then $I_F(\alpha^t) = (1-t)\Psi_{\alpha^0}(\psi) + t\Psi_{\alpha^1}(\psi) \leq (1-t)I_F(\alpha^0) + tI_F(\alpha^1)$, and equality would make both α^0 and α^1 stationary for G^ψ .

(d) For $\alpha \in \Delta_{F'}$ only the rows in F' enter, and the terms with $j \in F \setminus F'$ are subtracted. Dropping them gives $I_F(\alpha) \leq I_{F'}(\alpha)$, and $u_j = \varepsilon \rightarrow 0$ for $j \notin F'$ gives equality. Then $I_{F'} \geq \lambda_{F'}$ by the lower bound, and $\lambda_{F'} > \lambda_F$ because the Perron root of an irreducible nonnegative matrix, here $Q_F + cI$, is larger than that of each proper principal submatrix [36, Ch. 2].

(a) If $\alpha \neq \alpha_F^*$ lies on the boundary of Δ_F , (d) gives $I_F(\alpha) > \lambda_F$. If $\alpha \in \Delta_F^\circ$ and $I_F(\alpha) = \lambda_F$, strict convexity gives $I_F(\frac{1}{2}(\alpha + \alpha_F^*)) < \lambda_F$, against the lower bound.

(c) Since $q_i = \sum_{j \in F \setminus \{i\}} q_{ij} + q_i^{\text{out}}(F)$, the objective is $\sum_i \alpha_i q_i^{\text{out}}(F)$ plus, for each pair $\{i, j\}$, the term $\alpha_i q_{ij}(1 - u_j/u_i) + \alpha_j q_{ji}(1 - u_i/u_j)$. By the AM-GM inequality this term is at most $(\sqrt{\alpha_i q_{ij}} - \sqrt{\alpha_j q_{ji}})^2$. The choice $u_i = \sqrt{\alpha_i/\nu_i}$, with $u_i = \varepsilon \rightarrow 0$ when $\alpha_i = 0$, attains the bound in the limit by reversibility. $\square$

In words, the most likely way to stay on F is to spend the fractions α_F^* of the time in its states, and the rate at any other fractions is strictly larger. Part (d) says that fractions on the boundary of Δ_F , which miss some states of F , have the rate of the smaller face that they use, and this rate is larger than λ_F . The lower bound in (a), the value at α_F^* and the interior form of (c) are checked in Lean. For symmetric Q_F , $l_F = r_F$ and α_F^* is the entrywise square of the Perron vector normalized in ℓ^2 .

3.3 Face selection

We also need an upper bound that holds on a whole face, including near its boundary, where the local law of Section 4 does not apply. For an irreducible face F put $V_F = \{\theta \in \mathbb{R}^F : \varrho(A_F - \text{diag } \theta) = 0\}$. Note that $q - \lambda_F \mathbf{1} \in V_F$, since $A_F - \text{diag}(q - \lambda_F \mathbf{1}) = Q_F + \lambda_F I$. Each $\theta \in V_F$ gives an exponential bound with the linear rate $\alpha \mapsto \langle q - \theta, \alpha \rangle$, and I_F is the supremum of these linear rates.

Lemma 3.2 (a uniform upper bound). *Let F be irreducible with $k = |F| \geq 2$, and let $\theta \in V_F$. There is C_θ , depending on θ , Q and μ , such that for every $n \in \mathbb{Z}_{\geq 0}^F$ with $\text{supp } n = F$ and every h with $h q_{\max} \leq \frac{1}{2}$, the probability that $X_1, \dots, X_{|n|}$ have composition n is at most $C_\theta h^{k-1} \exp(-h \langle n, q - \theta \rangle + C_\theta h^2 |n|)$.*

The proof is in Appendix A. With $\theta = q - \lambda_F \mathbf{1}$ it gives $P_N(n) \leq C h^{k-1} e^{-T \lambda_F + C T^2/N}$ for every n with support F . As $h^{k-1} = T^{k-1} N^{-(k-1)}$, this is the upper half of the rule $M_F \approx N^{-(k-1)} e^{-T \lambda_F}$ of Section 3.1, up to a power of T . The lemma is stated for a path of any length $|n|$, not only N , so it also applies to the blocks of a path.

The next theorem uses the local law (Theorem 4.4) and the face maxima (Theorem 4.6) of Section 4, which comes later. There is no circularity, since those two theorems are proved there without Theorem 3.3.

Theorem 3.3 (first-order face selection). *Fix Q and μ .*

- (i) *Let F satisfy (H), let $k = |F|$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. There is N_0 such that, uniformly for $N \geq N_0$, $1 \leq T \leq N^{1/3}$ and $\text{supp } n = F$ with $n/N \in \mathcal{K}$,*

$$\log P_N(n) = -(k-1)L - T I_F(n/N) + O_{\mathcal{K}}(1 + \log T).$$

- (ii) *Let F be admissible with components $C_1, \dots, C_r$, and let k^* be the least size of a component C_j with $\lambda_{C_j} = \lambda_F$. There are κ , T_0 and N_0 such that for $N \geq N_0$ and $T_0 \leq T \leq N^{1/3}$*

$$\begin{aligned} \log M_F &\geq -(k_F - 1)L + (k_F - k^*) \log T - T\lambda_F - \kappa, \\ \log M_F &\leq -(k_F - 1)L + (k_F - 1) \log T - T\lambda_F + \kappa. \end{aligned}$$

If F is not admissible, $M_F = 0$.

- (iii) *Let $T = \tau L$ with $\tau > 0$ fixed, and let $\mathcal{F}_\tau$ be the set of admissible faces that minimize Φ_τ . Then $\log \max_n P_N(n) = -L \min_F \Phi_\tau(F) + O(\log L)$, with the minimum over admissible faces, and for large N every global mode $\hat{n}$ has $\text{supp } \hat{n} \in \mathcal{F}_\tau$. This holds for every start μ .*
- (iv) *If in (iii) every face in $\mathcal{F}_\tau$ is irreducible, then every global mode $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^* + O(1/L)$ with $F = \text{supp } \hat{n}$. This is the case when Q has symmetric support or μ has full support (Theorem 3.4(c)).*

In words, (i) says that a composition with fractions α inside F has probability $N^{-(k-1)}e^{-TI_F(\alpha)}$ up to a power of T . Part (ii) says that the best composition on an admissible face has probability $N^{-(k_F-1)}e^{-T\lambda_F}$ times a power of T with exponent between 0 and $k_F - 1$, up to a bounded factor. So by (iii), with $T = \tau L$ the exponent of N is $-\Phi_\tau(F)$, and the mode lies on a face that minimizes it, whatever the start. By (iv), inside the winning face the mode sits near the quasi-ergodic law.

Idea of the proof. For faces with (H), part (i) and the face maxima come from the local law of Section 4, which we prove independently of this theorem. A reducible face is visited in blocks, one for each strongly connected component. For the upper bound in (ii) we apply Lemma 3.2 to each block, and for the lower bound we write down explicit paths that spend almost all their time in a component of least exit rate. Part (iii) compares the faces, and (iv) follows from Theorem 4.6.

Proof. (i) For $k = 1$ this is (2), since $I_{\{i\}} = q_i$. For $k \geq 2$, Lemma 4.5 gives $P_N(n) = N^{-(k-1)}f_T(n/N)(1 + O(T^2/N))$ with $T^2/N \leq N^{-1/3}$. We claim that $\log f_T(\alpha) = \frac{k-1}{2} \log T + \log C_F(\alpha) - TI_F(\alpha) + O_{\mathcal{K}}(1)$ for $\alpha \in \mathcal{K}$ and $T \geq 1$. For $T \geq T_1$, with T_1 large, this is Theorem 4.4. For $1 \leq T \leq T_1$ it holds because $f_T(\alpha)$ is positive and continuous on the compact set $[1, T_1] \times \mathcal{K}$ (see the remarks after Theorem 4.4). Finally C_F is positive and continuous on $\mathcal{K}$, so $\log C_F$ is bounded there. This gives (i).

(ii) If F is not admissible, $M_F = 0$. If F is admissible and irreducible, it satisfies (H) by Lemma 2.2(b), and $k^* = k_F$. Then Theorem 4.6 gives both bounds, since $0 \leq \frac{k_F-1}{2} \log T \leq (k_F - 1) \log T$. For a reducible face, a path visits each component in one block of consecutive times (proof of Lemma 2.2). For the upper bound, Lemma 3.2 (or a direct count if $|C| = 1$) bounds a block of length N_C in the component C by $O(h^{|C|-1}e^{-h\lambda_C N_C})$, and each of the $r - 1$ switches between blocks gives a factor h . As $\lambda_C \geq \lambda_F$, this gives $M_F = O(h^{k_F-1}e^{-T\lambda_F})$. For the lower

bound the path spends about N/T visits in each component except one of least exit rate, which loses a bounded factor per component, and we divide by the number of compositions. The details are in Appendix A.6.

(iii) For large N we have $T_0 \leq T = \tau L \leq N^{1/3}$, and by (ii) $\log M_F = -L \Phi_\tau(F) + O(\log L)$ for every admissible face, while $M_F = 0$ for the others. Since $\max_n P_N(n) = \max_F M_F$, the first claim follows. Let $m_\tau = \min_F \Phi_\tau(F)$. An admissible $F \notin \mathcal{F}_\tau$ has $\Phi_\tau(F) \geq m_\tau + \varepsilon_0$ for a fixed $\varepsilon_0 > 0$, so $M_F \leq N^{-m_\tau - \varepsilon_0 + o(1)}$, which is smaller than $\max_n P_N(n)$ for large N . A global mode has positive probability, so its support is admissible and lies in $\mathcal{F}_\tau$.

(iv) By (iii), for large N a global mode $\hat{n}$ maximizes P_N on its face $F \in \mathcal{F}_\tau$, which is irreducible and admissible and so satisfies (H) by Lemma 2.2(b). So Theorem 4.6 with $T = \tau L \leq N^{1/3}$ gives $\hat{n}/N = \alpha_F^* + O(1/L)$. $\square$

For a reducible face we only have the bounds in (ii), and Proposition 4.13 shows that the power of T can differ from $(k_F - 1)/2$.

3.4 Reading the winners off a convex hull

For an admissible face F put $P_F = (\lambda_F, |F| - 1)$, and let S be the set of these points. Then $\Phi_\tau(F) = \tau x + y$ at $(x, y) = P_F$. Picture a line of slope $-\tau$ moving up from far below. The first points of S that it touches are the winners at τ . For each $\tau > 0$ the points of $\text{conv } S$ where $\tau x + y$ is least form a vertex or an edge of $\text{conv } S$, and we call the union of these over all $\tau > 0$ the lower-left boundary of $\text{conv } S$. Figure 3(b) shows it for the 4-cycle.

Theorem 3.4 (geometry of the winners). *Let $\mathcal{F}_\tau$ be the set of admissible faces that minimize Φ_τ .*

- (a) *The lower-left boundary of $\text{conv } S$ is a path whose vertices lie in S . Read from right to left, they are $v_i = (\lambda^{(i)}, y^{(i)})$, $0 \leq i \leq m$, with $\lambda^{(0)} > \dots > \lambda^{(m)}$ and $y^{(0)} < \dots < y^{(m)}$. Put $\tau_i = (y^{(i)} - y^{(i-1)})/(\lambda^{(i-1)} - \lambda^{(i)})$ for $1 \leq i \leq m$, $\tau_0 = 0$ and $\tau_{m+1} = \infty$. Then $\tau_1 < \dots < \tau_m$. For $\tau_i < \tau < \tau_{i+1}$, $\mathcal{F}_\tau$ is the set of faces F with $P_F = v_i$. At $\tau = \tau_i$ it is the set of faces whose point lies on the closed edge $[v_{i-1}, v_i]$.*
- (b) *If $0 < \tau < \tau'$, $F \in \mathcal{F}_\tau$ and $G \in \mathcal{F}_{\tau'}$, then $\lambda_F \geq \lambda_G$ and $|F| \leq |G|$. Both inequalities are strict when $\mathcal{F}_\tau \cap \mathcal{F}_{\tau'} = \emptyset$.*
- (c) *If Q has symmetric support, or μ has full support, then every face in $\mathcal{F}_\tau$ is irreducible, for every $\tau > 0$.*

In words, as τ grows the winners walk up the lower-left boundary of the hull, from small faces with large exit rates to large faces with small exit rates, and they change at the values τ_i , minus the slopes of its edges.

Idea of the proof. The least value $g(\tau)$ of Φ_τ is a minimum of finitely many affine functions of τ , so it is concave and piecewise affine. Its pieces are the vertices v_i and its kinks are the values τ_i . Part (b) adds two inequalities, and part (c) replaces a reducible face by one of its components.

Proof. (a) Let $g(\tau) = \min_{(x,y) \in S} (\tau x + y)$. It is a minimum of finitely many affine functions, so it is concave and piecewise affine. Let $\tau_1 < \dots < \tau_m$ be the points where its slope changes. A point of S that attains g inside (τ_i, τ_{i+1}) attains it on the whole interval, so exactly one point v_i does. The slope decreases, so $\lambda^{(0)} > \dots > \lambda^{(m)}$, and continuity at τ_i gives the formula for τ_i and $y^{(i)} > y^{(i-1)}$. A point of S that attains $g(\tau_i)$ lies on the line through v_{i-1} and v_i , and comparing with g slightly

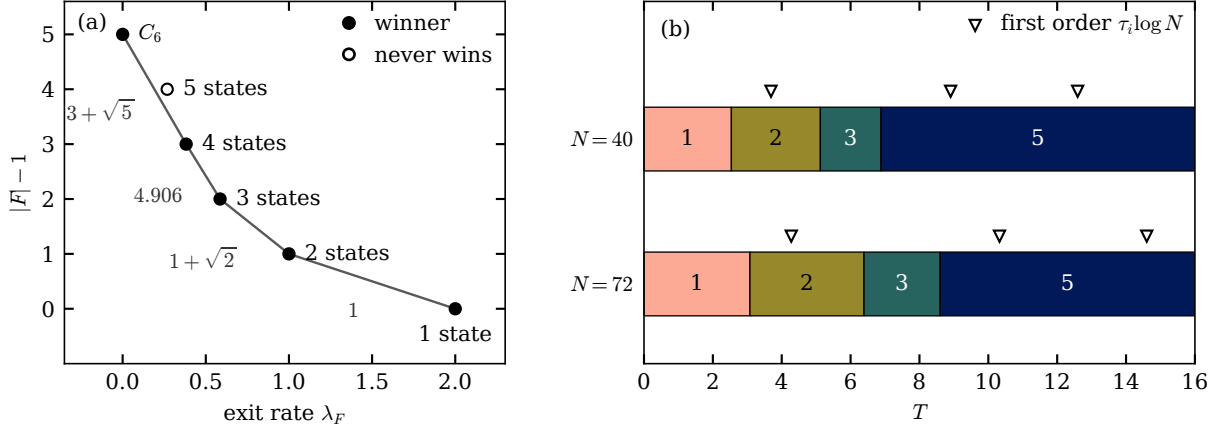


Figure 4: (a) The points $(\lambda_F, |F| - 1)$ of the arcs of the 6-cycle and of C_6 , with the lower-left boundary of their hull. The numbers beside the edges are the switch values τ_i of Theorem 3.4. The winners (filled) are the arcs of one to four states and C_6 , and the arc of five states (open) never wins. (b) The support size of the exact mode on C_5 with the uniform start, against T , at $N = 40$ and $N = 72$. The mode skips the arc of four states. The triangles mark the first-order switch points $\tau_i \log N$ with $\tau_i = 1, 1 + \sqrt{2}$ and $2 + \sqrt{2}$ (Table 1).

below and above τ_i puts it on the closed edge. The minimizers on $\text{conv } S$ are the hull of those on S , so the lower-left boundary is the path $v_0 \cdots v_m$.

(b) Adding $\Phi_\tau(F) \leq \Phi_\tau(G)$ and $\Phi_{\tau'}(G) \leq \Phi_{\tau'}(F)$ gives $(\tau' - \tau)(\lambda_F - \lambda_G) \geq 0$, and then $|F| - |G| \leq \tau(\lambda_G - \lambda_F) \leq 0$. If $\lambda_F = \lambda_G$, then $|F| = |G|$ and $F \in \mathcal{F}_{\tau'}$. So if the two sets are disjoint, $\lambda_F > \lambda_G$ and then $|F| < |G|$.

(c) With symmetric support use Lemma 2.2(c). If μ has full support and F is admissible and reducible, a component C with $\lambda_C = \lambda_F$ is admissible by Lemma 2.2(b), and $\Phi_\tau(C) \leq \Phi_\tau(F) - 1$. $\square$

The finite form of Theorem 3.4 is checked in Lean. A face with small λ_F for its size is a set of states that the chain rarely leaves, a well-separated community. As τ grows the exit rate counts for more, and the mode moves to larger communities. It need not grow around its first vertex, and it can move to a disjoint community (Table 1). For symmetric Q , λ_F is the least value of $\sum_{i < j} q_{ij}(f_i - f_j)^2$, the sum over all pairs in $[d]$, over unit vectors f supported in F (Rayleigh–Ritz), and it decreases strictly as a connected face grows, by Proposition 3.1(d). So only the least exit rate of each size matters. Figure 4(a) shows S for the 6-cycle.

4 The second-order theory

First order locates a crossing only up to a factor $1 + o(1)$ in T . To locate it up to $o(1)$ we need the constant in front of $N^{-(k-1)}e^{-T\lambda_F}$ for every face that satisfies (H), together with the power of T . The plan is as follows. Section 4.1 defines the constant K_F . Section 4.2 proves a local limit theorem inside a face and deduces M_F up to a factor $1 + O(1/T + T^2/N)$. Section 4.3 compares two faces, Section 4.4 recovers Papers A and B, and Section 4.5 shows what can change without (H). Section 4.6 adds the next term, of relative order $1/T$. In this section gradients and Hessians in α are taken in the coordinates $\alpha' = (\alpha_2, \dots, \alpha_k)$, $|\cdot|$ is the Euclidean norm, and $q = (q_i)_{i \in F}$.

4.1 The tilted chain and the constant

To find the probability of a composition with fractions α we change the rates so that α becomes typical. The optimal vector r_α in the definition of $I_F(\alpha)$ gives a new generator $\widehat{Q}_\alpha$ on F , with the same switching graph, whose stationary law is exactly α . Under it the visit fractions satisfy a central limit theorem with covariance Γ_α/T , and the price of the change of rates is $e^{-T I_F(\alpha)}$ times boundary factors for the start and the end of the path. The constant $C_F(\alpha)$ collects these factors and the Gaussian normalization.

Definition 4.1 (tilt and constants). Let F be irreducible with $k \geq 2$. For $\alpha \in \Delta_F^\circ$ let r_α attain the supremum defining $I_F(\alpha)$ (Proposition 3.1(b)), and put $\theta(\alpha) = A_F r_\alpha / r_\alpha$ and $l_\alpha = \alpha / r_\alpha$ (entrywise). The *tilted generator* is $\widehat{Q}_\alpha = R^{-1}(A_F - \text{diag } \theta(\alpha))R$ with $R = \text{diag } r_\alpha$. With $D = \text{diag } \alpha$ let

$$Z_\alpha = (\mathbf{1}\alpha^\top - \widehat{Q}_\alpha)^{-1} - \mathbf{1}\alpha^\top, \quad \Gamma_\alpha = DZ_\alpha + Z_\alpha^\top D,$$

and let Σ_α be Γ_α with the row and column of state 1 deleted. Finally let $K_{\{i\}} = \mu_i$ for a single state, and

$$C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(2\pi)^{(k-1)/2} \sqrt{\det \Sigma_\alpha}}, \quad K_F = C_F(\alpha_F^*).$$

Nothing in Definition 4.1 changes when r_α is scaled. Note that $l_\alpha \cdot r_\alpha = 1$ and $I_F(\alpha) = \langle q - \theta(\alpha), \alpha \rangle$. Since r_α minimizes $u \mapsto \sum_{i \neq j} \alpha_i q_{ij} u_j / u_i$, its derivative in u_j , which is $\sum_i l_{\alpha,i} q_{ij} - l_{\alpha,j} \theta_j(\alpha)$, vanishes at r_α . Hence

$$l_\alpha^\top (A_F - \text{diag } \theta(\alpha)) = 0. \quad (3)$$

Also $(A_F - \text{diag } \theta(\alpha))r_\alpha = 0$ with $r_\alpha > 0$, so $\theta(\alpha) \in V_F$ [36]. So $\widehat{Q}_\alpha$ is an irreducible generator with the pattern of A_F and, by (3), stationary law α . If $(\mathbf{1}\alpha^\top - \widehat{Q}_\alpha)v = 0$, multiplying by $\alpha^\top$ gives $\alpha^\top v = 0$, so $\widehat{Q}_\alpha v = 0$ and $v = 0$. Thus Z_α exists, and

$$Z_\alpha \mathbf{1} = 0, \quad \alpha^\top Z_\alpha = 0, \quad -\widehat{Q}_\alpha Z_\alpha = I - \mathbf{1}\alpha^\top. \quad (4)$$

So $\Gamma_\alpha \mathbf{1} = 0$. As Γ_α is symmetric, its adjugate is then a multiple of $\mathbf{1}\mathbf{1}^\top$, so deleting another state instead of state 1 gives the same $\det \Sigma_\alpha$. At α_F^* the vector $u = r_F$ gives the value $\lambda_F = I_F(\alpha_F^*)$, so $r_{\alpha_F^*}$ is a multiple of r_F , $\theta(\alpha_F^*) = q - \lambda_F \mathbf{1}$, and $l_{\alpha_F^*}$ is the matching multiple of l_F . In the local law below, $\mu \cdot r_\alpha$ accounts for the start of the path, $l_\alpha \cdot \mathbf{1}$ for its free end, and Σ_α for the Gaussian spread of the occupation fractions.

For conductances $c_{ij} = c_{ji}$ on F let $\text{tree}(c) = \sum_{\mathcal{T}} \prod_{\{i,j\} \in \mathcal{T}} c_{ij}$, the sum over the spanning trees $\mathcal{T}$ of the complete graph on F .

Proposition 4.2 (the constant). *Let F be irreducible with $k \geq 2$.*

- (a) *The maps $\alpha \mapsto \theta(\alpha)$, $I_F(\alpha)$, Σ_α , r_α and l_α (with $\sum_i r_{\alpha,i} = 1$) are real-analytic on Δ_F° , Σ_α is positive definite, and $\nabla^2 I_F(\alpha) = \Sigma_\alpha^{-1}$. Hence $\det \Sigma_\alpha \cdot \det \nabla^2 I_F(\alpha) = 1$ and $C_F(\alpha) = (\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1}) \sqrt{\det \nabla^2 I_F(\alpha) / (2\pi)^{k-1}}$.*
- (b) *Suppose that $\nu_i q_{ij} = \nu_j q_{ji}$ for $i, j \in F$ and some $\nu \in (0, \infty)^F$. Then r_α is a multiple of $\sqrt{\alpha/\nu}$ and l_α of $\sqrt{\alpha\nu}$, and $\widehat{Q}_\alpha$ is reversible with conductances $c_{ij}(\alpha) = \sqrt{\alpha_i \alpha_j} q_{ij} \sqrt{\nu_i / \nu_j}$. Also $I_F(\alpha) = \sum_i q_i \alpha_i - \sum_{i \neq j} c_{ij}(\alpha)$, and*

$$\det \Sigma_\alpha = \frac{2^{k-1} (\prod_i \alpha_i)^2}{\text{tree}(c(\alpha))}, \quad C_F(\alpha) = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{(4\pi)^{(k-1)/2}} \frac{\sqrt{\text{tree}(c(\alpha))}}{\prod_i \alpha_i}.$$

In words, the Gaussian factor in C_F is the square root of the determinant of the Hessian of the rate. For a reversible face it is given by a weighted count of spanning trees.

Proof. The proof of (a) is in Appendix A and uses Lemma A.1.

(b) Put $u = \sqrt{\alpha/\nu}$ and $l = \alpha/u = \sqrt{\alpha\nu}$. Then $\sum_i l_i q_{ij} = \sum_i \sqrt{\alpha_i \nu_i} q_{ij}$ and $l_j (A_F u)_j / u_j = \nu_j \sum_i q_{ji} \sqrt{\alpha_i / \nu_i}$ agree because $\nu_j q_{ji} = \nu_i q_{ij}$. So u is a critical point of $\sum_{i \neq j} \alpha_i q_{ij} u_j / u_i$, which is convex in $\log u$, and r_α is a multiple of u . Then $\alpha_i (\hat{Q}_\alpha)_{ij} = l_i q_{ij} u_j = c_{ij}(\alpha)$, which is symmetric by reversibility, and $\alpha_i \theta_i(\alpha) = \sum_j c_{ij}(\alpha)$ gives I_F .

For the determinant let $\mathcal{L} = -D\hat{Q}_\alpha$, the Laplacian of the conductances. Fix η with $\eta_1 = 0$, and let $u = Z_\alpha \eta$ and $\zeta = D\eta - (\alpha \cdot \eta)\alpha$. By (4), $\mathcal{L}u = \zeta$ and $\alpha \cdot u = 0$, so $\eta^\top \Gamma_\alpha \eta = 2(D\eta)^\top u = 2\zeta^\top u$. Let $v = u - u_1 \mathbf{1}$. As $\zeta \cdot \mathbf{1} = 0$ and $\mathcal{L}\mathbf{1} = 0$, we get $\zeta^\top u = \zeta^\top v$ and $\mathcal{L}v = \zeta$, and since $v_1 = 0$ the rows $2, \dots, k$ give $\mathcal{L}_{(1)} v' = \zeta'$. Here $\mathcal{L}_{(1)}$ is $\mathcal{L}$ without row and column 1, primes denote the coordinates $2, \dots, k$, and $\zeta' = (D' - \alpha' \alpha'^\top) \eta'$ with $D' = \text{diag } \alpha'$. Hence $\Sigma_\alpha = 2(D' - \alpha' \alpha'^\top) \mathcal{L}_{(1)}^{-1} (D' - \alpha' \alpha'^\top)$. The matrix determinant lemma gives $\det(D' - \alpha' \alpha'^\top) = \prod_{i \geq 2} \alpha_i (1 - \sum_{i \geq 2} \alpha_i) = \prod_i \alpha_i$, and the matrix-tree theorem [37] gives $\det \mathcal{L}_{(1)} = \text{tree}(c(\alpha))$. $\square$

Corollary 4.3 (symmetric faces). *Let F satisfy (H) with $k \geq 2$, and let Q_F be symmetric with constant row sums. Then $\lambda_F = q_i - \sum_{j \in F \setminus \{i\}} q_{ij}$ for every $i \in F$, α_F^* is uniform on F , and*

$$K_F = \mu(F) k^{(k+1)/2} \sqrt{\text{tree}(q_F)} (4\pi)^{-(k-1)/2},$$

where $\text{tree}(q_F)$ is $\text{tree}(c)$ for the conductances $c_{ij} = q_{ij}$, $i, j \in F$.

Proof. The positive vector $\mathbf{1}$ is a right and left eigenvector of Q_F , so it belongs to the Perron root. So λ_F is minus the row sum, r_F and l_F are multiples of $\mathbf{1}$, and $\alpha_F^* = \mathbf{1}/k$. At $\alpha = \mathbf{1}/k$, Proposition 4.2(b) with $\nu = \mathbf{1}$ gives $r_\alpha = \mathbf{1}$, $l_\alpha = \alpha$ and $(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1}) = \mu(F)$. Also $c_{ij}(\alpha) = q_{ij}/k$, so $\text{tree}(c(\alpha)) = k^{-(k-1)} \text{tree}(q_F)$, and $\prod_i \alpha_i = k^{-k}$. $\square$

4.2 The local law and the face maxima

Inside the face the occupation fractions fluctuate on the scale $T^{-1/2}$ in $k-1$ directions. So a single composition gets the factor $T^{(k-1)/2}$ times the volume $N^{-(k-1)}$ of a lattice cell.

Theorem 4.4 (local law). *Let F satisfy (H) with $k \geq 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. Uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,*

$$f_T(\alpha) = T^{(k-1)/2} C_F(\alpha) e^{-T I_F(\alpha)} (1 + O_{\mathcal{K}}(1/T)).$$

For $F = \{i\}$, $f_T = \mu_i e^{-q_i T}$.

In words, the density of the visit fractions on the event that the chain stays in F is a Gaussian factor $T^{(k-1)/2}$ times the large-deviation factor $e^{-T I_F(\alpha)}$ times the constant $C_F(\alpha)$, with a relative error $O(1/T)$ that is uniform away from the boundary of the face.

Idea of the proof. The proof is in Appendix A. The sum over words that defines H is a matrix resolvent, so Cauchy's formula writes H as an integral over a torus of radii $\theta(\alpha)$ (Lemma A.2). All the coefficients are nonnegative, so the integrand is largest at the real point, and a Laplace lemma with a parameter (Lemma B.1) gives the expansion. The Hessian at the real point is computed from the Perron root of a perturbed generator (Lemma A.1), which brings in Σ_α . The hypothesis $\mu(F) > 0$ is needed, since $f_T = C_F = 0$ when $\mu(F) = 0$. Also $f_T(\alpha)$ is continuous and positive in (T, α) , as H is a convergent power series with some $W(m) > 0$ by (H).

Lemma 4.5 (discrete and continuum). *Let F satisfy (H) with $k \geq 2$, and let $\mathcal{K} \subset \Delta_F^\circ$ be compact. Uniformly for $\text{supp } n = F$ with $n/N \in \mathcal{K}$ and $1 \leq T \leq N^{1/2}$, $P_N(n) = N^{-(k-1)} f_T(n/N)(1 + O_{\mathcal{K}}(T^2/N))$.*

In words, the probability of a composition is the continuum density times the volume $N^{-(k-1)}$ of a lattice cell. The proof is in Appendix A. With Theorem 4.4 the lemma replaces the term $O_{\mathcal{K}}(1 + \log T)$ in Theorem 3.3(i) by $\frac{k-1}{2} \log T + \log C_F(n/N) + O_{\mathcal{K}}(1/T + T^2/N)$.

Theorem 4.6 (face maxima). *Let F satisfy (H). If $k \geq 2$, then uniformly for $1 \leq T \leq N^{1/3}$,*

$$M_F = N^{-(k-1)} T^{(k-1)/2} K_F e^{-T\lambda_F} (1 + O(1/T + T^2/N)),$$

and every maximizer $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^* + O(1/T)$. For $k = 1$, $M_{\{i\}}$ is given by (2). In both cases, uniformly for $1 \leq T \leq N^{1/3}$,

$$\log M_F = -(k-1)L + \frac{k-1}{2} \log T + \log K_F - T\lambda_F + O(1/T + T^2/N). \quad (5)$$

In words, the best composition on F sits within $O(N/T)$ of $N\alpha_F^*$, and its probability is the Gaussian peak of the local law at α_F^* . The factor $N^{-(k-1)}$ is the volume of a lattice cell, and $T^{(k-1)/2}$ is the height of a Gaussian density whose spread in each free direction is of order $T^{-1/2}$.

Idea of the proof. Near α_F^* we use the local law and Lemma 4.5, and maximize a Gaussian with a small linear perturbation. Away from α_F^* the rate exceeds λ_F by a fixed amount, and Lemma 3.2, applied with finitely many vectors θ , shows that these compositions are exponentially less likely in T . For bounded T positivity and continuity suffice.

Proof. For $k = 1$ this is (2). Let $k \geq 2$, put $X = N^{-(k-1)} T^{(k-1)/2} K_F e^{-T\lambda_F}$ and $x = \alpha' - \alpha_F^*$. By Propositions 3.1(a) and 4.2(a), I_F and $\log C_F$ are real-analytic near α_F^* , I_F has its minimum λ_F there and $\nabla^2 I_F > 0$. So there is a closed ball $U = \{|x| \leq \rho_0\} \subset \Delta_F^\circ$ on which $c_U |x|^2 \leq I_F - \lambda_F \leq C_U |x|^2$, $|\log(C_F/K_F) - \gamma \cdot x| \leq C_1 |x|^2$ with $\gamma = \nabla \log C_F(\alpha_F^*)$, and $\log C_F$ is C_3 -Lipschitz.

Outside U . As I_F is lower semicontinuous, Proposition 3.1(a) gives $I_F \geq \lambda_F + 3\eta$ on $\Delta_F \setminus U^\circ$ for some $\eta > 0$. Since $I_F(\beta) = \sup_{u>0} \langle q - A_F u/u, \beta \rangle$ and $A_F u/u \in V_F$, compactness gives finitely many $\theta_j \in V_F$ with $\max_j \langle q - \theta_j, \beta \rangle > \lambda_F + 2\eta$ on $\Delta_F \setminus U^\circ$. Lemma 3.2 then gives $P_N(n) \leq C(T/N)^{k-1} e^{-T(\lambda_F + 2\eta)} = X \cdot O(T^{(k-1)/2} e^{-2\eta T})$ for $\text{supp } n = F$ and $n/N \notin U^\circ$.

Inside U . By Theorem 4.4 and Lemma 4.5 with $\mathcal{K} = U$, for $\alpha = n/N \in U$,

$$P_N(n) = X \frac{C_F(\alpha)}{K_F} e^{-T(I_F(\alpha) - \lambda_F)} (1 + E), \quad |E| \leq C_2(1/T + T^2/N). \quad (6)$$

For $T \geq 2C_1/c_U$ the middle factors are at most $\exp(\sup_x [\gamma \cdot x - \frac{c_U T}{2} |x|^2]) = e^{|\gamma|^2/(2c_U T)}$, so $M_F \leq X(1 + O(1/T + T^2/N))$. For the lower bound let n^0 have support F and $|n_i^0 - N\alpha_{F,i}^*| < 1$ for all i , which exists for $N \geq N_0$ since every $\alpha_{F,i}^* > 0$. Then $C_F(n^0/N) \geq K_F e^{-kC_3/N}$ and $T(I_F(n^0/N) - \lambda_F) \leq C_U k^2 T/N^2$, so (6) gives $P_N(n^0) \geq X(1 - O(1/T + T^2/N))$. Fix T_2 so large that the two bounds above give $M_F/X \in [\frac{1}{2}, \frac{3}{2}]$ for $T \geq T_2$. For $1 \leq T \leq T_2$, f_T is positive and continuous on $[1, T_2] \times U$. So Lemma 4.5 and the bound outside U give $M_F/X \in [c, C]$ with constants $0 < c < C$. This gives the first display and (5).

The maximizer. For $T \geq T_3$ with T_3 large, the bound outside U is below $X/2 \leq P_N(n^0)$, so $\hat{\alpha} = \hat{n}/N \in U$ and $|E| \leq \frac{1}{2}$. Taking logarithms in $P_N(\hat{n}) \geq P_N(n^0)$ gives $c_U T |\hat{x}|^2 \leq C_3 |\hat{x}| + C_4(1/T + T^2/N) \leq C_3 |\hat{x}| + 2C_4/T$ with $\hat{x} = \hat{\alpha}' - \alpha_F^*$, because $T \leq N^{1/3}$ gives $T^2/N \leq 1/T$. This step uses the exponent $\frac{1}{3}$, and it gives $|\hat{x}| = O(1/T)$. For $1 \leq T < T_3$ the claim holds since $|\hat{x}|$ is bounded. $\square$

4.3 Crossings and tie windows

We now compare two faces. Taking logarithms in Theorem 4.6, the difference $\log M_F - \log M_G$ is an explicit increasing function of T up to $O(1/T + T^2/N)$, and its zero is the crossing.

Theorem 4.7 (the crossing equation). *Let F and G satisfy (H), with $\Delta = k_F - k_G \geq 1$ and $\delta = \lambda_G - \lambda_F > 0$, and let $D_N(T) = \log M_F - \log M_G$.*

(i) *For large N , D_N has a zero in $[\frac{\Delta}{2\delta}L, \frac{2\Delta}{\delta}L]$.*

(ii) *For large N , every zero $T_N \in [1, N^{1/3}]$ of D_N satisfies*

$$\delta T_N + \frac{\Delta}{2} \log T_N = \Delta L - \log \frac{K_F}{K_G} + O\left(\frac{1}{L}\right). \quad (7)$$

(iii) *All such zeros lie in one interval of length $O(1/L)$, and*

$$T_N = \frac{\Delta}{\delta} \left[L - \frac{1}{2} \log L + b_{FG} \right] + O\left(\frac{\log L}{L}\right), \quad b_{FG} = -\frac{1}{2} \log \frac{\Delta}{\delta} - \frac{1}{\Delta} \log \frac{K_F}{K_G}.$$

Proof. By (5), uniformly on $[1, N^{1/3}]$, $D_N(T) = -\Delta L + \frac{\Delta}{2} \log T + \delta T + \log \frac{K_F}{K_G} + O(1/T + T^2/N)$.

(i) At $T = \frac{\Delta}{2\delta}L$ this is $-\frac{\Delta}{2}L + O(\log L) < 0$, and at $T = \frac{2\Delta}{\delta}L$ it is $\Delta L + O(\log L) > 0$. For fixed N , M_F is a maximum of finitely many polynomials in h , and it is positive because F is admissible. So D_N is continuous and has a zero in between.

(ii) Let $\psi(T) = \delta T + \frac{\Delta}{2} \log T$, which is increasing. At a zero $T_N \in [1, N^{1/3}]$, $\psi(T_N) = \Delta L - \log(K_F/K_G) + O(1)$. This is larger than $\psi(\frac{\Delta}{2\delta}L) = \frac{\Delta}{2}L + O(\log L)$, and $\delta T_N \leq \psi(T_N)$. So $\frac{\Delta}{2\delta}L < T_N \leq \frac{2\Delta}{\delta}L$, and then $O(1/T_N + T_N^2/N) = O(1/L)$.

(iii) Let $\psi(T^\circ) = \Delta L - \log(K_F/K_G)$. As $\psi' \geq \delta$, every zero is within $O(1/L)$ of T° . Also $T^\circ = \frac{\Delta}{\delta}L + O(\log L)$, so $\log T^\circ = \log L + \log \frac{\Delta}{\delta} + O(\log L/L)$, which gives the explicit form. $\square$

In words, the larger face F pays $N^{-\Delta}$ for spreading and gains $e^{\delta T}$ for staying, and its Gaussian peak gives an extra factor $T^{\Delta/2}$. The crossing balances the three. The explicit form in (iii) converges slowly, so computations should solve (7) (see Corollary 5.1). We do not prove that the zero of D_N is unique in general.

At a switch value τ^* of Theorem 3.4 several faces minimize Φ_{τ^*} , sometimes more than two, as all faces do at $\tau^* = 1$ for the model of Paper B. Then the competition plays out in a window $T = \tau^*(L - \frac{1}{2} \log L + t)$ with t of order 1. In this window each tied face has a line $\ell_F(t)$, and the mode follows the highest line.

Theorem 4.8 (tie windows). *Fix $\tau^* > 0$. Let m^* be the least value of Φ_{τ^*} over the admissible faces and $\mathcal{F}^*$ the set of minimizers. Assume that every face in $\mathcal{F}^*$ is irreducible. By Theorem 3.4(c) this holds if Q has symmetric support or μ has full support. Put $T = \tau^*(L - \frac{1}{2} \log L + t)$, and for $F \in \mathcal{F}^*$ let $\ell_F(t) = \log K_F + \frac{k_F-1}{2} \log \tau^* - \tau^* \lambda_F t$ and $E(t) = \max_{G \in \mathcal{F}^*} \ell_G(t)$.*

(i) *Uniformly for t in compact sets, for $F \in \mathcal{F}^*$, $\log M_F = -m^*L + \frac{m^*}{2} \log L + \ell_F(t) + O(\log L/L)$.*

(ii) *There are $\varepsilon_1, \varepsilon_0 > 0$ such that for large N and $|T/L - \tau^*| \leq \varepsilon_1$, every admissible $F \notin \mathcal{F}^*$ has $M_F \leq N^{-\varepsilon_0} \max_{G \in \mathcal{F}^*} M_G$.*

(iii) *There are $\varepsilon, t_0 > 0$ such that for large N and $t_0 \leq |t| \leq \varepsilon L$, every global mode lies on a face $F \in \mathcal{F}^*$ with $\ell_F = E$ on $[t_0, \infty)$ if $t > 0$, or on $(-\infty, -t_0]$ if $t < 0$. Such a face has the least λ_F in $\mathcal{F}^*$ if $t > 0$ and the largest if $t < 0$.*

Consequently, for every $\eta > 0$ and large N , if $|t| \leq \varepsilon L$ is at distance at least η from every kink of E , then every global mode lies on a face whose line equals E near t . So the line of the face that carries the global mode changes only at $T = \tau^*(L - \frac{1}{2} \log L + t_j) + o(1)$, where t_j runs over the kinks of E , and it changes near each kink. Faces with identical lines (the same k_F , λ_F and K_F) tie at this order and give no crossing.

In words, inside the window the logarithm of every tied face maximum is the same leading term plus the line $\ell_F(t)$, faces that do not tie are smaller by a power of N , and far out in the window the face with the least (for $t > 0$) or largest (for $t < 0$) exit rate wins.

Proof. Every $F \in \mathcal{F}^*$ is admissible and irreducible, so it satisfies (H) by Lemma 2.2(b). Write $T = \tau^*L(1 + y)$ with $y = (t - \frac{1}{2} \log L)/L$, so that $|y| \leq \frac{1}{2}$ for $|t| \leq L/4$ and large N . By (5) and $\tau^*\lambda_F + k_F - 1 = m^*$, uniformly for $|t| \leq L/4$,

$$\log M_F = -m^*L + \frac{m^*}{2} \log L + \ell_F(t) + \frac{k_F-1}{2} \log(1+y) + O(1/L) \quad (F \in \mathcal{F}^*). \quad (8)$$

For t in a compact set, $\log(1+y) = O(\log L/L)$, which gives (i).

(ii) Let $g > 0$ be the least value of $\Phi_{\tau^*}(F) - m^*$ over the admissible $F \notin \mathcal{F}^*$. As $0 \leq \lambda_F \leq q_{\max}$, for $|\tau - \tau^*| \leq \varepsilon_1 = g/(4q_{\max} + 4)$ we have $\Phi_{\tau}(F) - \Phi_{\tau}(G) \geq g/2$ for such F and all $G \in \mathcal{F}^*$. With $\tau = T/L$, Theorem 3.3(ii) gives $\log M_F - \log M_G \leq -\frac{g}{2}L + O(\log L) \leq -\frac{g}{4}L$.

(iii) Take $t_0 \geq 1$ larger than every $|t_j|$ and $\varepsilon \leq \min(\frac{1}{4}, \varepsilon_1/(2\tau^*))$, so that (ii) applies. Let $t \geq t_0$, let $\ell_F = E$ on $[t_0, \infty)$, and let $G \in \mathcal{F}^*$ with $\ell_G \neq \ell_F$, so that $\lambda_F \leq \lambda_G$. As both faces are tied, $\Delta = k_F - k_G = \tau^*(\lambda_G - \lambda_F) \geq 0$. If $\Delta = 0$, (8) gives $\log(M_F/M_G) = \log(K_F/K_G) + O(1/L)$ with $K_F > K_G$. If $\Delta > 0$, $\log(M_F/M_G) = \Delta t + \ell_F(0) - \ell_G(0) + \frac{\Delta}{2} \log(1+y) + O(1/L)$, where $|\log(1+y)| \leq 2|y| \leq t/2$ for large N , so it is positive if t_0 is large. The case $t \leq -t_0$ is the same.

For the consequence, fix $\eta > 0$ and let $S_{\eta} = [-t_0, t_0] \setminus \bigcup_j (t_j - \eta, t_j + \eta)$. For $G \in \mathcal{F}^*$, $E - \ell_G$ is convex, piecewise linear and nonnegative with kinks only at kinks of E . So its zero set is empty, a single kink of E , or an interval whose finite endpoints are kinks of E , and on each component of S_{η} either $\ell_G = E$ or $E - \ell_G \geq g_{\eta} > 0$. Then (i) and (ii) give the claim on S_{η} , and (iii) for $t_0 \leq |t| \leq \varepsilon L$. $\square$

4.4 Papers A and B

We use the polynomials $S_{a,b}$ and S_N , the root u_N , the crossing p_N , $z_N = N(1 - p_N)$ and $c_0 = \frac{1}{2} \log(\pi/8)$ of Section 2.1. The first result shows that a single state against a symmetric pair is Kagey's board at every N .

Proposition 4.9 (an exact pair identity). *Let $i \neq j$ with $q_{ij} = q_{ji} = a$, $q_i = q_j = q$ and $\mu_i = \mu_j > 0$, and let $F = \{i, j\}$. For every $N \geq 2$, $1 \leq m \leq N - 1$ and $h > 0$ such that $I + hQ$ is stochastic and $qh < 1$,*

$$\frac{P_N(me_i + (N - m)e_j)}{P_N(Ne_i)} = S_{m, N-m}(u), \quad u = \frac{ah}{1 - qh}.$$

So the balanced compositions, with $m = \lfloor N/2 \rfloor$ or $m = \lceil N/2 \rceil$, are the most likely ones with support F , $M_F = M_{\{i\}}S_N(u)$, and $M_F > M_{\{i\}}$ exactly when $u > u_N$.

Proof. A path with this composition is a word in i, j with m letters i . Each stay has probability $1 - qh$ and each switch ah , so a word with s switches has probability $\mu_i(1 - qh)^{N-1-s}(ah)^s = \mu_i(1 - qh)^{N-1}u^s$, while $P_N(Ne_i) = \mu_i(1 - qh)^{N-1}$. Sum over the words. By [2, Proposition 4.1] the

bin polynomials grow coefficientwise as the bin moves toward the middle, and $S_{m,N-m} = S_{N-m,m}$. So the balanced m give the largest $S_{m,N-m}(u)$. Finally S_N increases strictly and equals 1 only at u_N (Section 2.1). $\square$

The identity in Proposition 4.9 is checked in Lean.

Corollary 4.10 (Paper A). *Let $d = 2$, $q_{12} = q_{21} = 1$ and $\mu = (\frac{1}{2}, \frac{1}{2})$, the walk of [2] with $T = N(1 - p)$. Then $K_{\{1\}} = K_{\{2\}} = \frac{1}{2}$ and $K_{\{1,2\}} = \sqrt{2/\pi}$, so $b_{FG} = c_0$ for $F = \{1, 2\}$ and $G = \{1\}$. For every $N \geq 2$, $M_{\{1,2\}} = M_{\{1\}}$ holds exactly at $T = z_N$, and $z_N + \frac{1}{2} \log z_N = L + c_0 + O(1/L)$.*

Proof. Corollary 4.3 with $k = 2$, $\text{tree}(q_F) = 1$ and $\mu(F) = 1$ gives $K_{\{1,2\}}$, and then $b_{FG} = -\log(K_{\{1,2\}}/K_{\{1\}}) = c_0$. Proposition 4.9 with $a = q = 1$ gives $M_{\{1,2\}} = M_{\{1\}} S_N(h/(1-h))$, and $h/(1-h) = u_N$ exactly when $p = p_N$. So for every N the only zero of D_N is z_N , and by Theorem 4.7(i) it lies in $[L/2, 2L] \subset [1, N^{1/3}]$ for large N . Then Theorem 4.7(ii) with $\Delta = \delta = 1$ gives the equation. $\square$

This is the crossing equation of [2] without its terms $1/(8z_N)$ and R_N . Proposition 4.14 recovers the term $1/(8z_N)$. In its notation this term is $(c_{\{1\}} - c_{\{1,2\}})/z_N$ with $c_{\{1\}} = 0$ and $c_{\{1,2\}} = -\frac{1}{8}$, and (10) with $\Delta = \delta = 1$ gives $z_N + \frac{1}{2} \log z_N = L + c_0 + \frac{1}{8z_N} + O(L^{-2})$.

Corollary 4.11 (Paper B). *Let $Q = J - dI$, where J is the all-ones matrix, and let μ be uniform, the model of [3]. A face of k states has $\lambda_F = d - k$, α_F^* uniform and $K_F = \frac{1}{d} k^{k+1/2} (4\pi)^{-(k-1)/2}$. So $\Phi_1(F) = d - 1$ for every face, and all faces tie at $\tau = 1$. The crossing of a face of $k \geq 2$ states with a vertex has $\Delta = \delta = k - 1$ and $b_{FG} = a_k = \frac{1}{2} \log(4\pi) - \frac{k+1/2}{k-1} \log k$, the constant of the vertex crossing of [3].*

Proof. Each Q_F is symmetric with row sums $-(d - k)$, and the complete graph on F has k^{k-2} spanning trees by Cayley's formula. So Corollary 4.3 with $\mu(F) = k/d$ gives λ_F , α_F^* and K_F . A vertex has $K = \frac{1}{d}$ and $\lambda = d - 1$, so $b_{FG} = -\frac{1}{k-1} \log(dK_F) = a_k$. The arithmetic that gives b_{FG} from K_F here and in Corollary 4.10 is checked in Lean. At $\tau^* = 1$, Theorem 4.8 describes the tie window, and [3] resolves this window in full. $\square$

4.5 Reducible faces

Remark 4.12 (scope of the $\frac{1}{2}$). The coefficient $\frac{1}{2}$ of $\log \log N$ in Theorems 4.7 and 4.8 is due to the power $T^{(k-1)/2}$ in Theorem 4.6, so it needs only (H). At a kink of E where the line of G gives way to that of F as t increases, $\Delta = \tau^* \delta$ and $t_j = b_{FG}$. So every change in a tie window is a crossing of Theorem 4.7. The best composition of a reducible face can visit a state only once, with no Gaussian window in that direction. Then the $\frac{1}{2}$ can fail, and in the next example it is 1 and then 0.

Proposition 4.13 (a directed 3-cycle). *Let $d = 3$, $q_{12} = b > 2$, $q_{23} = q_{31} = 1$, all other rates 0, and $\mu_1 = 1$.*

- (i) *The admissible faces are $\{1\}$, $\{1, 2\}$ (reducible, $\lambda = 1$) and $\{1, 2, 3\}$ (irreducible, $\lambda = 0$). Exactly $M_{\{1\}} = (1 - bh)^{N-1}$ and $M_{\{1,2\}} = bh(1 - h)^{N-2}$, and the latter is attained only at $n = (1, N - 1, 0)$. For the full face, $\alpha^* = (1, b, b)/(1 + 2b)$, $\det \nabla^2 I = (1 + 2b)^4/(3b^2)$ and $K_{\{1,2,3\}} = (1 + 2b)^2/(2\sqrt{3}\pi b)$.*

- (ii) $M_{\{1\}} = M_{\{1,2\}}$ has exactly one root $T_N^{(1)}$ with $bh < 1$. It satisfies $(b-1)T + \log T = L - \log b + O(T^2/N)$, so $T_N^{(1)} = \frac{1}{b-1}[L - \log b + \log \frac{b-1}{b}] + O(\log L/L)$, and the coefficient of $\log \log N$ in the bracket is 1.
- (iii) Every root $T \in [L/2, 2L]$ of $M_{\{1,2\}} = M_{\{1,2,3\}}$ satisfies $T = L + \log \frac{2\sqrt{3}\pi b^2}{(1+2b)^2} + O(1/L)$, so the coefficient of $\log \log N$ is 0. These roots lie in an interval of length $O(1/L)$.
- (iv) Both are changes of the global mode. For large N and $T \leq N^{1/3}$, the mode is $(N, 0, 0)$ for $T < T_N^{(1)}$, it is $(1, N-1, 0)$ between $T_N^{(1)}$ and the smallest root in (iii), and it lies on the full face above the largest root in (iii).

In words, the face $\{1, 2\}$ is reducible, and its best composition $(1, N-1, 0)$ visits state 1 once and then stays in state 2. Its maximum is about $b(T/N)e^{-T}$, with the power T^1 where an irreducible pair would have $T^{1/2}$. This changes the coefficient of $\log \log N$ at both crossings.

Proof. (i) The chain moves only along $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ from 1, which gives the supports and the components $\{1\}$, $\{2\}$ of $\{1, 2\}$. A path with support $\{1, 2\}$ is $1^{n_1}2^{n_2}$, of probability $(1-bh)^{n_1-1}bh(1-h)^{n_2-1}$, and raising n_1 by one multiplies it by $(1-bh)/(1-h) < 1$. For the full face $\lambda = 0$, $r = \mathbf{1}$, and $l = \alpha^* = (1, b, b)/(1+2b)$ is the stationary law, so $(\mu \cdot r)(l \cdot \mathbf{1}) = 1$. The ratios u_2/u_1 , u_3/u_2 and u_1/u_3 have product 1, so the AM–GM inequality gives $I(\alpha) = \sum_i q_i \alpha_i - 3(b\alpha_1\alpha_2\alpha_3)^{1/3}$. Let $g = (\alpha_1\alpha_2\alpha_3)^{1/3}$, $w = 1/\alpha$ and $s = 1 + 2b$. The Hessian of g in all three coordinates is $-gM$ with $M = \frac{1}{3} \text{diag}(w)^2 - \frac{1}{9}ww^\top$, so $\nabla^2 I = 3b^{1/3}g J^\top M J$, where J is the Jacobian of $(\alpha_2, \alpha_3) \mapsto (1 - \alpha_2 - \alpha_3, \alpha_2, \alpha_3)$. At α^* , $3b^{1/3}g = 3b/s$ and $w = s(1, y, y)$ with $y = 1/b$. So $J^\top M J$ has diagonal entries $\frac{2}{9}s^2(1 + y + y^2)$ and off-diagonal entries $\frac{1}{9}s^2(2 + 2y - y^2)$, and $\det J^\top M J = \frac{s^4}{81} \cdot 3y^2(2 + y)^2 = \frac{s^6}{27b^4}$. Hence $\det \nabla^2 I = s^4/(3b^2)$, and Proposition 4.2(a) gives $K_{\{1,2,3\}}$. The exact parts of (i) are checked in Lean.

(ii) Let $D_1 = \log M_{\{1,2\}} - \log M_{\{1\}} = \log(bh) + (N-2)\log(1-h) - (N-1)\log(1-bh)$. Then $D_1'(h) = \frac{1}{h} - \frac{N-2}{1-h} + \frac{b(N-1)}{1-bh} > 0$ for $bh < 1$, while $D_1 \rightarrow -\infty$ as $h \rightarrow 0$ and $D_1 \rightarrow \infty$ as $bh \rightarrow 1$. So there is exactly one root (this is checked in Lean). Expanding, $D_1 = \log b + \log T - L + (b-1)T + O(T^2/N)$, which is positive at $T = L/(b-1)$ and negative at $T = L/(2(b-1))$. So the root lies between them, and $\log T = \log L - \log(b-1) + O(\log L/L)$ gives the expansion.

(iii) Here $\log M_{\{1,2\}} = \log b + \log T - L - T + O(T^2/N)$, and Theorem 4.6 for the full face ($k = 3$, $\lambda = 0$, $\mu(F) = 1$) gives $\log M_{\{1,2,3\}} = -2L + \log T + \log K_{\{1,2,3\}} + O(1/T + T^2/N)$. So on $[L/2, 2L]$ the difference is $L - T + \log b - \log K_{\{1,2,3\}} + O(1/L)$, which gives (iii).

(iv) Lemma 3.2 with $\theta = q$ gives $M_{\{1,2,3\}} \leq C(T/N)^2$. For $T < T_N^{(1)}$, $M_{\{1\}} > M_{\{1,2\}}$ by (ii), and $M_{\{1\}} \geq N^{-b/(b-1)-o(1)}$ since $T_N^{(1)} \leq L/(b-1)$. This beats $C(T/N)^2$ because $b/(b-1) < 2$. For $T > T_N^{(1)}$, $M_{\{1,2\}} > M_{\{1\}}$. For $T \leq L/2$, $\log M_{\{1,2\}} - \log M_{\{1,2,3\}} \geq L - T - \log T - C > 0$. On $[L/2, 2L]$ the sign is that of the formula in (iii), which is positive at $L/2$ and negative at $2L$. For $2L \leq T \leq N^{1/3}$ the same formula is at most $-L + C < 0$. $\square$

4.6 The next term

The local law, the face maxima and the crossing equation each have a next term, of relative order $1/T$. For a face F that satisfies (H) with $k \geq 2$ we write $\Sigma_F = \Sigma_{\alpha_F^*}$ and $\gamma_F = \nabla \log C_F(\alpha_F^*)$, in the coordinates α' . The function c and the constants c_F below are not related to the conductances $c_{ij}(\alpha)$ of Proposition 4.2.

Proposition 4.14 (the next term). *Let F satisfy (H).*

(i) Let $k \geq 2$. There is a real-analytic function c on Δ_F° , given by (23), such that for every compact $\mathcal{K} \subset \Delta_F^\circ$, uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,

$$f_T(\alpha) = T^{(k-1)/2} C_F(\alpha) e^{-T I_F(\alpha)} \left(1 + \frac{c(\alpha)}{T} + O_{\mathcal{K}}\left(\frac{1}{T^2}\right) \right).$$

(ii) Put $c_F = c(\alpha_F^*) + \frac{1}{2} \gamma_F^\top \Sigma_F \gamma_F$ if $k \geq 2$, and $c_F = 0$ if $k = 1$. For large N , uniformly for $1 \leq T \leq N^{1/3}$,

$$\log M_F = -(k-1)L + \frac{k-1}{2} \log T + \log K_F - T \lambda_F + \frac{c_F}{T} + O\left(\frac{1}{T^2} + \frac{T^2}{N}\right). \quad (9)$$

(iii) Let G also satisfy (H), with Δ , δ and D_N as in Theorem 4.7. For large N , every zero $T_N \in [1, N^{1/3}]$ of D_N satisfies

$$T_N \left(\Delta L - \frac{\Delta}{2} \log T_N - \delta T_N - \log \frac{K_F}{K_G} \right) = c_F - c_G + O\left(\frac{1}{L}\right). \quad (10)$$

(iv) Let $F = \{i, j\}$ with $q_{ij} = q_{ji} > 0$, $q_i = q_j$ and $\mu_i = \mu_j > 0$. Then $c_F = -1/(8q_{ij})$. Let $Q = J - dI$, and let F have $k \geq 2$ states with μ constant and positive on F . Then $c_F = -(k-1)/8$.

In words, the local law has a correction $c(\alpha)/T$, which is a Gaussian moment of the next Taylor coefficients of the torus integrand. The best composition does not sit exactly at $N\alpha_F^*$. It moves by about $N\Sigma_F\gamma_F/T$ toward larger C_F , and this gains $\frac{1}{2}\gamma_F^\top\Sigma_F\gamma_F/T$ in the logarithm. The crossing equation (7) then gains the term $(c_G - c_F)/T_N$ on its right side, with an error $O(L^{-2})$.

Idea of the proof. The proof is in Appendix B. For (i) we carry the Laplace lemma one step further (Lemma B.2). The terms of order $T^{-1/2}$ and $T^{-3/2}$ are odd and integrate to 0, so the error is $O(T^{-2})$. For (ii) the logarithm of the local law near α_F^* is a concave quadratic plus the linear term $\gamma_F \cdot x$, and we complete the square. The maximizer is within $O(1/T)$ of α_F^* , and moving it to the nearest lattice point changes the logarithm only by $O(T/N^2)$. Part (iii) subtracts (9) for the two faces. In (iv) we do not evaluate c from its formula. We compare our expansion with a second expansion of the same probability.

The two values in (iv) have different sources. The value for a symmetric pair is proved here. It uses only Proposition 4.9, the Bessel bound (12) of [2] and the classical expansion of $I_0 + I_1$. These give

$$\log M_{\{i,j\}} = -L + \frac{1}{2} \log T + \log K_{\{i,j\}} - T \lambda_{\{i,j\}} - \frac{1}{8q_{ij}T} + O\left(\frac{1}{T^2} + \frac{T^2}{N}\right), \quad (11)$$

for large N and uniformly for $1 \leq T \leq N^{1/3}$. This expansion does not use the Laplace lemma. The value for the complete graph rests on the expansion of the balanced bins in [3, Theorem 3.1(i)], which we quote and do not prove again. In both cases the formula (23) gives the same value. For the pair this is a short computation at the end of Appendix B, and for the full faces of K_3 and K_4 the script `verify_second_order.py` evaluates (23) exactly. For other faces c_F has to be computed from (23), γ_F and Σ_F , as we do for the 4-cycle in Section 5.

5 The 4-cycle and other graphs

In this section Q is symmetric. So the admissible faces are the connected faces with positive start mass, and they satisfy (H) (Lemma 2.2). By Theorem 3.4(c), Theorem 4.8 applies at every tie. We call the faces in $\mathcal{F}_\tau$ the *winners* at τ . We first complete the worked example of Section 3.1 with all its constants, then show on K_3 that the start can decide the winner, and then list the outcomes for other graphs.

Corollary 5.1 (the 4-cycle cascade). *Let Q be the 4-cycle C_4 with $q_{i,i\pm 1} = 1$ (indices mod 4), and let μ be uniform.*

- (a) *The admissible faces are the vertices, the adjacent pairs, the arcs of three states and C_4 , with exit rates 2, 1, $2 - \sqrt{2}$ and 0. The winners are the vertices for $\tau < 1$, the adjacent pairs for $1 < \tau < 2$ and C_4 for $\tau > 2$. The arcs of three states never win.*
- (b) *$K_{\text{vertex}} = \frac{1}{4}$, $K_{\text{pair}} = 1/\sqrt{2\pi}$, $K_{\text{arc3}} = (4 + 3\sqrt{2})/(4\pi)$ and $K_{C_4} = 8/\pi^{3/2}$, and $\det \Sigma = 1/512$ on C_4 .*
- (c) *For every $N \geq 2$ and $0 < h < \frac{1}{2}$, $M_{\text{pair}} = M_{\text{vertex}} S_N(u)$ with $u = h/(1 - 2h)$. So the face maxima of a vertex and an adjacent pair cross exactly at $u = u_N$, the crossing of [2]. For large N this crossing satisfies $T + \frac{1}{2} \log T = L + c_0 + O(1/L)$.*
- (d) *For large N , every crossing $T_N \in [1, N^{1/3}]$ of the face maxima of an adjacent pair and C_4 satisfies $T_N + \log T_N = 2L - \log(8\sqrt{2}/\pi) + O(1/L)$.*
- (e) *Put $b = -\frac{1}{2} \log 2 - \frac{1}{2} \log(8\sqrt{2}/\pi)$. In the tie window at $\tau = 1$ the face that carries the global mode changes from the vertices to the adjacent pairs only at $T = L - \frac{1}{2} \log L + c_0 + o(1)$. In the tie window at $\tau = 2$ it changes from the adjacent pairs to C_4 only at $T = 2(L - \frac{1}{2} \log L + b) + o(1)$.*

In words, the mode moves from a vertex to an adjacent pair at the crossing of Kagey's board, exactly at every N , and from the pairs to the whole cycle at a crossing whose equation has the coefficient $1 = \frac{\Delta}{2}$ of $\log T$, since the cycle has two more states than a pair.

Proof. (a) A proper face that is not an arc is disconnected, so it is not admissible. For an adjacent pair $-Q_F$ has rows $(2, -1)$ and $(-1, 2)$, with least eigenvalue 1. For an arc of three states $-Q_F = 2I - A$, where A is the adjacency matrix of the path on three states, and $r = (1/\sqrt{2}, 1, 1/\sqrt{2})$ is a positive eigenvector with eigenvalue $2 - \sqrt{2}$. Also $Q\mathbf{1} = 0$. So Φ_τ equals 2τ , $\tau + 1$, $(2 - \sqrt{2})\tau + 2$ and 3, and comparing the first, second and fourth gives the winners. Since $2 - \sqrt{2} > \frac{1}{2}$, the 3-arc has $\Phi_\tau > \frac{1}{2}(\tau + 1) + \frac{1}{2} \cdot 3$, which is at least the smaller of the exponents of the pair and of C_4 . So it never wins.

(b) A vertex has $K = \mu_i = \frac{1}{4}$. The pair and C_4 have symmetric Q_F with constant row sums, so Corollary 4.3 applies, with $\mu(F) = \frac{1}{2}$ and one spanning tree for the pair, and $\mu(F) = 1$ and four spanning trees for C_4 . On C_4 the law α^* is uniform and every edge has conductance $\frac{1}{4}$, so Proposition 4.2(b) gives $\det \Sigma = 2^3 \cdot 4^{-8}/(4 \cdot 4^{-3}) = 1/512$. The 3-arc has unequal row sums, and we use Proposition 4.2(b) with $\nu = \mathbf{1}$. Here $\alpha^* = r \circ r/|r|^2 = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ and $r_\alpha = l_\alpha = \sqrt{\alpha^*}$, so $(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1}) = \frac{1}{4}(1 + 1/\sqrt{2})^2 = (3 + 2\sqrt{2})/8$. The only spanning tree is the path, with $\text{tree}(c) = c_{12}c_{23} = \frac{1}{8}$, and $\prod_i \alpha_i^* = \frac{1}{32}$. So $K_{\text{arc3}} = \frac{3+2\sqrt{2}}{8} \cdot \frac{32}{2\sqrt{2}} \cdot \frac{1}{4\pi} = \frac{4+3\sqrt{2}}{4\pi}$.

(c) The identity is Proposition 4.9 with $a = 1$ and $q = 2$. So the two face maxima are equal only at $u = u_N$, and by Theorem 4.7(i) this crossing lies in $[L/2, 2L]$ for large N . As $K_{\text{pair}}/K_{\text{vertex}} = 2\sqrt{2}/\pi = e^{-c_0}$, Theorem 4.7(ii) with $\Delta = \delta = 1$ gives the equation.

(d) This is Theorem 4.7 with $F = C_4$, G an adjacent pair, $\Delta = 2$, $\delta = 1$ and $K_F/K_G = 8\sqrt{2\pi}/\pi^{3/2} = 8\sqrt{2}/\pi$.

(e) At $\tau = 1$ the tied faces are the vertices and the adjacent pairs, and at $\tau = 2$ they are the adjacent pairs and C_4 . Faces of one type have the same line. By Remark 4.12 the kink of E is at $t = b_{FG}$, which is c_0 at $\tau = 1$ by (c) and b at $\tau = 2$ by (d). So Theorem 4.8 gives (e). $\square$

The finite parts of (a) are checked in Lean. Lean also checks that a closed form for the constants of all arcs, which we do not use, gives the values of K_{pair} and K_{arc3} above.

Proposition 4.14 gives the next term c_F/T of the face maxima, with the explicit coefficient $c_F = c(\alpha_F^*) + \frac{1}{2}\gamma_F^\top \Sigma_F \gamma_F$, where c is given by (23). For an adjacent pair, Proposition 4.14(iv) gives $c_{\text{pair}} = -\frac{1}{8}$. For the 3-arc and for C_4 we evaluated the coefficient exactly by computer algebra (the script `verify_second_order.py`), and the values are $c_{\text{arc3}} = \frac{5}{4}\sqrt{2} - 2$ and $c_{C_4} = -\frac{9}{16}$. So these two constants are exact evaluations of the formula of Proposition 4.14, done by computer and not by hand. With the evaluated constant c_{C_4} , Proposition 4.14(iii) gives the next term at the crossings in (d), namely

$$T_N \left(2L - \log T_N - T_N - \log \frac{8\sqrt{2}}{\pi} \right) = c_{C_4} - c_{\text{pair}} + O\left(\frac{1}{L}\right) = -\frac{7}{16} + O\left(\frac{1}{L}\right).$$

Theorem 4.7 also applies to a face that never carries the mode, and this is a good test of the implicit equation (7). Take the 3-arc F against a vertex G , so that $\Delta = 2$, $\delta = \sqrt{2}$ and $\log(K_F/K_G) = \log((4 + 3\sqrt{2})/\pi) = 0.964591$. At $N = 10^{14}$ a numerical computation of the face maxima, over the symmetric compositions $(a, N - 2a, a)$, puts the crossing at $T_N = 42.2633$, where $\Delta L - \frac{\Delta}{2} \log T_N - \delta T_N = 0.959070$. The root of the implicit equation is within 0.004 of T_N , while the explicit form in Theorem 4.7 gives 42.2059. Also $T_N(0.959070 - 0.964591) = -0.2333$, close to the limit $c_{\text{arc3}} - c_{\text{vertex}} = -0.2322$ that Proposition 4.14(iii) gives with the evaluated constant c_{arc3} (a vertex has $c_{\text{vertex}} = 0$).

By Corollary 4.3 the start enters K_F only through $\mu(F)$, and this can change which face wins inside a tie window.

Corollary 5.2 (the start on K_3). *Let $d = 3$ and $Q = J - 3I$, put $T = L - \frac{1}{2} \log L + t$, and let $w = \log \frac{16}{9\sqrt{3}} = 0.026$.*

(a) *Let $\mu = (\frac{1}{2}, \frac{1}{2}, 0)$. For every $\eta > 0$ and large N , if $c_0 + \eta \leq t \leq c_0 + w - \eta$, then every global mode lies on the face $\{1, 2\}$.*

(b) *Let μ be uniform. For every compact set of t and large N , no global mode lies on a face of two states.*

Proof. All faces tie at $\tau^* = 1$ (Corollary 4.11), and Theorem 4.8 applies since Q is symmetric. By Corollary 4.3 the lines of a vertex i , an edge $\{i, j\}$ and $[3]$ are $\log \mu_i - 2t$, $\log((\mu_i + \mu_j)\sqrt{2/\pi}) - t$ and $\log K_{[3]}$ with $K_{[3]} = 3^{5/2}/(4\pi)$. In (a) the line of $\{1, 2\}$ lies strictly above the lines of the other admissible faces exactly when $c_0 < t < \frac{1}{2} \log \frac{2}{\pi} - \log K_{[3]} = c_0 + w$. So E has no kink in this interval, and the consequence in Theorem 4.8 gives (a). In (b) the gap between $\max(\log \frac{1}{3} - 2t, \log K_{[3]})$ and an edge line is least where the first two lines meet, and there it is $\frac{1}{2} \log(3^{7/2}/32) > 0.18$. So Theorem 4.8(i) gives (b). $\square$

The left end c_0 is the crossing of Paper A. Indeed Proposition 4.9 with $a = 1$ and $q = 2$ shows that for every N the face maxima of $\{1, 2\}$ and of a vertex cross exactly at $u = u_N$. The comparison of the lines in (a) and the value of w are checked in Lean. With the uniform start the

graph	winners in order, with the value of τ where each one starts	status
K_d	$\lambda_F = d - F $. Vertices for $\tau < 1$ and $[d]$ for $\tau > 1$. Every face ties at $\tau = 1$, and [3] resolves the tie at second order.	Lean
symmetric Q	$\lambda_{[d] \setminus \{x\}} \leq q_x / (d - 1)$, with equality exactly when $q_{yx} = q_x / (d - 1)$ for all $y \neq x$. With unit rates on a connected graph, the winners go from vertices straight to $[d]$ exactly when the graph is complete.	Lean
$C_d, d \geq 4$	Arcs of $k = 1, \dots, k_d$ states with $k_d = \lfloor 2d/3 \rfloor$, then C_d . The arc of $k \geq 2$ states wins from $1/(\lambda_{k-1} - \lambda_k)$ with $\lambda_k = 2 - 2 \cos \frac{\pi}{k+1}$, and C_d from $\lfloor d/3 \rfloor / \lambda_{k_d}$. These values are 1, $1 + \sqrt{2}$, 4.906 and 8.771 for $k = 2, \dots, 5$, and they grow like $k^3 / (2\pi^2)$. Arcs of $k \geq 3$ states win exactly when $d \geq \lceil 3k/2 \rceil$.	Lean, asymptotics by hand
$P_d, d \geq 3$	End-arcs of $1, \dots, \lfloor 2d/3 \rfloor$ states, then P_d . The end-arc of k states has exit rate $2 - 2 \cos \frac{\pi}{2k+1}$, and the first switch is at the golden ratio $(1 + \sqrt{5})/2$.	Proposition D.3
P_4 at finite N	The end-arc of 3 states never wins, and its exit rate is only 0.0071 too large to win. Yet it is the exact mode for $T \in (9.381, 12.502)$ at $N = 40$, and on a window at every N tested up to 300.	Proposition D.3, finite N numerical
Θ_ε	Triangles $\{1, 2, 3\}$ and $\{4, 5, 6\}$ joined by a link 3–4 of rate $\varepsilon < \varepsilon_1 = (64 + \sqrt{46})/60$. The states 1, 2, 5, 6, then $\{1, 2\}$ and $\{5, 6\}$ from 1, the triangles from $1 + \varepsilon/3 + O(\varepsilon^2)$, and $[6]$ from $9/\varepsilon + 2 + O(\varepsilon)$.	Proposition D.4, with 2 exact resultants
paw	A triangle $\{1, 2, 3\}$ with a leaf 4 at 1. $\{4\}$, then $\{1, 4\}$ from $1 + \sqrt{2}$, $\{1, 2, 3\}$ from $\sqrt{2} + \sqrt{3}$, and all 4 states from $2 + \sqrt{3}$.	Lean
house	The square 1, 2, 3, 4 with a roof state 5 joined to 1, 2. $\{3\}, \{4\}, \{5\}$, then $\{3, 4\}$ from 1, $\{1, 2, 5\}$ from $1 + \sqrt{2}$, and all 5 states from $2 + \sqrt{2}$.	Lean
K_4 plus a vertex	A state 5 joined to 1, 2 of K_4 . $\{5\}$, then $\{1, 2, 3, 4\}$ from $3(\sqrt{17} + 1)/8 = 1.921$, and all 5 states from $(5 + \sqrt{17})/4 = 2.281$.	Lean

Table 1: First-order winners on some graphs with unit rates and a start of full support (the second row allows any symmetric Q). The last column says where each row is proved (Remark 5.3).

mode jumps from the vertices straight to $[3]$, as [3] shows for every complete graph. Paper B [3] also compares the window in (a) with exact computations at finite N , where it approaches its width w slowly.

Remark 5.3 (the outcomes table). Table 1 lists the outcomes for other graphs. Each row comes from Theorem 3.4 applied to the exit rates of the connected faces. So a row needs two things, namely the exit rates and the points that lie on the lower-left boundary of the hull. The rows marked Lean are proved in Lean (Section 8). On C_d two steps are by hand. A proper face that is not an arc is disconnected, hence not admissible, and the growth of the switch values follows from $2 - 2 \cos x = x^2 + O(x^4)$. The rows P_d and Θ_ε are proved in Appendix D, for every $d \geq 3$ and for every $\varepsilon < \varepsilon_1$ (Propositions D.3 and D.4). The proof for Θ_ε uses two resultants that we computed by computer algebra, and it also shows that the bound ε_1 is sharp. In the row P_4 the first sentence is the case $d = 4$ of Proposition D.3, and the rest is an exact computation of the law at the values of N listed. A well-separated community wins for a long range of τ , such as a triangle of Θ_ε from $\tau \approx 1$ to $\tau \approx 9/\varepsilon$. The mode need not grow around its first vertex. It leaves the leaf of the paw, and on the house and on K_4 plus a vertex it jumps to a disjoint community. The second row

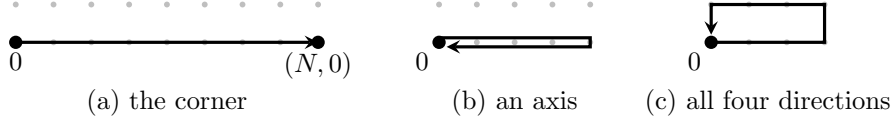


Figure 5: Paths of $N = 8$ steps. (a) The only path to the corner $(N, 0)$ never switches. A path to the origin either (b) stays on one axis and only reverses, or (c) turns and then uses all four directions.

explains the last one. The state 5 feeds $\{1, 2, 3, 4\}$ unequally, so the point of $\{1, 2, 3, 4\}$ lies strictly left of the chord from $\{5\}$ to $[5]$, and the winners cannot jump from $\{5\}$ straight to $[5]$. At feasible N the switch values are approached slowly (Figure 4(b)), and on P_4 even the order of the phases differs.

6 The planar persistent walk

We now return to the question of the introduction, Kagey's board in two dimensions. A ball moves on $\mathbb{Z}^2$ in the directions east, north, west and south, numbered $0, 1, 2, 3$ modulo 4, with unit vectors $e_0 = (1, 0)$, $e_1 = (0, 1)$, $e_2 = -e_0$ and $e_3 = -e_1$. Let $r, s \geq 0$ with $\beta = 2r + s > 0$, and let $0 < h < 1/\beta$. The first direction is uniform. After that the ball keeps its direction with probability $p_0 = 1 - \beta h$, turns left or right with probability rh each, and reverses with probability sh . This is the chain of Example 2.5(iv), with $q_{j,j\pm 1} = r$ and $q_{j,j+2} = s$, and its position after N steps is $Y_N = (n_0 - n_2, n_1 - n_3)$. So the bin is now a linear function of the composition. In this section N is even, since otherwise the origin cannot be reached, and we allow every $h < 1/\beta$. As before $T = Nh$. Only the path that always goes east reaches the corner Ne_0 , so

$$c_N = \mathbb{P}(Y_N = Ne_0) = \frac{1}{4}p_0^{N-1}.$$

We compare it with $o_N = \mathbb{P}(Y_N = 0)$. Figure 5 shows the three kinds of paths that matter.

We write $u = sh/p_0$, $v = rh/p_0$ and $w = \beta h/p_0 = 2v + u$, so $1 + w = 1/p_0$. These are the odds of a reversal, of a turn to a given side and of any switch, as in the shared notation $u = q/p$. We use S_N , u_N , p_N , $z_N = N(1 - p_N)$ and c_0 from Section 2.1. As N is even, $S_N(u)$ is the sum of u^j over the words with $N/2$ letters of each of two kinds, where j is the number of switches. It has nonnegative coefficients and no constant term, so it increases strictly on $[0, \infty)$.

The answer at first order. Face selection predicts the answer, up to powers of $\log N$. Let $h = \tau \log N/N$. The corner is the vertex $\{0\}$, which has exit rate β , so c_N is about $N^{-\beta\tau}$. A vertex, a face with three directions, or a face with two adjacent ones misses the opposite of one of its directions, so it cannot reach the origin. On the axis $\{0, 2\}$, which has exit rate $2r$, the origin is the single composition $(N/2, 0, N/2, 0)$, of probability about $N^{-(2r\tau+1)}$. On the full face a single composition has probability about N^{-3} , but about $N/2$ compositions (a, b, a, b) end at the origin, and together they give about N^{-2} . So the origin should beat the corner when $\beta\tau > \min(2r\tau + 1, 2)$, that is, when $\tau > \tau^* = \min(2/\beta, 1/s)$. At τ^* the axes win if $s > 2r$ and the full face wins if $s < 2r$. This is only a heuristic, since it applies Theorem 3.3 to a linear function of the composition, which we have not proved in general.

The rest of the section proves this and then locates the crossing. We split the origin mass into an axis part and a part with all four directions (Proposition 6.2) and prove the first-order answer (Theorem 6.3). Then we find the crossing up to $O(N^{-\gamma})$ when reversals dominate (Theorem 6.4)

and up to $o(1)$ when turns dominate (Theorem 6.5 and Corollary 6.6). Section 6.1 treats two cases in which the walk is, exactly or nearly, two copies of Kagey's board.

Three levels. A path is given by the number of times it switches, the directions of its runs, and the lengths of its runs.

Lemma 6.1 (three levels). *Write a sequence of directions as its number m of switches, its skeleton $Z = (Z_0, \dots, Z_m)$, which lists the directions of the runs so that $Z_t \neq Z_{t-1}$, and its run lengths $\ell = (\ell_0, \dots, \ell_m)$, which form a composition of N into $m+1$ positive parts. This is a bijection, and $Y_N = \sum_{t=0}^m \ell_t e_{Z_t}$. Under the walk, (a) m has the law $\text{Bin}(N-1, \beta h)$. Given m , (b) Z and ℓ are independent, (c) Z is the random walk on $\mathbb{Z}_4$ with uniform Z_0 whose independent increments are ± 1 with probability r/β each and 2 with probability s/β , and (d) ℓ is uniform on the $\binom{N-1}{m}$ compositions.*

Proof. Put $q(\pm 1) = r$ and $q(2) = s$. The sequence with data (m, Z, ℓ) has probability

$$\frac{1}{4} p_0^{N-1-m} \prod_{t=1}^m h q(Z_t - Z_{t-1}) = \binom{N-1}{m} (\beta h)^m p_0^{N-1-m} \cdot \frac{1}{4} \prod_{t=1}^m \frac{q(Z_t - Z_{t-1})}{\beta} \cdot \binom{N-1}{m}^{-1}.$$

For fixed m the middle factor is the probability of Z under the walk in (c), so it sums to 1 over the skeletons, and the last factor sums to 1 over the compositions. So the three factors are the laws in (a), (c) and (d), and the product form gives (b). $\square$

The next proposition separates the skeletons that stay on one axis from those that use both.

Proposition 6.2 (exact decomposition and a unique crossing). *Let $N \geq 2$ be even and $0 < h < 1/\beta$. Let $o_N^{(0)}$ and $o_N^{(4)}$ be the probabilities that $Y_N = 0$ and the walk never turns, and that $Y_N = 0$ and the walk uses all four directions. Then $o_N = o_N^{(0)} + o_N^{(4)}$,*

$$o_N^{(0)} = 2S_N(u) c_N, \quad 0 \leq o_N^{(4)} \leq \frac{w^2}{2p_0} (1 - p_0^N).$$

Equivalently, $o_N/c_N = 2S_N(u) + E_N$ with $0 \leq E_N \leq 2w^2((1+w)^N - 1)$. Also, o_N/c_N is a polynomial in v and u with nonnegative coefficients and no constant term. If $s > 0$, or if $s = 0$ and $N \geq 4$, it increases strictly in h from 0 at $h = 0$ to ∞ as $h \uparrow 1/\beta$. So there is exactly one $h_N^ \in (0, 1/\beta)$ with $o_N = c_N$, and the corner is more likely than the origin exactly when $h < h_N^*$.*

In words, the origin mass splits exactly into the axis part, which is twice the middle bin of Kagey's board measured against its endpoint, and a part from paths with all four directions, which is small when w is small. The ratio o_N/c_N grows with the switching rate, so the corner and the origin cross exactly once, at every even $N \geq 4$.

Proof. An origin path with no turn stays on the axis of its first step. An origin path that turns uses both axes, and since $n_0 = n_2$ and $n_1 = n_3$ at the origin, it uses all four directions. So every origin path is of exactly one of the two kinds. The origin paths on the axis $\{0, 2\}$ are the paths with $n_0 = n_2 = N/2$, and the path that always goes east has probability c_N . So Proposition 4.9 with $a = s$ and $q = \beta$, for the pairs $\{0, 2\}$ and $\{1, 3\}$, gives $o_N^{(0)} = 2S_N(u) c_N$. The bound on $o_N^{(4)}$ follows from Lemma 6.1 and a count of the run lengths that bring a skeleton back to the origin. We prove it in Appendix C. Dividing by c_N gives the form with E_N .

A path with j turns and k reversals has probability $\frac{1}{4} p_0^{N-1-j-k} (rh)^j (sh)^k = c_N v^j u^k$. So o_N/c_N is the sum of $v^j u^k$ over the origin paths, and $j + k \geq 1$ for each of them. Both u and v are

nondecreasing in h , and each of them with a positive rate increases strictly to ∞ as $h \uparrow 1/\beta$. If $s > 0$, the path that goes east $N/2$ times and then west gives the term u . If $s = 0$ and $N \geq 4$, then $r > 0$, and a path whose four runs go east, north, west and south, with lengths $\ell_0 = \ell_2 \geq 1$ and $\ell_1 = \ell_3 \geq 1$, gives the term v^3 . This proves the claims on monotonicity and the limits. For $s = 0$ and $N = 2$ the origin cannot be reached. $\square$

The identity and the unique crossing for $s > 0$ are checked in Lean (Section 8). We write $T_N^* = Nh_N^*$ for the crossing and $u_N^* = sh_N^*/(1 - \beta h_N^*)$ for the odds of a reversal there. At the crossing $o_N = c_N$, so $2S_N(u_N^*) = o_N^{(0)}/o_N$ there. We call this number the zero-turn share of the origin mass.

We compare S_N with Bessel functions by the uniform Bessel bound of [2, Theorem 3.3(i)]. Put $\psi(z) = z(I_0(z) + I_1(z))$. In our notation that bound says that for every $N \geq 2$ (even or odd) and every $z > 0$

$$\left(1 - \frac{z(z+2)}{2N}\right)\psi(z) \leq \frac{N}{2} S_N\left(\frac{z}{N}\right) \leq \psi(z). \quad (12)$$

Since $I_0' = I_1$ and $(zI_1)' = zI_0$, we have $\psi' = \psi + I_0$, so $(\log \psi)' \geq 1$. We use this to turn a bound on $\psi(z)$ into a bound on z .

Theorem 6.3 (first order). *Fix $r > 0$ and $s \geq 0$, put $\tau^* = \min(2/\beta, 1/s)$ with $1/0 = \infty$, and let $N \rightarrow \infty$ through even integers.*

- (a) *Fix $\tau > 0$ and put $h = \tau \log N/N$. If $\tau < \tau^*$, then $o_N/c_N \leq N^{-\gamma}$ for some $\gamma > 0$ and all large N . If $\tau > \tau^*$, then $o_N/c_N \geq N^\gamma$ for some $\gamma > 0$ and all large N .*
- (b) *$T_N^*/\log N \rightarrow \tau^*$.*
- (c) *If $s > 2r$, then $2S_N(u_N^*) \rightarrow 1$, so the origin mass at the crossing comes from the two axes, half from each. If $s < 2r$, then $2S_N(u_N^*) \leq N^{-\gamma}$ for some $\gamma > 0$ and all large N , so it comes from the paths that use all four directions.*

In words, the heuristic above is right. The crossing sits at $T \approx \tau^* \log N$, and at the crossing the origin is reached through the axes when reversals dominate and through all four directions when turns dominate. (For $r = 0$ the walk never leaves the axis of its first step, and Theorem 6.4(c) gives the same limit $1/s$.)

Idea of the proof. The proof is in Appendix C. The upper bound in (a) follows from Proposition 6.2 and (12). For the lower bound when $s\tau > 1$ the axis part is enough. For the lower bound when $\beta\tau > 2$ we count balanced skeletons, those with as many runs east as west and as many north as south, with Lemma 6.1. They give $o_N^{(4)}$ of order at least $N^{-2}T^{-1/2}$. Part (b) follows from (a) and the monotonicity in Proposition 6.2, and part (c) from (b) and the bounds on the two parts.

Reversals dominate. Let $s > 2r$. Then the origin mass at the crossing comes from the axes, and on an axis the walk is Kagey's board. The crossing is then the crossing of [2] for $N/2$ steps. We write $z_{N/2} = \frac{N}{2}(1 - p_{N/2})$ for it, and put

$$\mathcal{A}(\Lambda) = \Lambda - \frac{1}{2} \log \Lambda + c_0 + \frac{\frac{1}{4} \log \Lambda - \frac{1}{2} c_0 + \frac{1}{8}}{\Lambda},$$

so that the third-order expansion of z_N in [2, Corollary 3.7(c)] reads $z_N = \mathcal{A}(L) + O((\log L)^2/L^2)$.

Theorem 6.4 (reversals dominate). (a) Let $s > 0$. For all even $N \geq 2$ and $0 < h < 1/\beta$,

$$\frac{o_N}{c_N} = 2S_N(u)(1 + \varepsilon_N(h)), \quad 0 \leq \varepsilon_N(h) \leq \frac{w^2(1+w)^N}{S_N(u)}.$$

(b) Let $r \geq 0$ and $s > 2r$, let $N \rightarrow \infty$ through even integers, and let $0 < \gamma < \min(1, (s - 2r)/s)$. Then

$$\varepsilon_N(h_N^*) = O((\log N)^{3/2-r/s} N^{-(s-2r)/s}), \quad Nu_N^* = z_{N/2} + O(N^{-\gamma}).$$

(c) Under the hypotheses of (b), $sT_N^* = z_{N/2} + O(N^{-\gamma})$. Hence

$$sT_N^* = \mathcal{A}(L - \log 2) + O\left(\frac{(\log L)^2}{L^2}\right), \quad sT_N^* - z_N = -\log 2 + \frac{\log 2}{2L} + O\left(\frac{(\log L)^2}{L^2}\right).$$

In words, when reversals dominate, the planar walk at its crossing behaves like Kagey's board with half as many steps, and sT_N^* is smaller than Kagey's crossing z_N by about $\log 2$. Here is why. At the crossing $o_N \approx o_N^{(0)} = 2S_N(u)c_N$, so the crossing solves $2S_N(u) \approx 1$. By (12), applied with N and with $N/2$ steps, the equations $2S_N(z/N) = 1$ and $S_{N/2}(z/(N/2)) = 1$ both say that $\psi(z) \approx N/4$. So doubling S_N has the same effect as halving N .

Proof. (a) Since $s > 0$, $S_N(u) \geq 2u > 0$. Divide Proposition 6.2 by $2S_N(u)$. Then $\varepsilon_N(h) = E_N/(2S_N(u))$ and $E_N \leq 2w^2(1+w)^N$.

(b) Write $u^* = u_N^*$, $z^* = Nu^*$ and $E^* = E_N(h_N^*)$. At the crossing Proposition 6.2 gives $2S_N(u^*) + E^* = 1$. So $S_N(u^*) \leq \frac{1}{2} < S_N(u_N)$, and $u^* < u_N$. Put $w_N = \beta u_N/s$, which is the value of w when $u = u_N$. Since $w \mapsto 2w^2(1+w)^N$ is increasing, $E^* \leq \eta_N := 2w_N^2(1+w_N)^N$. By the second-order expansion of z_N in [2, Corollary 3.7(b)], $Nu_N = z_N/p_N = L - \frac{1}{2} \log L + O(1)$. So $(1+w_N)^N \leq e^{Nw_N} = O(N^{\beta/s} L^{-\beta/(2s)})$ and $w_N^2 = O(L^2/N^2)$. As $\beta/s - 2 = -(s - 2r)/s$ and $2 - \beta/(2s) = \frac{3}{2} - \frac{r}{s}$, this gives $\eta_N = O(L^{3/2-r/s} N^{-(s-2r)/s})$, which tends to 0. Then $\varepsilon_N(h_N^*) = E^*/(1 - E^*) \leq \eta_N/(1 - \eta_N)$.

Next let $z' = \frac{N}{2}u_{N/2}$, so that $S_{N/2}(2z'/N) = 1$. By (12) with $N/2$ steps, $\frac{N}{4} \leq \psi(z') \leq \frac{N}{4}(1 - \delta'_N)^{-1}$ with $\delta'_N = z'(z' + 2)/N$. By (12) with N steps and $\frac{1}{2}(1 - \eta_N) \leq S_N(u^*) \leq \frac{1}{2}$, we get $\frac{N}{4}(1 - \eta_N) \leq \psi(z^*) \leq \frac{N}{4}(1 - \delta_N)^{-1}$ with $\delta_N = z^*(z^* + 2)/(2N)$. Both z' and $z^* < Nu_N$ are $O(L)$, so $\delta_N, \delta'_N = O(L^2/N)$. As $(\log \psi)' \geq 1$, $|z^* - z'| \leq |\log \psi(z^*) - \log \psi(z')| = O(\eta_N + L^2/N)$. Finally $z_{N/2} = z'/(1 + u_{N/2}) = z' + O(L^2/N)$. So $z^* - z_{N/2} = O(\eta_N + L^2/N)$, which is $O(N^{-\gamma})$.

(c) Since $h_N^* = u^*/(s + \beta u^*)$, we have $sT_N^* = z^*/(1 + \beta u^*/s) = z^* + O(L^2/N)$, as $u^* < u_N = O(L/N)$. With (b) this gives $sT_N^* = z_{N/2} + O(N^{-\gamma})$. The third-order expansion of z_N , for $N/2$ steps and for N steps, gives $z_{N/2} = \mathcal{A}(L - \log 2) + O((\log L)^2/L^2)$ and $z_N = \mathcal{A}(L) + O((\log L)^2/L^2)$. The last term of $\mathcal{A}$ has derivative $O(\log L/L^2)$ near L , so

$$\mathcal{A}(L - \log 2) - \mathcal{A}(L) = -\log 2 - \frac{1}{2} \log\left(1 - \frac{\log 2}{L}\right) + O\left(\frac{\log L}{L^2}\right) = -\log 2 + \frac{\log 2}{2L} + O\left(\frac{\log L}{L^2}\right). \quad \square$$

For $(r, s) = (0.1, 1)$ the difference $sT_N^* - z_N$ is -0.6317 at $N = 640$ and -0.6493 at $N = 10240$, while the first two terms in (c) give -0.6395 and -0.6556 (Figure 6(a)).

Turns dominate. Let $s < 2r$. Now the origin mass at the crossing is carried by the paths with all four directions, and Proposition 6.2 only bounds $o_N^{(4)}$ from above. We need its size, which is a local limit theorem in two dimensions.

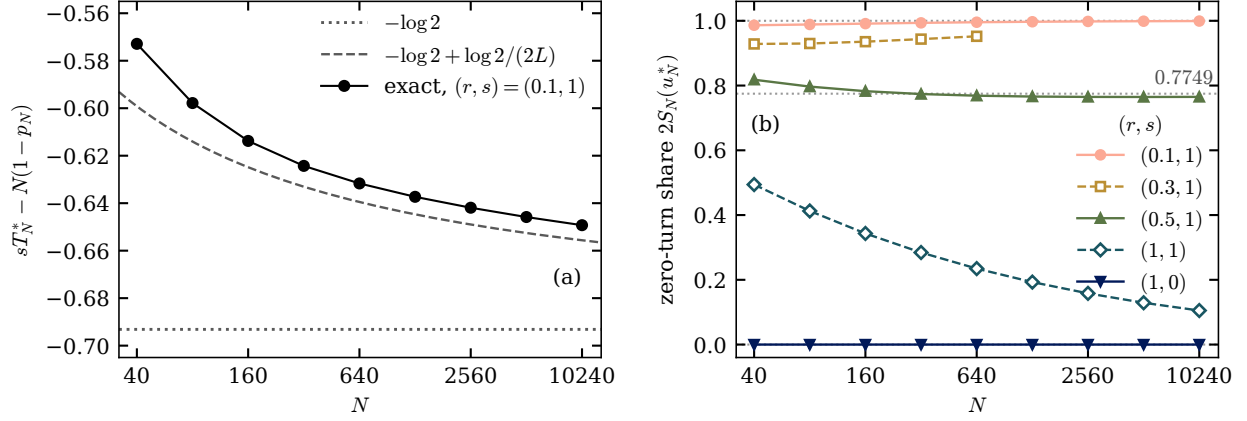


Figure 6: (a) The exact crossing of the planar walk with $(r, s) = (0.1, 1)$, written as $sT_N^* - z_N$ with $z_N = N(1 - p_N)$, for $N = 40, 80, \dots, 10240$. The dashed curve is $-\log 2 + \log 2/(2L)$ from Theorem 6.4(c), and the dotted line is its limit $-\log 2$. (b) The zero-turn share $2S_N(u_N^*)$ of the origin mass at the crossing. It tends to 1 for $s > 2r$ and to 0 for $s < 2r$ (Theorem 6.3(c)), and to $2\sqrt{2}/(\sqrt{2} + \sqrt{5}) = 0.7749$ at the switch $s = 2r$ (Corollary 6.6(b)). For $(r, s) = (1, 0)$ there are no reversals and the share is 0 for every N . This is the walk with turns only of Theorem 6.9. The pair $(0.3, 1)$ was computed up to $N = 640$.

Theorem 6.5 (four-direction local limit). *Fix $r > 0$ and $s \geq 0$. Let $N \rightarrow \infty$ through even integers, let $T = T_N \rightarrow \infty$ with $T_N \leq N^{1/8}$, and put $h = T/N$. Then*

$$N^2 o_N^{(4)} = \frac{2(r+s)}{\pi} T (1 + o(1)).$$

In words, $o_N^{(4)}$ is twice the density at 0 of a centered Gaussian vector with covariance $\frac{N^2}{2(r+s)T} I_2$ (the factor 2 appears because Y_N lies on a sublattice of index 2). The variance can be found as follows. The functions $\cos(\pi j/2)$ and $\sin(\pi j/2)$ of the direction j are eigenvectors of Q with eigenvalue $-2(r+s)$, so each coordinate of the velocity forgets its sign after about $N/(2(r+s)T)$ steps. A coordinate of Y_N is then a sum of about $2(r+s)T$ nearly independent pieces of that length, which move along its axis half of the time, and this gives the variance $N^2/(2(r+s)T)$. The range $T \leq N^{1/8}$ is what our proof gives, and below we only need T of order $\log N$.

Idea of the proof. The proof is in Appendix C. By Lemma 6.1 we may fix the skeleton and let the run lengths be uniform. The total lengths in the four directions are then close to N times a Dirichlet vector, and the chance that east cancels west and north cancels south is close to $N^{-2}g$, with an explicit function g of the numbers of runs in the four directions. Divided by m , this g is a Gaussian function of $W_m/\sqrt{m}$, where W_m is the sum of the unit vectors of the runs, and a martingale central limit theorem for W_m gives the constant.

At the crossing $o_N^{(4)} = c_N(1 - 2S_N(u_N^*))$, and with Theorem 6.5 this becomes an equation for T_N^* .

Corollary 6.6 (turns dominate, and the switch). *Let $r > 0$ and let $N \rightarrow \infty$ through even integers.*

(a) *If $0 \leq s < 2r$, then*

$$\beta T_N^* + \log T_N^* = 2L + \log \frac{\pi}{8(r+s)} + o(1),$$

or equivalently $\beta T_N^ = 2z_N + \log \frac{\beta}{2(r+s)} + o(1)$.*

(b) If $s = 2r$, then $2S_N(u_N^*) \rightarrow \frac{2\sqrt{2}}{\sqrt{2}+\sqrt{5}} = 0.774852\dots$ and

$$\beta T_N^* + \log T_N^* = 2L + \log \frac{\pi}{8(r+s)} + \log \frac{\sqrt{5} - \sqrt{2}}{\sqrt{5} + \sqrt{2}} + o(1).$$

In words, when turns dominate the crossing equation has $\log T$ with the coefficient 1, where Kagey's board has $\frac{1}{2}$. The reason is that the origin is now a point of a local limit in two dimensions. In one dimension the density at the center is of order $T^{1/2}/N$, and in two dimensions it is of order T/N^2 . At the switch $s = 2r$ neither kind of path wins, and the two shares of the origin mass tend to explicit constants.

Proof. Let $0 \leq s \leq 2r$. Then $\tau^* = 2/\beta$, so $T_N^*/L \rightarrow 2/\beta$ by Theorem 6.3(b), and Theorem 6.5 applies with $T_N = T_N^*$. At the crossing Proposition 6.2 gives $o_N^{(4)} = c_N(1 - 2S_N(u_N^*))$, and both sides are positive. Also $\log c_N = -\log 4 + (N-1)\log(1 - \beta h_N^*) = -\log 4 - \beta T_N^* + O(L^2/N)$. Taking logarithms and using Theorem 6.5,

$$\beta T_N^* + \log T_N^* = 2L + \log \frac{\pi}{8(r+s)} + \log(1 - 2S_N(u_N^*)) + o(1). \quad (13)$$

(a) By Theorem 6.3(c), $2S_N(u_N^*) \rightarrow 0$, so the last logarithm in (13) is $o(1)$. For the second form note that $\log T_N^* = \log(2L/\beta) + o(1)$, and that $2z_N = 2L - \log L + \log \frac{\pi}{8} + o(1)$ by the crossing equation of [2] recalled in Section 2.1.

(b) Now $\beta = 2s$ and $2(r+s) = 3s$. Put $z^* = Nu_N^*$, so that $z^* = sT_N^* + O(L^2/N)$ and $z^* \rightarrow \infty$. As in the proof of Theorem 6.3, (12) and $\psi(z) \sim \sqrt{2z/\pi} e^z$ give $2S_N(u_N^*) = \frac{4}{N} \sqrt{2z^*/\pi} e^{z^*} (1 + o(1))$. Let $\varrho_N = Ne^{-sT_N^*}/\sqrt{T_N^*}$, $a = 8\sqrt{s/(2\pi)}$ and $b = 12s/\pi$. Then $2S_N(u_N^*) = (a/\varrho_N)(1 + o(1))$, and Theorem 6.5 with $c_N^{-1} = 4e^{2sT_N^*}(1 + o(1))$ gives $o_N^{(4)}/c_N = (b/\varrho_N^2)(1 + o(1))$. These two positive terms add up to 1. So ϱ_N is bounded and bounded away from 0, and each of its limit points $\tilde{\varrho}$ solves $\tilde{\varrho}^2 - a\tilde{\varrho} - b = 0$. As $a^2 = 32s/\pi$ and $a^2 + 4b = 80s/\pi$, the only positive root is $\tilde{\varrho} = 2\sqrt{s/\pi}(\sqrt{2} + \sqrt{5})$. Hence $2S_N(u_N^*) \rightarrow a/\tilde{\varrho} = 2\sqrt{2}/(\sqrt{2} + \sqrt{5})$, so $1 - 2S_N(u_N^*) \rightarrow (\sqrt{5} - \sqrt{2})/(\sqrt{5} + \sqrt{2})$, and (13) gives the display. $\square$

Dividing (a) by β gives $T_N^* = \frac{2}{\beta}[L - \frac{1}{2}\log L + O(1)]$. With Theorem 6.4(c) this shows that $T_N^* = \tau^*[L - \frac{1}{2}\log L + O(1)]$ for all rates with $\beta > 0$. So the coefficient $\frac{1}{2}$ of $\log \log N$ in Theorem 4.7 appears here too, even when the origin mass comes from a line of compositions and not from a single one. For $r = s = 1$ the bold curve of Figure 2 solves the first equation of (a) without its $o(1)$ term. The approach to these limits is slow. At $N = 10240$ the zero-turn share is 0.7650 for $(r, s) = (0.5, 1)$, where its limit is 0.7749, and still 0.1051 for $(1, 1)$, where its limit is 0 (Figure 6(b)). The errors in Theorem 6.5 and Corollary 6.6 are only $o(1)$, and we do not know the next term.

6.1 The product curve and turns only

The results so far are asymptotic when turns matter. There is one curve in the plane of the two switching probabilities on which everything is exact, because there the planar walk is two independent copies of Kagey's board. Rotate the lattice by 45° , that is, write $Y_N = (Y^{(1)}, Y^{(2)})$ and use $U = Y^{(1)} + Y^{(2)}$ and $V = Y^{(1)} - Y^{(2)}$. Each step changes U by ± 1 and V by ± 1 , so a path of the planar walk is a pair of words in the letters $\pm$, namely the sign word of U and the sign word of V . A left turn changes the sign of one of the two and a right turn changes the other, while a reversal changes both.

Proposition 6.7 (the product curve). *Let $N \geq 1$ and $0 < \rho < 1$. Suppose the planar walk keeps its direction with probability $(1 - \rho)^2$, turns to each side with probability $\rho(1 - \rho)$ and reverses with probability ρ^2 (so $v = x$ and $u = x^2$ with $x = \rho/(1 - \rho)$). Then the sign words of U and V are independent copies of the walk of [2] with switching probability ρ . Hence, with P_N^A the law of the bin on Kagey's board with N steps and $q = \rho$,*

$$\mathbb{P}(Y_N = (y_1, y_2)) = P_N^A\left(\frac{N + y_1 + y_2}{2}\right) P_N^A\left(\frac{N + y_1 - y_2}{2}\right)$$

for all integers y_1, y_2 (the right side is 0 unless both arguments are integers in $[0, N]$). In particular, for even N ,

$$\frac{o_N}{c_N} = S_N(x)^2,$$

and the corner and the origin are equally likely exactly when $\rho = 1 - p_N$.

In words, on this curve the two diagonal coordinates of the planar walk do not see each other. So the planar crossing is Kagey's crossing, exactly and at every N , and every bound that Paper A proves for p_N holds for it.

Proof. The steps e_0, e_1, e_2, e_3 change (U, V) by $(+, +)$, $(+, -)$, $(-, -)$ and $(-, +)$. So a direction is the same thing as a pair of signs, and a uniform first direction is a pair of independent fair signs. From any direction one turn changes only the sign of U , the other turn changes only the sign of V , and a reversal changes both. Under the hypothesis, at each step neither sign changes with probability $(1 - \rho)^2$, only the first with probability $\rho(1 - \rho)$, only the second with probability $\rho(1 - \rho)$, and both with probability ρ^2 , whatever the past. These are the probabilities of two independent events of probability ρ each. So the two sign words are independent, and each is a fair first letter followed by switches of probability ρ . The word of U has $(N + U)/2$ letters $+$, and likewise for V , which gives the product formula. The corner Ne_0 has $U = V = N$ and the origin has $U = V = 0$, so $o_N/c_N = (P_N^A(N/2)/P_N^A(N))^2 = S_N(x)^2$, and this equals 1 exactly when $x = u_N$, that is, when $\rho = 1 - p_N$. $\square$

On the curve $\beta h = 1 - (1 - \rho)^2 = \rho(2 - \rho)$, so the crossing is at $\beta T_N^* = 2z_N - z_N^2/N$. Corollary 6.6(a) is stated for fixed rates and does not cover this case, but the two agree in form. At the crossing on the curve, $s/r = \rho/(1 - \rho)$ tends to 0, and then $\log \frac{\beta}{2(r+s)}$ tends to 0. In continuous time the walk with turns only is a rotation of two independent telegraph processes (Orsingher [30], Cinque and Orsingher [29, Theorem 2.2], Smith, Le Doussal, Majumdar and Schehr [32], see also [28, Eq. (36)]), and with reversals the two processes are dependent [29, Section 3]. On the lattice two independent switches happen at the same step with probability ρ^2 , which is why the curve has reversals. At $\rho = \frac{1}{2}$ this is the classical fact that the simple random walk in the plane is a rotation of two independent walks on the line. We did not find the lattice statement for other ρ .

Off the curve the two sign words are dependent. But o_N/c_N is increasing in each of the odds v and u , so the curve still gives bounds at every N .

Corollary 6.8 (comparison with the product curve). *Let $N \geq 2$ be even and $0 < h < 1/\beta$. Then*

$$S_N(\min(v, \sqrt{u}))^2 \leq \frac{o_N}{c_N} \leq S_N(\max(v, \sqrt{u}))^2.$$

So at the crossing u_N lies between v and $\sqrt{u}$.

In words, at the planar crossing the odds u_N of a switch at Kagey's crossing lie between the odds of a turn to a given side and the square root of the odds of a reversal.

Proof. By Proposition 6.2, o_N/c_N is a polynomial $R_N(v, u)$ with nonnegative coefficients, so it is nondecreasing in v and in u . By Proposition 6.7, $R_N(x, x^2) = S_N(x)^2$ for every $x > 0$. With $x = \max(v, \sqrt{u})$ we have $v \leq x$ and $u \leq x^2$, so $R_N(v, u) \leq R_N(x, x^2)$. With $x = \min(v, \sqrt{u})$ we get the lower bound in the same way (if $x = 0$ it is trivial). At the crossing $R_N = 1 = S_N(u_N)^2$, and S_N is increasing. $\square$

With turns only ($s = 0$) the sign words of U and V are two copies of the walk of [2] that never switch at the same step. The upper bound of Corollary 6.8 then reads $o_N/c_N \leq S_N(v)^2$. The next theorem shows that forbidding common switches changes the crossing very little.

Theorem 6.9 (turns only). *Let $s = 0 < r$ and let $N \geq 4$ be even. Call a word of length N in the letters $\pm$ balanced if it has $N/2$ letters of each sign, and let $\Delta_N(x)$ be the sum of $x^{j+j'}$ over the pairs of balanced words, with j and j' switches, that switch at a common step.*

(a) $o_N/c_N = S_N(v)^2 - \Delta_N(v)$ and $\Delta_N \geq 0$. So the value v_N^* of v at the crossing satisfies $v_N^* \geq u_N$.

(b) If $z = Nx$ satisfies $z(z+2) \leq N$, then $\Delta_N(x) \leq (2x + 8xS_N(x))(2z+2)S_N(x)$.

(c) As $N \rightarrow \infty$,

$$Nv_N^* = Nu_N + O\left(\frac{L^2}{N}\right) \quad \text{and} \quad rT_N^* = z_N + O\left(\frac{L^2}{N}\right).$$

In words, with turns only the planar crossing is Kagey's crossing up to $O(L^2/N)$, and measured by the odds of a turn it is never earlier. Note that v_N^* and $rT_N^* = Nv_N^*/(1 + 2v_N^*)$ do not depend on r . At $N = 10240$ we find $v_N^*/u_N - 1 = 4.0 \times 10^{-4}$ and $rT_N^* - z_N = -0.0027$, while $L^2/N = 0.0083$. (The second number is negative although $v_N^* \geq u_N$, because the odds v correspond to $rT = Nv/(1 + 2v)$ and the odds u_N to $z_N = Nu_N/(1 + u_N)$.)

Idea of the proof. Part (a) is a count of paths. For (b) we use a union bound over the step of a common switch. Given that a word is balanced, it switches at a given step with probability of order x (Lemma C.4), and it has about z switches in all. So a second balanced word shares a switch with it with probability of order $xz = z^2/N$. Part (c) follows from (a), (b) and the uniform Bessel bound (12). The proofs of (b) and (c) are in Appendix C.

Proof of (a). A path with turns only is a pair of words that never switch at the same step, and it has probability $c_N v^{j+j'}$ if the words have j and j' switches. It ends at the origin exactly when both words are balanced. Since $S_N(v)$ is the sum of v^j over the balanced words, $S_N(v)^2$ is the sum of $v^{j+j'}$ over all pairs of balanced words, and removing the pairs with a common switch leaves o_N/c_N . So $S_N(v_N^*)^2 \geq 1 = S_N(u_N)^2$, and S_N is increasing. $\square$

7 Application: sticky priors for hidden Markov models

This section is an application for readers from statistics. Nothing earlier depends on it, and a reader who is interested only in the general theory and the planar walk can skip to Section 8.

In a hidden Markov model the hidden states $X_1, \dots, X_N$ form a Markov chain on d states. Before any data are seen, the model already says how the N time steps are likely to be shared among the states. This is the law P_N of the count vector $n = (n_1, \dots, n_d)$, where n_i is the number of time steps in state i (the composition of Section 2). A sticky prior makes the chain leave its current state rarely, which is the regime of this paper. We ask which states the most likely count vector uses.

Here is what we use from the earlier sections. Suppose the chain leaves its state with probability $h = \tau \log N/N$ at each step and then picks the next state from a kernel Π . For a set F of states (a face) let ρ_F be the Perron root of Π restricted to F . The chain stays in F for all N steps with probability about $N^{-\tau(1-\rho_F)}$, and about $N^{|F|-1}$ count vectors use exactly the states of F . Theorem 3.3 says that for large N the most likely count vector uses a set F that minimizes $\Phi_\tau(F) = \tau(1 - \rho_F) + |F| - 1$, and that its fractions n/N are then close to the law α_F^* , the entrywise product of the left and right Perron vectors of Π on F . Besides this we use the description of a tie between several minimizers (Theorem 4.8) and the exact identity for a symmetric pair of states (Proposition 4.9), once each.

The answer is as follows. As the stickiness decreases, that is, as τ grows, the most likely count vector moves from one state to larger sets of states. Among symmetric kernels only the uniform one jumps from one state straight to all of them.

Fox et al. [22] add a self-transition bias $\kappa \geq 0$ to the HDP hidden Markov model (HDP stands for hierarchical Dirichlet process), so that the hidden chain stays in each state longer. In their weak-limit approximation with base weights $\bar{\beta}$ and concentration ω , row j of the transition matrix is Dirichlet with parameters $\omega\bar{\beta}_1, \dots, \omega\bar{\beta}_j + \kappa, \dots, \omega\bar{\beta}_d$. By neutrality, row j off the diagonal, divided by $1 - \pi_{jj}$, is Dirichlet with parameters $(\omega\bar{\beta}_l)_{l \neq j}$ and independent of π_{jj} . So κ only sets how sticky the chain is, and the kernel of the switches does not depend on it. In this section we fix every exit probability $1 - \pi_{jj}$ at the same value $h = \tau L/N$, the scale of Kagey's crossing, and study the kernel. The prior itself mixes over all values of the exit probabilities, and we do not treat this mixture.

So let $\Pi = (\pi_{ij})$ be a stochastic matrix on $[d]$ with zero diagonal, $\tau > 0$ and $h = \tau L/N < 1$. The hidden chain has transition matrix $(1 - h)I + h\Pi = I + hQ$ with $Q = \Pi - I$, so $q_i = 1$ and $T = \tau L$ (Example 2.5(iii)). We assume that $\pi_{ij} > 0$ for all $i \neq j$, which we call (A+), and that the start μ has full support. Under the prior (A+) holds almost surely. Let $\rho_F = \varrho(\Pi_F)$ for the principal submatrix Π_F on F . Then the exit rate of F is $\lambda_F = 1 - \rho_F$, and Q_F and Π_F have the same Perron vectors. A vertex (a single state) has $\rho_F = 0$, and $\rho_{[d]} = 1$. If $2 \leq |F| \leq d - 1$, then Π_F is irreducible by (A+), and $0 < \rho_F < 1$ since ρ_F is at least the least row sum of Π_F and the Perron root of the irreducible Π exceeds that of each proper principal submatrix [36]. So $\Phi_\tau(F) = \tau(1 - \rho_F) + |F| - 1$, and we put

$$\tau_- = \min_{|F| \geq 2} \frac{|F| - 1}{\rho_F}, \quad \tau_+ = \max_{F \neq [d]} \frac{d - |F|}{1 - \rho_F}.$$

We call a global mode of P_N , that is, a most likely value of the count vector before any data are seen, a *prior mode*.

Corollary 7.1 (support and location of the prior mode). *Assume (A+), let μ have full support, and fix $\tau > 0$. Let $\mathcal{F}_\tau$ be the set of faces that minimize Φ_τ . There is $N_0(\tau)$ such that for $N \geq N_0(\tau)$ every prior mode n^* satisfies $\text{supp } n^* \in \mathcal{F}_\tau$ and $n^*/N = \alpha_F^* + O(1/L)$ with $F = \text{supp } n^*$. If $\mathcal{F}_\tau = \{F^*\}$, then $n^*/N \rightarrow \alpha_{F^*}^* = l \circ r$, where l and r are the left and right Perron vectors of Π_{F^*} with $l \cdot r = 1$. No reversibility is needed.*

In words, for large N a prior mode uses exactly the states of a set that minimizes Φ_τ , and it shares the N time steps among them in the proportions α_F^* .

Proof. By (A+) every face is irreducible and, as μ has full support, admissible. So Theorem 3.3(iii) and (iv) give the claims, with $\alpha_F^* = l \circ r$. $\square$

The threshold $N_0(\tau)$ is not uniform in τ . At $\tau = \tau_-$ the vertices tie with the faces that attain τ_- , and Theorem 4.8(iii) with $\tau^* = \tau_-$ and a fixed large $t = t_0$ shows that for large N the prior mode is not a single state at $\tau = \tau_-(1 - \frac{\log L - 2t_0}{2L}) < \tau_-$. So $N_0(\tau) \rightarrow \infty$ as $\tau \uparrow \tau_-$.

The next proposition says for which τ the prior mode uses one state, all states, or a set in between.

Proposition 7.2 (thresholds). *Assume (A+) and $d \geq 3$.*

- (a) *The minimizers of Φ_τ are exactly the vertices if and only if $\tau < \tau_-$, and no vertex minimizes Φ_τ if $\tau > \tau_-$. The face $[d]$ is the unique minimizer if and only if $\tau > \tau_+$, and $[d]$ does not minimize Φ_τ if $\tau < \tau_+$.*
- (b) $1 < \tau_- \leq d - 1 \leq \tau_+$.
- (c) $\tau_- = d - 1$ if and only if $\tau_+ = d - 1$, and both hold if and only if

$$\rho_F \leq \frac{|F| - 1}{d - 1} \quad \text{for every face } F. \quad (14)$$

If (14) fails, then $\tau_- < d - 1 < \tau_+$, and for $\tau \in (\tau_-, \tau_+)$ every minimizer F of Φ_τ has $2 \leq |F| \leq d - 1$.

Consequently, let μ have full support, fix τ and let $N \geq N_0(\tau)$ as in Corollary 7.1. If $\tau < \tau_-$, the prior modes are the Ne_i with $\mu_i = \max_l \mu_l$. If $\tau > \tau_+$, every prior mode has full support. If $\tau_- < \tau < \tau_+$, every prior mode has a support F with $2 \leq |F| \leq d - 1$, and $F = F^*$ if F^* is the unique minimizer of Φ_τ .

Proof. (a) A vertex has $\Phi_\tau = \tau$. For $|F| \geq 2$, $\Phi_\tau(F) - \tau = \rho_F((|F| - 1)/\rho_F - \tau)$ with $\rho_F > 0$, and for $F \neq [d]$, $\Phi_\tau(F) - (d - 1) = (1 - \rho_F)(\tau - (d - |F|)/(1 - \rho_F))$ with $1 - \rho_F > 0$, while $\Phi_\tau([d]) = d - 1$. (b) For $2 \leq |F| \leq d - 1$, $\rho_F < 1$ gives $(|F| - 1)/\rho_F > 1$, and $[d]$ gives $d - 1 \geq 2$, so $\tau_- > 1$. Also $[d]$ gives $\tau_- \leq d - 1$, and a vertex gives $\tau_+ \geq d - 1$. (c) By (b), $\tau_- = d - 1$ exactly when $(|F| - 1)/\rho_F \geq d - 1$ for all $|F| \geq 2$, and $\tau_+ = d - 1$ exactly when $(d - |F|)/(1 - \rho_F) \leq d - 1$ for all $F \neq [d]$. Both say $(d - 1)\rho_F \leq |F| - 1$, which holds for the vertices and $[d]$. If (14) fails, we get $\tau_- < d - 1 < \tau_+$, and (a) excludes the vertices and $[d]$ on (τ_-, τ_+) . The consequence follows from Corollary 7.1 and $P_N(Ne_i) = \mu_i(1 - h)^{N-1}$. Parts (a) to (c) are checked in Lean. $\square$

We say that Π has a *direct jump* when (14) holds. Then at first order the prior mode goes from one state straight to all d states at $\tau = d - 1$. For the uniform kernel $\pi_{ij} = 1/(d - 1)$ the matrix Π_F has constant row sums, so $\rho_F = (|F| - 1)/(d - 1)$. Hence (14) says that no set of states is more closed than under the uniform kernel. That kernel has $Q = (J - dI)/(d - 1)$, and all its faces tie at $\tau = d - 1$. This is the degeneracy of [3] (Corollary 4.11). The next corollary shows that a direct jump is special.

Corollary 7.3 (only the uniform kernel). *Let $d \geq 3$ and assume (A+). (a) If Π is symmetric, then Π has a direct jump if and only if $\pi_{ij} = 1/(d - 1)$ for all $i \neq j$. (b) If $d = 3$, then Π has a direct jump if and only if $\pi_{ij}\pi_{ji} \leq \frac{1}{4}$ for all three pairs. In particular every cyclic kernel $\pi_{i,i+1} = a$, $\pi_{i,i-1} = 1 - a$ (indices mod 3) has a direct jump.*

Proof. A pair has $\rho_{\{i,j\}} = \sqrt{\pi_{ij}\pi_{ji}}$. (a) Here $\rho_{\{i,j\}} = \pi_{ij}$, and since the rows sum to 1 the average of π_{ij} over the pairs is $1/(d - 1)$. So (14) fails unless all π_{ij} are equal, and the uniform kernel satisfies it with equality. (b) For $d = 3$, (14) is automatic for the vertices and [3], and for a pair it reads $\pi_{ij}\pi_{ji} \leq \frac{1}{4}$. The cyclic kernel has $a(1 - a) \leq \frac{1}{4}$. Both parts are checked in Lean. $\square$

So for a symmetric kernel that is not uniform there is a range of τ in which the prior mode uses more than one state but not all of them (Proposition 7.2(c)). Without symmetry a direct jump is possible, and for three states we can say how likely it is under the prior.

Proposition 7.4 (the direct jump under the prior). *Let $d = 3$ and let the base weights be uniform. In the weak-limit sticky HDP-HMM the rows of Π are then independent, and the first off-diagonal entry of each row has the law $\text{Beta}(\omega/3, \omega/3)$. Write $\mathbb{P}_\omega$ for this law of Π and D for the event that Π has a direct jump. Then (a) $\mathbb{P}_\omega(D) \rightarrow \frac{1}{4}$ as $\omega \rightarrow 0$, and (b) $\mathbb{P}_\omega(D) = \frac{3\sqrt{3}}{8\pi\omega}(1 + o(1))$ as $\omega \rightarrow \infty$.*

Proof. Let B_1, B_2, B_3 be independent with law $\text{Beta}(\omega/3, \omega/3)$, and put $\pi_{12} = B_1$, $\pi_{21} = B_2$ and $\pi_{31} = B_3$, so that the other off-diagonal entries are $1 - B_j$. By Corollary 7.3(b), D is the event that B_1B_2 , $(1 - B_1)B_3$ and $(1 - B_2)(1 - B_3)$ are all at most $\frac{1}{4}$. (a) As $\omega \rightarrow 0$ each B_j tends in law to a fair coin on $\{0, 1\}$, and the three products then take only the values 0 and 1. So ∂D has limit probability 0, and $\mathbb{P}_\omega(D)$ tends to the limit probability of D . Only $(1, 0, 1)$ and $(0, 1, 0)$ make all three products 0 (checked in Lean), so the limit is $\frac{1}{4}$. Part (b) is proved in Appendix E. $\square$

In words, a direct jump has probability close to $\frac{1}{4}$ when the concentration is small and of order $1/\omega$ when it is large. For (b), for large ω the set D is a thin tube around the cyclic kernels, with cross-section of area about $\frac{9}{8}t^4$ at the logit coordinate t along the cyclic kernels (Appendix E), which gives the order $1/\omega$. So the uniform tie is not typical. At $\omega = 1$, Monte Carlo with 10^6 draws gives $\mathbb{P}_\omega(D) = 0.115$. So about 0.88 of random kernels have a range of τ in which, at first order, the prior mode sits on a pair (Proposition 7.2(c)).

The comparison of a single state with a symmetric pair is exact at every N .

Corollary 7.5 (an exact link to Kagey's crossing). *Let $i \neq j$ satisfy $\pi_{ij} = \pi_{ji} = \pi_0 > 0$ and $\mu_i = \mu_j > 0$. We do not assume (A+). Then for every $N \geq 2$ the balanced compositions are the most likely ones with support $\{i, j\}$, and they are more likely than Ne_i if and only if $\tau > \tau^{\text{pair}}(N) := Nu_N/((\pi_0 + u_N)L)$. If also $\mu_i = \mu_j = \max_l \mu_l$, then for every τ with $\tau^{\text{pair}}(N) < \tau < N/L$ no prior mode is a single state. Moreover,*

$$\tau^{\text{pair}}(N) = \frac{1}{\pi_0} \left[1 - \frac{\frac{1}{2} \log L - c_0}{L} + O\left(\frac{\log L}{L^2}\right) \right],$$

which tends to the first-order value $1/\rho_{\{i,j\}} = 1/\pi_0$.

In words, a single state against a symmetric pair is Kagey's board, so the value of τ at which the pair overtakes the state is known at every N and not only in the limit.

Proof. Every exit rate is 1, so this is Proposition 4.9 with $a = \pi_0$ and $q = 1$, where $u = \pi_0 h/(1-h) > u_N$ exactly when $h > u_N/(\pi_0 + u_N)$. For the prior modes, $P_N(Ne_l) = \mu_l(1-h)^{N-1} \leq P_N(Ne_i)$ for every l , and the range of τ is not empty as $u_N/(\pi_0 + u_N) < 1$. Finally $Nu_N = z_N/p_N$ and $u_N = O(L/N)$, so $\tau^{\text{pair}}(N) = \frac{z_N}{\pi_0 L}(1 + O(L/N))$, and the second-order expansion of z_N in [2, Corollary 3.7(b)] gives the claim. $\square$

So far we looked at the single most likely count vector. The last result gives the total probability that the chain uses only the states of a set F .

Proposition 7.6 (face masses). *Assume (A+) and $0 < h < 1$. For every face F with $|F| \geq 2$ and $\Lambda_F = 1 - h\lambda_F$,*

$$\frac{\min r_F}{\max r_F} \mu(F) \Lambda_F^{N-1} \leq \mathbb{P}(\text{supp } n \subseteq F) \leq \frac{\max r_F}{\min r_F} \mu(F) \Lambda_F^{N-1},$$

τ	prior mode	one state	$\{1, 2\}$	$[3]$
0.5	one state	0.0643	0.465	0.379
0.8	one state	0.0122	0.423	0.531
3	(120, 120, 0)	4.3×10^{-8}	0.129	0.871
8	(111, 111, 18)	1.2×10^{-21}	0.0081	0.992

Table 2: The kernel Π with rows $(0, 0.9, 0.1)$, $(0.9, 0, 0.1)$, $(0.5, 0.5, 0)$, the uniform start and $N = 240$, where $\tau_- = 10/9$ and $\tau_+ = 10$. The prior mode (from exact dynamic programming, with a tie of three compositions at $\tau = 0.5$ and 0.8) and the prior probabilities of the supports.

and for fixed τ , $\Lambda_F^{N-1} = N^{-\tau\lambda_F} e^{O(L^2/N)}$. Also $\mathbb{P}(\text{one state}) = (1-h)^{N-1} \sim N^{-\tau}$ and $\mathbb{P}(\text{supp } n = [d]) \rightarrow 1$ for fixed τ .

In words, the chain stays inside F with probability $\mu(F)N^{-\tau\lambda_F}$ up to a bounded factor, and the probability that it uses every state tends to 1.

Proof. Let $U_F = (1-h)I + h\Pi_F \geq 0$. Then $\mathbb{P}(\text{supp } n \subseteq F) = \mu_F^\top U_F^{N-1} \mathbf{1}$ and $U_F r_F = \Lambda_F r_F$. As $r_F / \max r_F \leq \mathbf{1} \leq r_F / \min r_F$ entrywise, this gives the bounds, and $(N-1)\log(1-h\lambda_F) = -\tau\lambda_F L + O(L^2/N)$. A vertex composition is a single path, and every $F \neq [d]$ has $\lambda_F > 0$, which gives the last two claims. The finite bounds are checked in Lean. $\square$

Remark 7.7 (prior mode versus mass). Let $d \geq 3$. For fixed $\tau < \tau_+$ and large N every prior mode has a proper support, while $\mathbb{P}(\text{supp } n = [d]) \rightarrow 1$. For $\tau < \tau_-$ the prior mode uses one state, an event of probability about $N^{-\tau}$. A face of size k holds about N^{k-1} compositions, so it can carry most of the mass while each of them is less likely than a vertex. Table 2 shows this. At $\tau = 0.8$ the prior mode is a single state, yet $\mathbb{P}(\text{one state}) = 0.0122$. At $\tau = 8$ first order puts the prior mode on $\{1, 2\}$, but at $N = 240$ it has full support. So the prior mode can be a poor summary of a sticky prior, and face selection tells us on which states it sits.

8 Verification and open questions

The theorems, propositions, lemmas and corollaries of this paper are proved in the text or in the appendices, with two kinds of outside input. We quote results of Papers A and B where we say so (for example the uniform Bessel bound (12), and the expansion behind the value for the complete graph in Proposition 4.14(iv)). And the rows of Table 1 marked Lean are proved in Lean and not in the text. No proof rests on a floating-point computation. Exact computer algebra enters in two places. First, Proposition D.4, which proves the row Θ_ϵ of Table 1, rests on two exact resultants of characteristic polynomials, computed in sympy by `verify_examples.py` (Section D.3). Second, the values of c_F for the 3-arc and for C_4 after Corollary 5.1 are exact evaluations, by `verify_second_order.py`, of the formula for c_F in Proposition 4.14(ii). The proposition is proved, and only these two values are computed.

The other numbers in the paper are of three kinds. Some are closed forms. Some are computations of the law at a finite N , namely the finite- N row of P_4 in Table 1, Figure 4(b), the crossing of the 3-arc at $N = 10^{14}$ in Section 5, the planar crossings and zero-turn shares in Section 6 and Figure 6, and Table 2. These are done in two independent ways where possible (for example by the run expansion and by a torus integral, or by a dynamic program and an exact

formula). They illustrate the theorems, and no statement is deduced from them. Finally, the value 0.115 in Section 7 is a Monte Carlo estimate.

For the planar walk, `verify_planar.py` computes o_N in four ways and finds the crossings. The script `verify_planar_product.py` checks the product formula of Proposition 6.7 in exact rational arithmetic for $N = 4, 6, 8$. In floating point it also checks the inequality $F(y/2) \leq 2F(y)$ of Appendix C, the bound on the number of switches in Lemma C.4, and Theorem 6.9(a) and (b) for N up to 96. These checks guard against slips. The proofs do not use them.

Many finite and algebraic steps, and the rows of Table 1 marked Lean, are proved in Lean 4 in the folder `kagey-problems/problem131/lean/PaperC`, and its file `MAP.md` matches each statement of the paper to its Lean theorems. For the planar walk Lean covers the identity and the unique crossing for $s > 0$ in Proposition 6.2. The asymptotic results, the product curve and the theorem on turns only are not in Lean.

Data and code availability. Everything is in the folder `kagey-problems/problem131` of <https://github.com/apemm/Kagey-Problems>, at the release `paperC-arxiv-v1`. To rerun the checks, run `python -B verify_all.py --paper c` in that folder. The file `SUPPLEMENT.md` lists each check, the two methods behind it and the predictions that we wrote down before most of the computations. The scripts are in the subfolder `paperC` and need `numpy`, `scipy`, `sympy` and `mpmath`. The largest cases, such as the planar crossings for $N \geq 1280$, run only with the option `--full`.

Open questions. We list them in the order of the paper.

The local law at the boundary. Theorem 4.4 holds on compact subsets of the open face, and a local law up to the boundary is open. The same formula cannot hold there. Take a star with center d and $\ell = d - 1 \geq 3$ leaves, and a start of full support. Then $P_N(n) = 0$ whenever $\text{supp } n = [d]$ and $n_d < \ell - 1$, since a path visits the center between any two runs at the leaves, although n/N lies in the open face. This zero set is checked in Lean.

Reducible faces. For a reducible face, Theorem 3.3(ii) places the power of T in M_F between $k_F - k^*$ and $k_F - 1$, and heuristically it is $(k^* - 1)/2 + k_F - k^*$. We do not prove this.

Uniqueness and constants. Theorem 4.7 puts all crossings of two faces in an interval of length $O(1/L)$, but uniqueness is proved only in special cases, such as Papers A and B, the pairs of Proposition 4.9 and the planar walk. We have no tree formula for K_F on non-reversible faces like Proposition 4.2(b), and no closed form for c_F . At finite N the order of the phases can differ from first order when a face lies close to the hull, as on the path P_4 (Table 1).

The planar walk. The first-order heuristic at the start of Section 6 applies face selection to a linear function of the composition. We prove its conclusion for the planar walk, but not such a rule in general. The errors in Theorem 6.5 and Corollary 6.6 are only $o(1)$, and Theorem 6.5 needs $T \leq N^{1/8}$. The next term of the crossing equation when turns dominate is open, and so is the crossover when $s - 2r$ tends to 0 as N grows. A heuristic suggests that the next term comes from replacing T by $T + \frac{1}{4r} - \frac{1}{2(r+s)}$ in Theorem 6.5. For $s = 0$ this agrees with the term $1/(8z_N)$ of Paper A, through Theorem 6.9(c). Off the product curve of Proposition 6.7 we have no closed form for o_N or for the crossing, and Corollary 6.8 only brackets the crossing. With turns only, the computations give $N(v_N^* - u_N) = 0.37 L^2/N$ at $N = 96$ and $0.38 L^2/N$ at $N = 10240$. So the order L^2/N in Theorem 6.9(c) seems to be right, but we do not prove a lower bound of this order.

Sticky priors. We do not treat random exit probabilities as in [22], or the probability of a direct jump for $d \geq 4$.

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A note on the use of AI

All mathematical claims, proofs and computations in this paper are my responsibility.

Statements and declarations

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Data availability. All code, data and Lean files supporting this paper are in the folder kagey-problems/problem131 of <https://github.com/apemm/Kagey-Problems>, at the release paperC-arxiv-v1. Section 8 describes what each check does and how to rerun it.

A Proofs of the local law and of the first-order bounds

This appendix proves Proposition 4.2(a), Lemma 3.2, Theorem 4.4, Lemma 4.5 and Theorem 3.3(ii) for reducible faces. Except in Section A.6, F is a face with $k \geq 2$ states, numbered $1, \dots, k$. For $x \in \mathbb{C}^F$ we write $x' = (x_2, \dots, x_k)$. We write i for the imaginary unit, since i denotes a state, and $\mathbb{T}^m = (\mathbb{R}/2\pi\mathbb{Z})^m$. We use two standard facts [36]. First, let $B \geq 0$ be square and S diagonal with positive diagonal. Then $\varrho(B - S) < 0$ if and only if $\varrho(S^{-1}B) < 1$, and also if and only if $(S - B)v > 0$ for some $v > 0$. In that case $(S - B)^{-1} = \sum_{j \geq 0} (S^{-1}B)^j S^{-1} \geq 0$. Second, if M is an irreducible Metzler matrix, then $\varrho(M)$ is larger than the Perron root of each proper principal submatrix of M , and every nonnegative eigenvector of M is positive and belongs to $\varrho(M)$.

A.1 The Perron root under a diagonal perturbation

An irreducible generator on F is a Metzler matrix G with $G\mathbf{1} = 0$ whose graph is strongly connected. It has a unique stationary law $\pi > 0$. The argument after Definition 4.1, with (G, π) in place of $(\hat{Q}_\alpha, \alpha)$, shows that $Z = (\mathbf{1}\pi^\top - G)^{-1} - \mathbf{1}\pi^\top$ exists and satisfies (4). Put $D = \text{diag } \pi$ and $\Gamma = DZ + Z^\top D$.

Lemma A.1 (the Perron root). *Let G , π , Z and Γ be as above. Then $\kappa(\eta) = \varrho(G + \text{diag } \eta)$ is real-analytic near 0, with $\kappa(0) = 0$, $\nabla \kappa(0) = \pi$ and $\nabla^2 \kappa(0) = \Gamma$. For $\eta \in \mathbb{R}^F$ and $u = Z\eta$,*

$$\eta^\top \Gamma \eta = \sum_{i \neq j} \pi_i G_{ij} (u_i - u_j)^2. \quad (15)$$

So Γ is positive semidefinite with kernel $\mathbb{R}\mathbf{1}$, and Σ , which is Γ without the row and column of state 1, is positive definite.

Proof. Since $G + \text{diag } \eta$ is irreducible, its Perron root is a simple eigenvalue. So by the implicit function theorem, κ and the right eigenvector $v(\eta)$ with $\pi^\top v = 1$ are real-analytic near 0, with $\kappa(0) = 0$ and $v(0) = \mathbf{1}$. With subscripts for derivatives at $\eta = 0$, differentiating $(G + \text{diag } \eta)v = \kappa v$ in η_j gives $Gv_j + e_j = \kappa_j \mathbf{1}$, and multiplying by $\pi^\top$ gives $\kappa_j = \pi_j$. Then $v_j = Ze_j$, because by (4) both vectors solve $Gx = \pi_j \mathbf{1} - e_j$ with $\pi^\top x = 0$, and the kernel of G is $\mathbb{R}\mathbf{1}$. Differentiating

again, in η_i , gives $Gv_{ij} + \text{diag}(e_i)v_j + \text{diag}(e_j)v_i = \kappa_{ij}\mathbf{1} + \kappa_i v_j + \kappa_j v_i$, and multiplying by $\pi^\top$ gives $\kappa_{ij} = \pi_i Z_{ij} + \pi_j Z_{ji} = \Gamma_{ij}$.

For (15) let $f = \eta - (\pi \cdot \eta)\mathbf{1}$. By (4), $-Gu = f$ and $\pi \cdot u = 0$, so $\eta^\top \Gamma \eta = 2 \sum_i \pi_i \eta_i u_i = 2 \sum_i \pi_i f_i u_i = -2 \sum_i \pi_i u_i (Gu)_i$. Expanding the squares in (15) and using $G\mathbf{1} = 0$ and $\pi^\top G = 0$ gives the same expression. Also $\Gamma\mathbf{1} = 0$ by (4). If $\eta^\top \Gamma \eta = 0$, then u is constant along every edge of the strongly connected graph of G , so u is constant. Then $u = 0$ because $\pi \cdot u = 0$, so $f = 0$ and $\eta \in \mathbb{R}\mathbf{1}$. A nonzero vector that vanishes at state 1 is not in $\mathbb{R}\mathbf{1}$, which gives the claim about Σ . $\square$

Proof of Proposition 4.2(a). For $\vartheta \in \mathbb{R}^F$ let $\Lambda(\vartheta) = \varrho(Q_F + \text{diag } \vartheta)$, so that $\Lambda(\vartheta + c\mathbf{1}) = \Lambda(\vartheta) + c$. It is a simple eigenvalue with positive right and left eigenvectors $\hat{r}, \hat{l}$, normalized by $\sum_i \hat{r}_i = 1$ and $\hat{l} \cdot \hat{r} = 1$, and by the implicit function theorem $\Lambda, \hat{r}$ and $\hat{l}$ are real-analytic in ϑ . Put $\pi = \hat{l} \circ \hat{r}$ and $G = \hat{R}^{-1}(Q_F + \text{diag } \vartheta - \Lambda(\vartheta)I)\hat{R}$ with $\hat{R} = \text{diag } \hat{r}$. Then G is an irreducible generator with stationary law π . Since $\hat{R}$ commutes with diagonal matrices, $\Lambda(\vartheta + \eta) = \Lambda(\vartheta) + \varrho(G + \text{diag } \eta)$. So by Lemma A.1, $\nabla \Lambda(\vartheta) = \pi$ and $\nabla^2 \Lambda(\vartheta) = \Gamma$, with Γ built from (G, π) .

Let $\Lambda_0(\beta) = \Lambda(0, \beta)$ for $\beta \in \mathbb{R}^{k-1}$. Its Hessian is Γ without the row and column of state 1, which is positive definite by Lemma A.1. So $\nabla \Lambda_0$ is injective, and by the inverse function theorem it has a real-analytic inverse on its open image. Fix $\alpha \in \Delta_F^\circ$ and put $\vartheta(\alpha) = q - \theta(\alpha)$. Then $Q_F + \text{diag } \vartheta(\alpha) = A_F - \text{diag } \theta(\alpha)$ has Perron root 0 and, by (3), Perron vectors r_α and l_α . So at $\vartheta(\alpha)$ we have $\pi = \alpha$, $G = \hat{Q}_\alpha$ and $\Gamma = \Gamma_\alpha$. Let $\beta(\alpha)$ be the last $k-1$ coordinates of $\vartheta(\alpha) - \vartheta_1(\alpha)\mathbf{1}$. By the shift property, $\nabla \Lambda_0(\beta(\alpha)) = \alpha'$ and $\nabla^2 \Lambda_0(\beta(\alpha)) = \Sigma_\alpha$. Hence $\beta(\alpha) = (\nabla \Lambda_0)^{-1}(\alpha')$ is real-analytic, and so are $\vartheta_1(\alpha) = -\Lambda_0(\beta(\alpha))$, $\theta(\alpha)$, its Perron vectors and Σ_α . Moreover $I_F(\alpha) = \langle \vartheta(\alpha), \alpha \rangle = \langle \beta(\alpha), \alpha' \rangle - \Lambda_0(\beta(\alpha))$. Differentiating and using $\nabla \Lambda_0(\beta) = \alpha'$ gives $\nabla I_F = \beta$. So $\nabla^2 I_F$ is the Jacobian of $(\nabla \Lambda_0)^{-1}$ at α' , which is Σ_α^{-1} . The formula for C_F follows from Definition 4.1. $\square$

A.2 The torus representation

We split

$$A_F = \begin{pmatrix} 0 & a_1^\top \\ b_1 & A' \end{pmatrix}, \quad \mu_F = (\mu_1, \mu'),$$

so $a_1 = (q_{1j})_{j \geq 2}$, $b_1 = (q_{i1})_{i \geq 2}$ and $A' = (q_{ij})_{i,j \geq 2}$. For $z' \in \mathbb{C}^{k-1}$ let $\mathcal{Z}' = \text{diag}(z')$. Where $\mathcal{Z}' - A'$ is invertible we put $X(z') = (\mathcal{Z}' - A')^{-1}$, $g(z') = a_1^\top X(z')b_1$ and

$$b(z') = (\mu_1 + \mu'^\top X(z')b_1)(1 + a_1^\top X(z')\mathbf{1}).$$

Lemma A.2 (torus representation). *Let $s' \in (0, \infty)^{k-1}$ satisfy $\varrho(A' - \text{diag } s') < 0$. On the set of $z' \in \mathbb{C}^{k-1}$ with $|z_i| \geq s_i$ for every $i \geq 2$, the matrix $\mathcal{Z}' - A'$ is invertible, and X, g and b are given by absolutely convergent power series in $1/z_2, \dots, 1/z_k$ with nonnegative coefficients. Hence $|g(z')| \leq g(s')$ and $|b(z')| \leq b(s')$ there. For every $t \in (0, \infty)^F$,*

$$H(t) = \frac{1}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} \exp\left(t_1 g(z') + \sum_{i \geq 2} t_i z_i\right) b(z') \prod_{i \geq 2} z_i d\varphi, \quad z_i = s_i e^{i\varphi_i}.$$

In words, $H(t)$ is a Cauchy integral over $k-1$ circles, one for each state of F except state 1. The circle of state 1 has already been integrated out, and it leaves the factor $e^{t_1 g(z')}$.

Proof. By the first fact, $\varrho(\text{diag}(s')^{-1}A') < 1$. So the series $X(z') = \sum_{j \geq 0} (\mathcal{Z}'^{-1}A')^j \mathcal{Z}'^{-1}$ converges absolutely for $|z_i| \geq s_i$, since its terms are bounded entrywise in modulus by those at $z' = s'$. As a_1, b_1 and μ are nonnegative, this gives the claims about X, g and b .

Choose $s_1 > g(s')$ and put $S = \text{diag}(s_1, s')$. We show that $\varrho(S^{-1}A_F) < 1$. For $\varepsilon > 0$ let $v = (1, v')$ with $v' = X(s')(b_1 + \varepsilon \mathbf{1}) > 0$. Then $((S - A_F)v)_i = \varepsilon$ for $i \geq 2$, and $((S - A_F)v)_1 = s_1 - g(s') - \varepsilon a_1^\top X(s') \mathbf{1}$. For small ε both are positive, and the first fact gives the claim.

Now let $\mathcal{Z} = \text{diag}(z)$ with $|z_i| = s_i$ for every $i \in F$. Since $\varrho(A_F S^{-1}) < 1$, expanding $(\mathcal{Z} - A_F)^{-1} = \mathcal{Z}^{-1}(I - A_F \mathcal{Z}^{-1})^{-1}$ gives

$$\mu_F^\top (\mathcal{Z} - A_F)^{-1} \mathbf{1} = \sum_c \mu_{c_1} \prod_{j < M} q_{c_j c_{j+1}} \prod_{i \in F} z_i^{-m_i(c)},$$

absolutely and uniformly on the torus. Here $c = c_1 \cdots c_M$ runs over the words in F with $M \geq 1$ and $c_j \neq c_{j+1}$, and $m_i(c)$ is the number of letters i in c . For an integer m and $t_i > 0$, Cauchy's formula gives $\frac{1}{2\pi} \int_0^{2\pi} e^{t_i z_i} z_i^{1-m} d\varphi_i = t_i^{m-1}/(m-1)!$ if $m \geq 1$ and 0 if $m \leq 0$. So integrating term by term removes the words that miss a state of F , and grouping the others by $m(c)$ gives

$$H(t) = \frac{1}{(2\pi)^k} \int_{\mathbb{T}^k} e^{\langle t, z \rangle} \mu_F^\top (\mathcal{Z} - A_F)^{-1} \mathbf{1} \prod_{i \in F} z_i d\varphi. \quad (16)$$

The Schur complement $z_1 - g(z')$ of $\mathcal{Z}' - A'$ in $\mathcal{Z} - A_F$ is not 0, since $|g(z')| \leq g(s') < s_1$. Block inversion gives

$$\mu_F^\top (\mathcal{Z} - A_F)^{-1} \mathbf{1} = \frac{b(z')}{z_1 - g(z')} + \mu'^\top X(z') \mathbf{1}.$$

We integrate over φ_1 in (16) and divide by 2π . The second term does not depend on z_1 , so it contributes $\frac{1}{2\pi} \int_0^{2\pi} e^{t_1 z_1} z_1 d\varphi_1 = 0$. In the first term $g(z')$ lies inside the circle $|z_1| = s_1$, so Cauchy's formula gives $e^{t_1 g(z')} b(z')$. The factors without z_1 are unchanged, and this proves the formula. $\square$

All coefficients are nonnegative, so the integrand has its largest modulus at $\varphi = 0$. This gives Lemma 3.2, and a Laplace expansion around $\varphi = 0$ gives Theorem 4.4.

A.3 Proof of Lemma 3.2

Let F be irreducible and $\theta \in V_F$. Then $A_F - \text{diag } \theta$ is an irreducible Metzler matrix with Perron root 0. So it has a right eigenvector $u > 0$ for 0, and $\theta_i = (A_F u)_i / u_i$. Each row of A_F has a positive entry, since G_F is strongly connected and $k \geq 2$, so $\theta > 0$. By the second fact, $\varrho(A' - \text{diag } \theta') < 0$. The rows $i \geq 2$ of $(A_F - \text{diag } \theta)u = 0$ give $u' = u_1 X(\theta') b_1$, and then row 1 gives $\theta_1 = g(\theta')$. Hence

$$\theta > 0, \quad \varrho(A' - \text{diag } \theta') < 0, \quad g(\theta') = \theta_1 \quad \text{for every } \theta \in V_F. \quad (17)$$

Now fix n with support F and h with $hq_{\max} \leq \frac{1}{2}$, and put $|n| = \sum_i n_i$, $t = hn$ and $\sigma = e^{hq_{\max}}$. Lemma 2.3 with $|n|$ in place of N gives the probability $P_{|n|}(n)$ in the lemma. Only terms with $m_i \leq n_i$ for every i are nonzero, and in them $h^{m_i-1} \binom{n_i-1}{m_i-1} \leq t_i^{m_i-1}/(m_i-1)!$ and $(1 - hq_i)^{n_i-m_i} \leq e^{-q_i t_i} \sigma^{m_i}$. Note that $h^{M-1} = h^{k-1} \prod_i h^{m_i-1}$ and $\sigma^M W(m) = \sigma W_\sigma(m)$, where W_σ is W for σA_F in place of A_F . So

$$P_{|n|}(n) \leq \sigma h^{k-1} e^{-\langle q, t \rangle} H_\sigma(t), \quad (18)$$

where H_σ is H for σA_F . We bound H_σ by Lemma A.2 for σA_F with the radii $\sigma \theta'$, which is allowed since $\varrho(\sigma A' - \text{diag } \sigma \theta') = \sigma \varrho(A' - \text{diag } \theta') < 0$. The functions of Lemma A.2 for σA_F satisfy $X_\sigma(\sigma z') = \sigma^{-1} X(z')$, so $g_\sigma(\sigma z') = \sigma g(z')$ and $b_\sigma(\sigma z') = b(z')$. By (17) the integrand has modulus at most $e^{\sigma \langle t, \theta \rangle} b(\theta') \sigma^{k-1} \prod_{i \geq 2} \theta_i$, so for every $\sigma > 0$

$$H_\sigma(t) \leq \sigma^{k-1} b(\theta') \prod_{i \geq 2} \theta_i e^{\sigma \langle t, \theta \rangle}. \quad (19)$$

With (18) this gives $P_{|n|}(n) \leq \sigma^k b(\theta') \prod_{i \geq 2} \theta_i h^{k-1} \exp(-h\langle n, q - \theta \rangle + (\sigma - 1)h\langle n, \theta \rangle)$. Finally $\sigma \leq e^{1/2}$ and $\sigma - 1 \leq 2hq_{\max}$, so the lemma holds with $C_\theta = \max(e^{k/2}b(\theta') \prod_{i \geq 2} \theta_i, 2q_{\max} \max_i \theta_i)$. The bound after the lemma is the case $\theta = q - \lambda_F \mathbf{1}$. $\square$

A.4 Proof of Theorem 4.4

For $k = 1$ the formula is part of Lemma 2.4. Let $k \geq 2$ and $\alpha \in \mathcal{K}$, and write $\theta = \theta(\alpha)$, $r = r_\alpha$, $l = l_\alpha$ and $\Theta' = \text{diag}(\theta')$. By (17) we may use Lemma A.2 with the radii θ' and $t = T\alpha$. With Lemma 2.4 this gives

$$f_T(\alpha) = \frac{T^{k-1} e^{-T\langle q, \alpha \rangle}}{(2\pi)^{k-1}} \int_{\mathbb{T}^{k-1}} e^{T\Psi(\alpha, \varphi)} \chi(\alpha, \varphi) d\varphi, \quad (20)$$

where $z_i = \theta_i e^{i\varphi_i}$, $\Psi(\alpha, \varphi) = \alpha_1 g(z') + \sum_{i \geq 2} \alpha_i z_i$ and $\chi(\alpha, \varphi) = b(z') \prod_{i \geq 2} z_i$. We check the hypotheses of Lemma B.1 with $m = k - 1$.

Hypothesis (iii). By (17), $\Psi(\alpha, 0) = \langle \alpha, \theta \rangle$ is real. By Lemma A.2, $|g(z')| \leq g(\theta') = \theta_1$ on the torus, so

$$\text{Re } \Psi(\alpha, \varphi) \leq \Psi(\alpha, 0) - \sum_{i \geq 2} \alpha_i \theta_i (1 - \cos \varphi_i).$$

All α_i and θ_i are positive, so the sum is positive unless $\varphi = 0$.

Hypotheses (i) and (ii). For $z \in \mathbb{R}^F$ near θ let $\lambda(z) = \varrho(A_F - \text{diag } z)$. With $R = \text{diag } r$ we have $A_F - \text{diag}(\theta - \eta) = R(\hat{Q}_\alpha + \text{diag } \eta)R^{-1}$, and $\hat{Q}_\alpha$ is an irreducible generator with stationary law α . So Lemma A.1 gives $\nabla \lambda(\theta) = -\alpha$ and $\nabla^2 \lambda(\theta) = \Gamma_\alpha$. For real z' near θ' we have $\varrho(A' - \text{diag } z') < 0$, so $(1, X(z')b_1) \geq 0$ by the first fact. By the definition of g this vector lies in the kernel of $A_F - \text{diag}(g(z'), z')$, so $\lambda(g(z'), z') = 0$ by the second fact. Write g_j, g_{ij} for the derivatives of g at θ' and put $w_j = e_j - (\alpha_j/\alpha_1)e_1 \in \mathbb{R}^F$ for $j \geq 2$. Differentiating $\lambda(g(z'), z') = 0$ once and twice at θ' gives

$$g_j = -\frac{\alpha_j}{\alpha_1}, \quad \alpha_1 g_{ij} = w_i^\top \Gamma_\alpha w_j.$$

So $\partial \Psi / \partial z_j = \alpha_1 g_j + \alpha_j = 0$ at θ' , and the Hessian $\mathcal{H}$ of Ψ in z' at θ' is $V^\top \Gamma_\alpha V$, where V has columns $w_2, \dots, w_k$. Since $\partial / \partial \varphi_j = iz_j \partial / \partial z_j$ and the first derivatives vanish, $\nabla_\varphi \Psi(\alpha, 0) = 0$ and $-\nabla_\varphi^2 \Psi(\alpha, 0) = \Theta' \mathcal{H} \Theta'$. Let $B = I + \mathbf{1} \alpha'^\top / \alpha_1$. Then $Vy + (\alpha'^\top y / \alpha_1) \mathbf{1} = (0, By)$ for $y \in \mathbb{R}^{k-1}$, and $\Gamma_\alpha \mathbf{1} = 0$, so $\mathcal{H} = B^\top \Sigma_\alpha B$. By Lemma A.1, Σ_α is positive definite, and hence so is $\mathcal{H}$. The matrix determinant lemma gives $\det B = 1/\alpha_1$, so $\det \mathcal{H} = \det \Sigma_\alpha / \alpha_1^2$.

Regularity. By Proposition 4.2(a), $\theta(\alpha)$ is real-analytic. Fix $\alpha_0 \in \mathcal{K}$. By (17) and continuity there is $\delta \in (0, \min_i \theta_i(\alpha_0))$ with $\varrho(A' - \text{diag}(\theta'(\alpha_0) - \delta \mathbf{1})) < 0$. By Lemma A.2, X, g and b are holomorphic where $|z_i| > \theta_i(\alpha_0) - \delta$ for every $i \geq 2$, and this holds on the torus of radii $\theta'(\alpha)$ for α near α_0 . So Ψ and χ are real-analytic near $\{\alpha_0\} \times \mathbb{T}^{k-1}$. Covering $\mathcal{K}$ by finitely many such sets shows that their φ -derivatives are jointly continuous on $\mathcal{K} \times \mathbb{T}^{k-1}$.

The constant. Lemma B.1 now gives, uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,

$$\int_{\mathbb{T}^{k-1}} e^{T\Psi} \chi d\varphi = e^{T\langle \alpha, \theta \rangle} \left(\frac{2\pi}{T} \right)^{\frac{k-1}{2}} \frac{\alpha_1}{\prod_{i \geq 2} \theta_i \sqrt{\det \Sigma_\alpha}} \left(\chi(\alpha, 0) + O\left(\frac{1}{T} \right) \right).$$

Here $\chi(\alpha, 0) = b(\theta') \prod_{i \geq 2} \theta_i$. As in the derivation of (17), $r' = r_1 X(\theta')b_1$. In the same way the columns $j \geq 2$ of $l'^\top (A_F - \text{diag } \theta) = 0$, which is (3), give $l' = l_1 X(\theta')^\top a_1$. Hence $\mu \cdot r = r_1(\mu_1 + \mu'^\top X(\theta')b_1)$ and $l \cdot \mathbf{1} = l_1(1 + a_1^\top X(\theta')\mathbf{1})$. Since $l_1 r_1 = \alpha_1$,

$$b(\theta') = \frac{(\mu \cdot r_\alpha)(l_\alpha \cdot \mathbf{1})}{\alpha_1}.$$

As $r_\alpha, l_\alpha > 0$ and $\mu(F) > 0$ by (H), $\chi(\alpha, 0)$ is positive and continuous, hence bounded below on $\mathcal{K}$. So the last bracket is $\chi(\alpha, 0)(1 + O(1/T))$. Putting this into (20) and using $I_F(\alpha) = \langle q - \theta, \alpha \rangle$ gives $f_T(\alpha) = T^{(k-1)/2} C_F(\alpha) e^{-T I_F(\alpha)} (1 + O(1/T))$, uniformly on $\mathcal{K}$. $\square$

A.5 Proof of Lemma 4.5

Under the continuum weights a path has of order T runs, and each run after the first in state i changes the number of placements by a factor $1 - O(m_i/n_i)$, which suggests a relative error of order T^2/N . Let $t = hn$ and $\alpha = n/N \in \mathcal{K}$. By Lemma 2.4, $N^{-(k-1)} f_T(n/N) = h^{k-1} e^{-\langle q, t \rangle} H(t) = \sum_m v_m$ with

$$v_m = W(m) h^{k-1} \prod_{i \in F} e^{-q_i t_i} \frac{t_i^{m_i-1}}{(m_i-1)!}, \quad m \in \mathbb{Z}_{\geq 1}^F.$$

Since $h^{m_i-1} \binom{n_i-1}{m_i-1} = \frac{t_i^{m_i-1}}{(m_i-1)!} \prod_{1 \leq j < m_i} (1 - j/n_i)$, the m -th term of Lemma 2.3 is $\kappa_m v_m$ with

$$\kappa_m = \prod_{i \in F} \left[\prod_{1 \leq j < m_i} \left(1 - \frac{j}{n_i} \right) \right] (1 - h q_i)^{n_i - m_i} e^{h q_i n_i}.$$

Let $n_{\min} = \min_i n_i$. If all $m_i \leq n_i$, then $(1 - h q_i)^{n_i - m_i} \leq e^{-h q_i (n_i - m_i)}$ gives $\kappa_m \leq e^{h q_{\max} M}$. Also $\log(1 - x) \geq -x - x^2$ for $0 \leq x \leq \frac{1}{2}$ gives $(1 - h q_i)^{n_i - m_i} e^{h q_i n_i} \geq e^{-n_i h^2 q_i^2}$, and the product of these over i is at least $e^{-q_{\max}^2 T^2/N}$. The factors $1 - j/n_i$ lie in $[0, 1]$, so their product is at least $1 - \sum_i m_i^2/(2n_i) \geq 1 - M^2/(2n_{\min})$. If some $m_i > n_i$, then $\kappa_m = 0$, while $M \geq n_{\min} + 1$ gives $M^2/(2n_{\min}) \geq 2$. In all cases

$$\left(1 - \frac{M^2}{2n_{\min}} \right) e^{-q_{\max}^2 T^2/N} \leq \kappa_m \leq e^{h q_{\max} M}. \quad (21)$$

Let $w_m = v_m / \sum_{m'} v_{m'}$, a probability since $H(t) > 0$ under (H). By Lemma 2.3 the ratio in the lemma is $\mathbb{E}_w \kappa_m$, so by (21) we need $\mathbb{E}_w e^{h q_{\max} M}$ and $\mathbb{E}_w M^2$. Since W for σA_F is $\sigma^{M-1} W(m)$, the function $\psi(c) = \log \mathbb{E}_w e^{c(M-1)} = \log H_{e^c}(t) - \log H(t)$ is convex with $\psi(0) = 0$. So $\psi(c) \leq c \psi(1)$ for $0 \leq c \leq 1$, and $\psi(1) \geq 1$ since $M \geq 2$. We show that $\psi(1) \leq C_{\mathcal{K}} T$. By (19) with $\sigma = e$ and $\theta = \theta(\alpha)$, $H_e(T\alpha) \leq e^{k-1} b(\theta') \prod_{i \geq 2} \theta_i e^{T \langle \alpha, \theta \rangle}$. For the lower bound note that $H(T\alpha) e^{-T \langle \alpha, \theta \rangle} = T^{-(k-1)} e^{T I_F(\alpha)} f_T(\alpha)$. By Theorem 4.4 this is at least $\frac{1}{2} T^{-(k-1)/2} C_F(\alpha)$ for $T \geq T_1$, and for $1 \leq T \leq T_1$ it is continuous and positive on $[1, T_1] \times \mathcal{K}$. So in both cases $H(T\alpha) \geq c_{\mathcal{K}} T^{-(k-1)/2} e^{T \langle \alpha, \theta \rangle}$. Since $\theta(\alpha)$ and $b(\theta'(\alpha))$ are bounded on $\mathcal{K}$, this gives $\psi(1) \leq C_{\mathcal{K}} T$.

Now let $x = h q_{\max} \leq \frac{1}{2}$. Then $\mathbb{E}_w e^{xM} = e^{x + \psi(x)} \leq e^{x(1 + C_{\mathcal{K}} T)} = 1 + O(T^2/N)$, because $T^2/N \leq 1$. The Chernoff bound $\mathbb{P}_w(M - 1 \geq y) \leq e^{\psi(1) - y}$ gives

$$\mathbb{E}_w (M - 1)^2 = \int_0^\infty 2y \mathbb{P}_w(M - 1 \geq y) dy \leq \psi(1)^2 + \int_{\psi(1)}^\infty 2y e^{\psi(1) - y} dy = \psi(1)^2 + 2\psi(1) + 2,$$

so $\mathbb{E}_w M^2 = O(T^2)$. Finally $n_{\min} \geq \varepsilon_{\mathcal{K}} N$ with $\varepsilon_{\mathcal{K}} = \min_{\alpha \in \mathcal{K}} \min_i \alpha_i > 0$. So by (21), $\mathbb{E}_w \kappa_m$ lies between $e^{-q_{\max}^2 T^2/N} (1 - \mathbb{E}_w M^2/(2\varepsilon_{\mathcal{K}} N))$ and $\mathbb{E}_w e^{h q_{\max} M}$, and both are $1 + O(T^2/N)$, uniformly for $\alpha \in \mathcal{K}$ and $1 \leq T \leq N^{1/2}$. $\square$

A.6 Proof of Theorem 3.3(ii) for reducible faces

Irreducible admissible faces satisfy (H) by Lemma 2.2(b), and Theorem 4.6 covers them. So let F be admissible and reducible, with components $C_1, \dots, C_r$. Let a be the least positive rate q_{ij} , and

take N_0 so large that $h \leq N^{-2/3}$ is as small as we need. Note that $T^2/N \leq 1$. As in the proof of Lemma 2.2, a path of positive probability with support F visits each component in one block of consecutive times, and it passes from one block to the next by one switch. We call the order of the blocks, the first state and these switches the pattern of the path. There are finitely many patterns.

Upper bound. Fix a pattern whose j -th block lies in the component D_j , starts at y_j and ends at x_j , and let $\text{supp } n = F$. Block j has length $N_j = \sum_{i \in D_j} n_i$. The paths with this pattern and composition n have total probability $\mu_{y_1} \prod_{j < r} h q_{x_j y_{j+1}}$ times the product over j of the weight of the block paths from y_j to x_j inside D_j with the right composition. This is at most the probability that $X_1, \dots, X_{N_j}$ have that composition for the chain started at y_j . For $C = D_j$ with $|C| \geq 2$, Lemma 3.2 with start y_j and $\theta = q - \lambda_C \mathbf{1}$ bounds it by $\kappa_1 h^{|C|-1} e^{-h\lambda_C N_j + \kappa_1 h^2 N_j}$, where κ_1 is the largest of these finitely many constants. For $C = \{i\}$ it is $(1 - h q_i)^{N_j-1} \leq e^{1/2 - h\lambda_C N_j}$. Since $\sum_j (|D_j| - 1) + r - 1 = k_F - 1$, $\sum_j N_j = N$ and $\lambda_C \geq \lambda_F$ for every component, summing over the patterns gives $P_N(n) \leq \kappa_2 (T/N)^{k_F-1} e^{-T\lambda_F + \kappa_1}$.

Lower bound. By Lemma 2.2(a) we can number the components so that there are $y_1 \in C_1$ with $\mu_{y_1} > 0$ and $x_j \in C_j$, $y_{j+1} \in C_{j+1}$ with $q_{x_j y_{j+1}} > 0$ for $j < r$. Take any $x_r \in C_r$, choose j^* with $\lambda_{C_{j^*}} = \lambda_F$ and $|C_{j^*}| = k^*$, and put $m = \lceil N/T \rceil$, so that $1 \leq hm \leq 2$.

Claim A. There is $c_A > 0$ such that for every component C and all $y, x \in C$, the chain started at y stays in C for its first m visits, visits every state of C and has $X_m = x$ with probability at least c_A . Take a walk $y = v_0, \dots, v_J = x$ through every state of C with $q_{v_s v_{s+1}} > 0$ and $J \leq |C|^2$. The paths that make exactly these switches, at any J of the $m-1$ transition times, have total probability at least $\binom{m-1}{J} (ha)^J (1 - h q_{\max})^m \geq (1 - Jh)^J a^J e^{-4q_{\max}/J!}$, since $1 \leq hm \leq 2$. Also $(1 - Jh)^J \geq 2^{-J}$ for small h .

Claim B. There is $c_B > 0$ such that for all $y, x \in C_{j^*}$ and $\ell \geq 2m$, the same event with C_{j^*} and ℓ visits has probability at least $c_B e^{-h\lambda_F \ell - h^2 \lambda_F^2 \ell}$. Let $w > 0$ satisfy $Q_{C_{j^*}} w = -\lambda_F w$ and let $B = I + h Q_{C_{j^*}} \geq 0$. Then $(B^s)_{zz'}$ is the probability of going from z to z' in s steps inside C_{j^*} , and $\sum_{z'} (B^s)_{zz'} \geq (B^s w)_z / \max w \geq (1 - h\lambda_F)^s \min w / \max w$. We use Claim A for the first m visits, from y back to y , then $s = \ell - 2m + 1$ steps inside C_{j^*} , then Claim A for the last m visits, ending at x . This gives at least $c_A^2 (\min w / \max w) (1 - h\lambda_F)^\ell$. Also $\lambda_F \leq q_{\max}$, because the Perron root of $Q_{C_{j^*}}$ is at least each diagonal entry. So $h\lambda_F \leq \frac{1}{2}$ and $\log(1 - h\lambda_F) \geq -h\lambda_F - h^2 \lambda_F^2$.

Give block $j \neq j^*$ the length m and block j^* the length $N - (r-1)m$, which is at least $2m$ for $T \geq T_0 = r + 2$ and $N \geq N_0$. The paths that go from y_j to x_j in block j as in the claims and pass from block j to block $j+1$ by the switch $x_j \rightarrow y_{j+1}$ all have support F . By the Markov property their total probability is at least $\mu_{y_1} (ha)^{r-1} c_A^{r-1} c_B e^{-T\lambda_F - q_{\max}^2}$. Their compositions are fixed by the restrictions to the blocks, so there are at most $(N+1)^{k^*-1} \prod_{j \neq j^*} (m+1)^{|C_j|-1}$ of them, and $m+1 \leq 3N/T$. Since $\sum_{j \neq j^*} (|C_j| - 1) = k_F - k^* - (r-1)$, dividing the probability by this number gives $M_F \geq c N^{-(k_F-1)} T^{k_F-k^*} e^{-T\lambda_F}$. $\square$

B A Laplace lemma with a parameter, and the next term

This appendix proves the Laplace lemma that Appendix A uses for Theorem 4.4. It then carries the lemma one step further (Lemma B.2) and proves Proposition 4.14. On $\mathbb{T}^m = (\mathbb{R}/2\pi\mathbb{Z})^m$ we use $\|\varphi\| = \max_i \text{dist}(\varphi_i, 2\pi\mathbb{Z})$. For $x \in \mathbb{R}^m$ we write $\|x\| = \max_i |x_i|$ and $|x|$ for the Euclidean norm. Repeated indices are summed from 1 to m .

Lemma B.1 (Laplace integrals with a parameter). *Let $\mathcal{K}$ be a compact metric space and $m \geq 1$. Let $\Psi, a : \mathcal{K} \times \mathbb{T}^m \rightarrow \mathbb{C}$ be such that the φ -derivatives of Ψ up to order 4 and of a up to order 2 exist and are jointly continuous on $\mathcal{K} \times \mathbb{T}^m$. Assume that for every $\alpha \in \mathcal{K}$*

- (i) $\Psi(\alpha, 0) \in \mathbb{R}$ and $\nabla_\varphi \Psi(\alpha, 0) = 0$,
- (ii) $\mathcal{J}(\alpha) = -\nabla_\varphi^2 \Psi(\alpha, 0)$ is real, symmetric and positive definite, and
- (iii) $\operatorname{Re} \Psi(\alpha, \varphi) < \Psi(\alpha, 0)$ for every $\varphi \neq 0$ in $\mathbb{T}^m$.

Then, uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,

$$\int_{\mathbb{T}^m} e^{T\Psi(\alpha, \varphi)} a(\alpha, \varphi) d\varphi = e^{T\Psi(\alpha, 0)} \left(\frac{2\pi}{T} \right)^{m/2} \det \mathcal{J}(\alpha)^{-1/2} \left(a(\alpha, 0) + O\left(\frac{1}{T}\right) \right).$$

In words, the integral is the value of the integrand at $\varphi = 0$ times the volume of a Gaussian peak, which is $(2\pi/T)^{m/2} \det \mathcal{J}^{-1/2}$. The error is uniform in the parameter α .

Idea of the proof. By (iii) and compactness the part of the torus away from $\varphi = 0$ is exponentially small. Near 0 we substitute $\varphi = x/\sqrt{T}$ and expand. The integrand becomes a Gaussian weight times $a(\alpha, 0)$, plus a term of order $T^{-1/2}$ that is odd in x and integrates to 0, plus a remainder of order $1/T$. All constants depend on α only through bounds that hold on the whole of $\mathcal{K}$.

Proof. By compactness $B_\Psi = \max_{|\beta| \leq 4} \sup |\partial_\varphi^\beta \Psi|$ and $B_a = \max_{|\beta| \leq 2} \sup |\partial_\varphi^\beta a|$ are finite, the least eigenvalue of $\mathcal{J}(\alpha)$ has a positive minimum $h_{\min}$ over $\mathcal{K}$, and $\det \mathcal{J}(\alpha) \leq (mB_\Psi)^m$. Below, C depends only on m , $h_{\min}$, B_Ψ , B_a and the numbers ε and η chosen below, and it may change from line to line. So every bound below is uniform on $\mathcal{K}$.

Localization. Choose $\varepsilon \in (0, 1)$ with $m^3 B_\Psi \varepsilon \leq \frac{3}{2} h_{\min}$. The set of (α, φ) with $\|\varphi\| \geq \varepsilon$ is compact, and by (iii) the continuous function $\operatorname{Re} \Psi(\alpha, \varphi) - \Psi(\alpha, 0)$ is negative on it. So it is at most $-\eta$ there for some $\eta > 0$, and the integral over this set has modulus at most $(2\pi)^m B_a e^{T(\Psi(\alpha, 0) - \eta)} \leq CT^{-1-m/2} e^{T\Psi(\alpha, 0)}$.

Scaling. Reduction modulo 2π identifies the cube $\|x\| \leq \varepsilon$ in $\mathbb{R}^m$ with the set $\|\varphi\| \leq \varepsilon$, and it preserves measure. Let $R(\alpha, \varphi) = \Psi(\alpha, \varphi) - \Psi(\alpha, 0) + \frac{1}{2} \varphi^\top \mathcal{J}(\alpha) \varphi$. By (i) and (ii), R and its derivatives of order 1 and 2 vanish at $\varphi = 0$. Taylor's theorem with integral remainder, which holds for complex values, gives on the cube $R = \frac{1}{6} \Psi_{ijk} \varphi^i \varphi^j \varphi^k + \omega_0$ with $|\omega_0| \leq C \|\varphi\|^4$, where Ψ_{ijk} are the third derivatives at $(\alpha, 0)$. It also gives $|R| \leq \frac{m^3}{6} B_\Psi \|\varphi\|^3 \leq \frac{1}{4} h_{\min} |\varphi|^2$, since $\sum_{|\beta|=3} 1/\beta! = m^3/6$ and $\|\varphi\|^3 \leq \varepsilon |\varphi|^2$. We substitute $\varphi = x/\sqrt{T}$ and put $w(x) = TR(\alpha, x/\sqrt{T})$. The integral over the cube is

$$T^{-m/2} e^{T\Psi(\alpha, 0)} \int_{\|x\| \leq \varepsilon \sqrt{T}} e^{-\frac{1}{2} x^\top \mathcal{J} x} e^{w(x)} a(\alpha, x/\sqrt{T}) dx,$$

and $|w(x)| \leq \frac{1}{4} h_{\min} |x|^2$.

Taylor expansion. Let $P(x) = \frac{1}{6} \Psi_{ijk} x^i x^j x^k$, $a_0 = a(\alpha, 0)$ and $a_i = \partial_{\varphi_i} a(\alpha, 0)$. Then $w = T^{-1/2} P + \omega_1$ with $|\omega_1| \leq CT^{-1} |x|^4$, and $|w| \leq CT^{-1/2} |x|^3$. Since $|e^w - 1 - w| \leq \frac{1}{2} |w|^2 e^{|w|}$, this gives $|e^w - 1 - T^{-1/2} P| \leq CT^{-1} (|x|^4 + |x|^6) e^{h_{\min} |x|^2/4}$. Taylor's theorem for a to orders 1 and 2 gives $|a(\alpha, x/\sqrt{T}) - a_0| \leq CT^{-1/2} |x|$ and $|a(\alpha, x/\sqrt{T}) - a_0 - T^{-1/2} a_i x^i| \leq CT^{-1} |x|^2$. Multiplying out,

$$e^w a(\alpha, x/\sqrt{T}) = a_0 + T^{-1/2} (a_0 P(x) + a_i x^i) + \omega(x), \quad |\omega(x)| \leq CT^{-1} (1 + |x|)^6 e^{h_{\min} |x|^2/4}.$$

Integration. Since $x^\top \mathcal{J} x \geq h_{\min} |x|^2$, the term ω contributes at most CT^{-1} to the integral, because $(1 + |x|)^6 e^{-h_{\min} |x|^2/4}$ is integrable. Outside the cube $|x| > \varepsilon \sqrt{T}$, so extending the integrals of the other two terms to $\mathbb{R}^m$ changes them by at most $C e^{-h_{\min} \varepsilon^2 T/4} \leq CT^{-1}$. On $\mathbb{R}^m$ we have $\int e^{-\frac{1}{2} x^\top \mathcal{J} x} dx = (2\pi)^{m/2} \det \mathcal{J}^{-1/2}$, and the odd polynomial $a_0 P(x) + a_i x^i$ integrates to 0 against this weight. This is why the error is $O(1/T)$ and not $O(T^{-1/2})$. So the integral over the cube is $e^{T\Psi(\alpha, 0)} T^{-m/2} [(2\pi)^{m/2} \det \mathcal{J}^{-1/2} a_0 + O(T^{-1})]$. Since $\det \mathcal{J} \leq (mB_\Psi)^m$, this error and the one from the localization are $O(1/T)$ relative to $e^{T\Psi(\alpha, 0)} (2\pi/T)^{m/2} \det \mathcal{J}^{-1/2}$. $\square$

B.1 The next term of the Laplace lemma

With two more derivatives the same proof gives the term of order $1/T$ and an error $O(T^{-2})$.

Lemma B.2 (the next term). *Let $\mathcal{K}$, m , Ψ and a satisfy the hypotheses of Lemma B.1. Assume also that the φ -derivatives of Ψ up to order 6 and of a up to order 4 exist and are jointly continuous on $\mathcal{K} \times \mathbb{T}^m$. Write $a_0 = a(\alpha, 0)$, write $\mathcal{J}^{ij}$ for the entries of $\mathcal{J}(\alpha)^{-1}$, and write Ψ_{ijk} , Ψ_{ijkl} , a_i and a_{ij} for the φ -derivatives at $(\alpha, 0)$. Put*

$$c_1(\alpha) = \frac{1}{2}a_{ij}\mathcal{J}^{ij} + \frac{1}{2}a_i\Psi_{jkl}\mathcal{J}^{ij}\mathcal{J}^{kl} \\ + a_0\left[\frac{1}{8}\Psi_{ijkl}\mathcal{J}^{ij}\mathcal{J}^{kl} + \frac{1}{12}\Psi_{ijk}\Psi_{lrs}\mathcal{J}^{il}\mathcal{J}^{jr}\mathcal{J}^{ks} + \frac{1}{8}\Psi_{ijk}\Psi_{lrs}\mathcal{J}^{ij}\mathcal{J}^{kl}\mathcal{J}^{rs}\right]. \quad (22)$$

Then c_1 is continuous on $\mathcal{K}$, and uniformly for $\alpha \in \mathcal{K}$ and $T \geq 1$,

$$\int_{\mathbb{T}^m} e^{T\Psi(\alpha, \varphi)} a(\alpha, \varphi) d\varphi = e^{T\Psi(\alpha, 0)} \left(\frac{2\pi}{T}\right)^{m/2} \det \mathcal{J}(\alpha)^{-1/2} \left(a_0 + \frac{c_1(\alpha)}{T} + O\left(\frac{1}{T^2}\right)\right).$$

In words, c_1 is the mean of an explicit polynomial of degree 6 under the Gaussian law with covariance $\mathcal{J}^{-1}$. A quick test of (22) is the integrand $e^{T(e^{i\varphi} - 1 - i\varphi)}$ with $m = 1$, $a = 1$ and integer T . Its integral is $2\pi e^{-T} T^T / T!$, and Stirling's formula gives the factor $1 - \frac{1}{12T}$. Here $\mathcal{J} = 1$, $\Psi_{111} = -i$ and $\Psi_{1111} = 1$, so (22) gives $c_1 = \frac{1}{8} - \frac{1}{12} - \frac{1}{8} = -\frac{1}{12}$ as well.

Idea of the proof. After the substitution $\varphi = x/\sqrt{T}$ the integrand is a Gaussian weight times a power series in $T^{-1/2}$ whose coefficients are polynomials in x . The coefficients of the odd powers of $T^{-1/2}$ are odd polynomials and integrate to 0. The coefficient of T^{-1} has the Gaussian mean c_1 , by Wick's formula.

Proof. We follow the proof of Lemma B.1 with its notation, and expand further. Now B_Ψ and B_a are the maxima over $|\beta| \leq 6$ and $|\beta| \leq 4$. The localization and the scaling are as before. The integral over $\|\varphi\| \geq \varepsilon$ is at most $(2\pi)^m B_a e^{T(\Psi(\alpha, 0) - \eta)} \leq CT^{-2-m/2} e^{T\Psi(\alpha, 0)}$.

Taylor expansion. Put $s = T^{-1/2}$, so $0 < s \leq 1$. For $j = 3, 4, 5$ let $P_j(x)$ be the part of degree j of the Taylor polynomial of $\Psi(\alpha, \cdot)$ at 0. So $P_3 = P$ and $P_4(x) = \frac{1}{24}\Psi_{ijkl}x^i x^j x^k x^l$. For $j = 1, 2, 3$ let $A_j(x)$ be the part of degree j of the Taylor polynomial of $a(\alpha, \cdot)$, so $A_1(x) = a_i x^i$ and $A_2(x) = \frac{1}{2}a_{ij}x^i x^j$. Taylor's theorem to order 6 for Ψ and to order 4 for a gives, on the cube $\|x\| \leq \varepsilon\sqrt{T}$,

$$w = w_0 + \omega_1, \quad w_0 = sP_3 + s^2P_4 + s^3P_5, \quad a(\alpha, sx) = a_0 + sA_1 + s^2A_2 + s^3A_3 + \omega_2,$$

with $|\omega_1| \leq Cs^4|x|^6$ and $|\omega_2| \leq Cs^4|x|^4$. Let $\mathbf{p}(v) = 1 + v + \frac{1}{2}v^2 + \frac{1}{6}v^3$. We compare e^w with $\mathbf{p}(w)$, then with $\mathbf{p}(w_0)$, and then with

$$Y = 1 + sP_3 + s^2(P_4 + \frac{1}{2}P_3^2) + s^3(P_5 + P_3P_4 + \frac{1}{6}P_3^3),$$

which collects the terms of $\mathbf{p}(w_0)$ of degree at most 3 in s . First, $|e^v - \mathbf{p}(v)| \leq \frac{1}{24}|v|^4 e^{|v|}$, and we know that $|w| \leq Cs|x|^3$ and $|w| \leq \frac{1}{4}h_{\min}|x|^2$. So $|e^w - \mathbf{p}(w)| \leq Cs^4|x|^{12}e^{h_{\min}|x|^2/4}$. Second, $\mathbf{p}(w) - \mathbf{p}(w_0) = \omega_1[1 + \frac{1}{2}(w + w_0) + \frac{1}{6}(w^2 + ww_0 + w_0^2)]$, and $|w|$ and $|w_0|$ are at most $C(1 + |x|)^5$, by $|w| \leq Cs|x|^3$ and $s \leq 1$. So $|\mathbf{p}(w) - \mathbf{p}(w_0)| \leq Cs^4(1 + |x|)^{16}$. Third, $\mathbf{p}(w_0) - Y$ is a sum of terms s^j with $4 \leq j \leq 9$ times polynomials in x of degree at most 15 with bounded coefficients, so $|\mathbf{p}(w_0) - Y| \leq Cs^4(1 + |x|)^{15}$. Together,

$$|e^w - Y| \leq Cs^4(1 + |x|)^{16}e^{h_{\min}|x|^2/4}.$$

Now write $e^w a = Y(a_0 + sA_1 + s^2A_2 + s^3A_3) + Y\omega_2 + (e^w - Y)a$. Here $|Y| \leq C(1 + |x|)^9$ and $|a| \leq B_a$. In the first product the terms of degree at least 4 in s are at most $Cs^4(1 + |x|)^{12}$. Hence

$$e^w a(\alpha, sx) = a_0 + s \mathcal{Q}_1(x) + s^2 \mathcal{Q}_2(x) + s^3 \mathcal{Q}_3(x) + \omega(x), \quad |\omega(x)| \leq CT^{-2}(1 + |x|)^{16}e^{h_{\min}|x|^2/4},$$

where

$$\begin{aligned} \mathcal{Q}_1 &= a_0 P_3 + A_1, & \mathcal{Q}_2 &= a_0(P_4 + \tfrac{1}{2}P_3^2) + A_1 P_3 + A_2, \\ \mathcal{Q}_3 &= a_0(P_5 + P_3 P_4 + \tfrac{1}{6}P_3^3) + A_1(P_4 + \tfrac{1}{2}P_3^2) + A_2 P_3 + A_3. \end{aligned}$$

Every term of $\mathcal{Q}_1$ and of $\mathcal{Q}_3$ has odd degree, so $\mathcal{Q}_1$ and $\mathcal{Q}_3$ are odd functions.

Integration. As before, ω contributes at most CT^{-2} to the integral over the cube, since $(1 + |x|)^{16}e^{-h_{\min}|x|^2/4}$ is integrable. Extending the integrals of the four polynomial terms to $\mathbb{R}^m$ changes them by at most $Ce^{-h_{\min}\varepsilon^2 T/4} \leq CT^{-2}$. On $\mathbb{R}^m$ the odd terms $\mathcal{Q}_1$ and $\mathcal{Q}_3$ integrate to 0 against $e^{-\frac{1}{2}x^\top \mathcal{J} x}$. This is why the error is $O(T^{-2})$ and not $O(T^{-3/2})$. So the integral over the cube is

$$e^{T\Psi(\alpha,0)} T^{-m/2} \left[(2\pi)^{m/2} \det \mathcal{J}^{-1/2} \left(a_0 + \frac{\mathbb{E}\mathcal{Q}_2(x)}{T} \right) + O(T^{-2}) \right],$$

where x is a centered Gaussian vector with $\mathbb{E}[x^i x^j] = \mathcal{J}^{ij}$. As in the proof of Lemma B.1, the errors are $O(T^{-2})$ relative to the main factor, because $\det \mathcal{J} \leq (mB_\Psi)^m$.

It remains to show that $\mathbb{E}\mathcal{Q}_2 = c_1$. By Wick's formula a Gaussian moment of order 4 or 6 is the sum, over the three or fifteen ways to pair the indices, of the products of the covariances of the pairs. The derivatives are symmetric in their indices. So $\mathbb{E}[A_2] = \frac{1}{2}a_{ij}\mathcal{J}^{ij}$, $\mathbb{E}[A_1 P_3] = \frac{1}{6} \cdot 3 a_i \Psi_{jkl} \mathcal{J}^{ij} \mathcal{J}^{kl}$ and $\mathbb{E}[P_4] = \frac{1}{24} \cdot 3 \Psi_{ijkl} \mathcal{J}^{ij} \mathcal{J}^{kl}$. In $\mathbb{E}[P_3^2] = \frac{1}{36} \Psi_{ijk} \Psi_{lrs} \mathbb{E}[x^i x^j x^k x^l x^r x^s]$, six pairings join each index of the first factor to an index of the second, and each gives $\Psi_{ijk} \Psi_{lrs} \mathcal{J}^{il} \mathcal{J}^{jr} \mathcal{J}^{ks}$. The other nine pair two indices inside each factor and join the remaining two, and each gives $\Psi_{ijk} \Psi_{lrs} \mathcal{J}^{ij} \mathcal{J}^{kl} \mathcal{J}^{rs}$. So $\frac{1}{2}\mathbb{E}[P_3^2]$ gives the last two terms of (22). Finally c_1 is a polynomial in a_0 , in the derivatives at $\varphi = 0$ and in the entries of $\mathcal{J}^{-1}$, which are continuous on $\mathcal{K}$. $\square$

B.2 Proof of Proposition 4.14

Throughout, F satisfies (H). If $F = \{i\}$, then (2) gives $\log M_F = \log \mu_i - q_i T + O(T^2/N)$, which is (9) with $c_F = 0$. So let $k \geq 2$ until the proof of (iii).

Part (i). We use the proof of Theorem 4.4 in Appendix A. It shows that Ψ and χ in (20) satisfy the hypotheses of Lemma B.1 with $m = k - 1$, and that their φ -derivatives of every order are jointly continuous on $\mathcal{K} \times \mathbb{T}^{k-1}$. So Lemma B.2 applies with $a = \chi$. Let $c_1(\alpha)$ be the function (22) for this Ψ and $a = \chi$, with $\mathcal{J}(\alpha) = -\nabla_\varphi^2 \Psi(\alpha, 0)$, and put

$$c(\alpha) = \frac{c_1(\alpha)}{\chi(\alpha, 0)}, \quad \chi(\alpha, 0) = b(\theta'(\alpha)) \prod_{i \geq 2} \theta_i(\alpha) > 0. \quad (23)$$

Then the bracket $\chi(\alpha, 0) + O(1/T)$ in the last step of that proof becomes $\chi(\alpha, 0) + c_1(\alpha)/T + O(T^{-2})$. Since $\chi(\alpha, 0)$ is bounded below on $\mathcal{K}$, this is $\chi(\alpha, 0)(1 + c(\alpha)/T + O(T^{-2}))$, and the rest of that proof gives the expansion in (i). By the regularity step of that proof, Ψ and χ are real-analytic in (α, φ) near $\{\alpha_0\} \times \mathbb{T}^{k-1}$ for every $\alpha_0 \in \Delta_F^\circ$. So their φ -derivatives at $\varphi = 0$ are real-analytic in α , and so are the entries of $\mathcal{J}(\alpha)^{-1}$ and c . Finally (i) gives $c(\alpha) = \lim_{T \rightarrow \infty} T[f_T(\alpha)T^{-(k-1)/2}C_F(\alpha)^{-1}e^{T I_F(\alpha)} - 1]$. So c is real, and it does not depend on which state of F we number 1.

Part (ii), the constants. Write $\Sigma = \Sigma_F$, $\gamma = \gamma_F$, $c^* = c(\alpha_F^*)$, $X = N^{-(k-1)}T^{(k-1)/2}K_F e^{-T\lambda_F}$ and $x = \alpha' - \alpha_F^*$. By Propositions 3.1(a) and 4.2(a) and by (i), the functions I_F , $\log C_F$ and c are

real-analytic near α_F^* , and I_F has its minimum λ_F at α_F^* with Hessian Σ^{-1} . So there are a closed ball $U = \{|x| \leq \rho_0\} \subset \Delta_F^\circ$ and constants $c_U > 0$ and L_1 such that on U

$$\begin{aligned} \left| \log \frac{C_F(\alpha)}{K_F} - \gamma \cdot x \right| &\leq L_1 |x|^2, & \left| I_F(\alpha) - \lambda_F - \frac{1}{2} x^\top \Sigma^{-1} x \right| &\leq L_1 |x|^3, \\ |c(\alpha) - c^*| &\leq L_1 |x|, & |c(\alpha)| &\leq L_1, & I_F(\alpha) - \lambda_F &\geq c_U |x|^2. \end{aligned}$$

Put $L_2 = |\gamma| + L_1 \rho_0$, so that $|\log(C_F(\alpha)/K_F)| \leq L_2 |x|$ on U . Fix ρ_1 so large that

$$\rho_1 \geq \frac{L_2}{c_U}, \quad \rho_1 \geq |\Sigma \gamma| + k, \quad c_U \rho_1^2 - L_2 \rho_1 - L_1 \geq |c_F| + 1.$$

For $\alpha \in U$ let

$$G_T(\alpha) = \log \frac{C_F(\alpha)}{K_F} - T(I_F(\alpha) - \lambda_F) + \frac{c(\alpha)}{T}.$$

By (i) with $\mathcal{K} = U$ and Lemma 4.5, there are T_0 and N_0 such that for $N \geq N_0$, $T_0 \leq T \leq N^{1/3}$ and $\text{supp } n = F$ with $\alpha = n/N \in U$,

$$\log \frac{P_N(n)}{X} = G_T(\alpha) + O\left(\frac{1}{T^2} + \frac{T^2}{N}\right). \quad (24)$$

Here we used $\log(1 + c/T + O(T^{-2})) = c/T + O(T^{-2})$, which holds because $|c| \leq L_1$ on U . We take $T_0 \geq \rho_1/\rho_0$, and we increase T_0 and N_0 finitely many times below.

Part (ii), two bounds for G_T . First let $\alpha \in U$ with $|x| \geq \rho_1/T$. Then $G_T(\alpha) \leq L_2 |x| - c_U T |x|^2 + L_1/T$. The function $v \mapsto L_2 v - c_U T v^2$ decreases for $v \geq L_2/(2c_U T)$, and ρ_1/T lies in this range. So

$$G_T(\alpha) \leq \frac{L_2 \rho_1 - c_U \rho_1^2 + L_1}{T} \leq \frac{c_F - 1}{T} \quad (\alpha \in U, |x| \geq \rho_1/T). \quad (25)$$

Next let $|x| \leq \rho_1/T$, and put $y = x - \Sigma \gamma/T$. Then $\alpha \in U$, and the bounds on U give $G_T(\alpha) = \gamma \cdot x - \frac{T}{2} x^\top \Sigma^{-1} x + c^*/T + \epsilon$ with $|\epsilon| \leq L_1 |x|^2 + L_1 T |x|^3 + L_1 |x|/T \leq L_1(\rho_1^2 + \rho_1^3 + \rho_1)T^{-2}$. Completing the square gives $\gamma \cdot x - \frac{T}{2} x^\top \Sigma^{-1} x = \frac{1}{2T} \gamma^\top \Sigma \gamma - \frac{T}{2} y^\top \Sigma^{-1} y$. So

$$G_T(\alpha) = \frac{c_F}{T} - \frac{T}{2} y^\top \Sigma^{-1} y + O\left(\frac{1}{T^2}\right) \quad (|x| \leq \rho_1/T), \quad (26)$$

uniformly in x . Since Σ^{-1} is positive definite, (25) and (26) give $G_T \leq c_F/T + O(T^{-2})$ on U .

Part (ii), the upper bound. The part of the proof of Theorem 4.6 that treats compositions outside U uses only that U is a closed ball around α_F^* inside Δ_F° . So there is $\eta > 0$ with $P_N(n) \leq X \cdot O(T^{(k-1)/2} e^{-2\eta T})$ for $\text{supp } n = F$ and $n/N \notin U$. After increasing T_0 this is at most $X e^{-\eta T} \leq X e^{(c_F-1)/T}$. For $n/N \in U$ we use (24) and the bound on G_T . Hence $\log(M_F/X) \leq c_F/T + O(T^{-2} + T^2/N)$.

Part (ii), the lower bound. Let α^0 be the point with $x = \Sigma \gamma/T$. Let n_i^0 be an integer nearest to $N \alpha_i^0$ for $i \geq 2$, and $n_1^0 = N - \sum_{i \geq 2} n_i^0$. Then $|n_i^0 - N \alpha_i^0| \leq k$ for every i . Since $\alpha^0 \rightarrow \alpha_F^*$ as $T \rightarrow \infty$ and every $\alpha_{F,i}^*$ is positive, n^0 has support F after increasing T_0 and N_0 . For $\alpha = n^0/N$ we have $|y| \leq k/N$ and $|x| \leq |\Sigma \gamma|/T + k/N \leq \rho_1/T$, because $T \leq N$. So (26) gives $G_T(n^0/N) \geq c_F/T - O(T/N^2 + T^{-2})$, and (24) gives $\log(M_F/X) \geq \log(P_N(n^0)/X) \geq c_F/T - O(T^{-2} + T^2/N)$. Note that a lattice point near α^0 loses only $O(T/N^2)$, because G_T is flat to first order at its maximizer.

The two bounds give (9) for $N \geq N_0$ and $T_0 \leq T \leq N^{1/3}$. For $1 \leq T \leq T_0$ it follows from (5), since c_F/T and $1/T$ are $O(T^{-2})$ there, with a constant that depends on T_0 . This proves (ii).

Part (iii). Now F and G are as in Theorem 4.7, and $k_G = 1$ is allowed. Let $T_N \in [1, N^{1/3}]$ be a zero of D_N . The proof of Theorem 4.7(ii) shows that $\frac{\Delta}{2\delta}L < T_N \leq \frac{2\Delta}{\delta}L$ for large N . Subtracting (9) for G from (9) for F at $T = T_N$ gives

$$0 = -\Delta L + \frac{\Delta}{2} \log T_N + \delta T_N + \log \frac{K_F}{K_G} + \frac{c_F - c_G}{T_N} + O\left(\frac{1}{T_N^2} + \frac{T_N^2}{N}\right).$$

We multiply by T_N . The error becomes $O(1/T_N + T_N^3/N) = O(1/L)$, which is (10).

Part (iv), a symmetric pair. Let $F = \{i, j\}$ with $q_{ij} = q_{ji} = a > 0$, $q_i = q_j = q$ and $\mu_i = \mu_j > 0$. Then F satisfies (H). We first prove (11), without the Laplace lemma. Let $1 \leq T \leq N^{1/3}$ and let N be so large that $qh \leq \frac{1}{2}$ and $I + hQ$ is stochastic. By Proposition 4.9, $M_F = \mu_i(1 - qh)^{N-1}S_N(u)$ with $u = ah/(1 - qh)$. Put $z = Nu = aT/(1 - qh)$. Then $aT \leq z \leq 2aT$ and $0 \leq z - aT \leq 2aqT^2/N$. By (12),

$$\log S_N(u) = \log \frac{2}{N} + \log \psi(z) + O\left(\frac{T^2}{N}\right),$$

since $z(z + 2)/(2N) = O(T^2/N)$ and this is at most $\frac{1}{2}$ for large N . By [38, Eq. 10.40.1], as $z \rightarrow \infty$, $I_0(z) = e^z(2\pi z)^{-1/2}(1 + \frac{1}{8z} + O(z^{-2}))$ and $I_1(z) = e^z(2\pi z)^{-1/2}(1 - \frac{3}{8z} + O(z^{-2}))$. So $I_0(z) + I_1(z) = 2e^z(2\pi z)^{-1/2}(1 - \frac{1}{8z} + O(z^{-2}))$ and

$$\log \psi(z) = z + \frac{1}{2} \log z + \frac{1}{2} \log \frac{2}{\pi} - \frac{1}{8z} + \rho(z), \quad |\rho(z)| \leq C_a z^{-2} \quad \text{for } z \geq a.$$

The bound on ρ holds for all $z \geq a$, and not only for large z , because ρ is continuous on $(0, \infty)$. The function $z + \frac{1}{2} \log z - \frac{1}{8z}$ has a bounded derivative on $[a, \infty)$. So replacing z by aT in it changes it by $O(T^2/N)$, and

$$\log \psi(z) = aT + \frac{1}{2} \log T + \frac{1}{2} \log \frac{2a}{\pi} - \frac{1}{8aT} + O\left(\frac{1}{T^2} + \frac{T^2}{N}\right).$$

Also $(N - 1) \log(1 - qh) = -qT + O(T^2/N)$. Adding the three parts,

$$\log M_F = -L + \frac{1}{2} \log T + \log\left(2\mu_i \sqrt{\frac{2a}{\pi}}\right) - (q - a)T - \frac{1}{8aT} + O\left(\frac{1}{T^2} + \frac{T^2}{N}\right).$$

By Corollary 4.3 with $k = 2$, $\mu(F) = 2\mu_i$ and $\text{tree}(q_F) = a$, we have $\lambda_F = q - a$ and $K_F = 2\mu_i \cdot 2^{3/2} \sqrt{a} (4\pi)^{-1/2} = 2\mu_i \sqrt{2a/\pi}$. This is (11). Now we compare it with (9) at $T = L$. Both expand $\log M_F$, so their difference $(c_F + \frac{1}{8a})/L$ is $O(L^{-2} + L^2/N) = O(L^{-2})$. Letting $N \rightarrow \infty$ gives $c_F = -1/(8a)$.

As a check we compute the same value from (23). Number the states so that i is state 1. Then $A' = 0$, so in the notation of Appendix A we have $g(z_2) = a^2/z_2$ and $b(z_2) = \mu_i(1 + a/z_2)^2$. At $\alpha = \alpha_F^* = (\frac{1}{2}, \frac{1}{2})$ the radius is $\theta_2 = a$, so $\Psi = a \cos \varphi$ and $\chi = \mu_i a(1 + e^{-i\varphi})^2 e^{i\varphi} = 2\mu_i a(1 + \cos \varphi)$. Hence $\mathcal{J} = a$, $\Psi_{111} = 0$, $\Psi_{1111} = a$, $\chi(\alpha_F^*, 0) = 4\mu_i a$, $\chi_1 = 0$ and $\chi_{11} = -2\mu_i a$, and (22) gives $c_1 = -\mu_i + \frac{1}{2}\mu_i = -\frac{1}{2}\mu_i$ and $c(\alpha_F^*) = -1/(8a)$. Also $\gamma_F = 0$, because C_F is unchanged when the two states are swapped (Proposition 4.2(b) with $\nu = 1$). So the two values agree.

Part (iv), the complete graph. Let $Q = J - dI$, let F have $k \geq 2$ states and let $\mu_i = \mu_0 > 0$ for $i \in F$. Then F satisfies (H), and Corollary 4.3 with $\mu(F) = k\mu_0$ and $\text{tree}(q_F) = k^{k-2}$ gives $\lambda_F = d - k$, α_F^* uniform on F and $K_F = \mu_0 k^{k+1/2} (4\pi)^{-(k-1)/2}$.

We first show that $\gamma_F = 0$, so that $c_F = c(\alpha_F^*)$. By Proposition 4.2(b) with $\nu = \mathbf{1}$ we may take $r_\alpha = l_\alpha = \sqrt{\alpha}$, and then

$$C_F(\alpha) = \frac{\mu_0 \left(\sum_{i \in F} \sqrt{\alpha_i} \right)^2}{(4\pi)^{(k-1)/2}} \frac{\sqrt{\mathcal{T}(\alpha)}}{\prod_{i \in F} \alpha_i},$$

where $\mathcal{T}(\alpha)$ is the tree sum of Proposition 4.2(b) for the conductances $\sqrt{\alpha_i \alpha_j}$, $i, j \in F$. A permutation of F permutes the spanning trees of the complete graph on F . So C_F is a symmetric function of $(\alpha_i)_{i \in F}$. Take $j \geq 2$ and swap the states 1 and j . This maps the point $\alpha_F^* + v(e_j - e_1)$ to $\alpha_F^* - v(e_j - e_1)$, so $v \mapsto \log C_F(\alpha_F^* + v(e_j - e_1))$ is even and its derivative at $v = 0$ vanishes. In the coordinates α' this derivative is the j -th entry of γ_F .

Now we compare with [3]. That paper treats the uniform start. It writes r for our h , so its $T = Nr$ is ours, and $p = 1 - (d-1)h$. Its V_N is p^{N-1}/d , and its $C_{N,k}$ is the probability, for the uniform start, of a balanced bin with k positive coordinates. Let n be such a bin with support F , so that $|n_i - N/k| < 1$ for $i \in F$. A path inside F has probability μ_0 times a product of transition probabilities, so for our start $P_N(n) = d\mu_0 C_{N,k}$. By [3, Theorem 3.1(i)], for large N and uniformly for $\frac{1}{2}L \leq T \leq 2L$,

$$\log \frac{C_{N,k}}{V_N} = (k-1) \left[T - L + \frac{1}{2} \log T - a_k \right] - \frac{k-1}{8T} + O\left(\frac{1}{T^2} + \frac{T^2}{N}\right),$$

with a_k as in Corollary 4.11. Also $\log V_N = -\log d - (d-1)T + O(T^2/N)$, and $\log \mu_0 - (k-1)a_k = \log K_F$. So

$$\log P_N(n) = -(k-1)L + \frac{k-1}{2} \log T + \log K_F - (d-k)T - \frac{k-1}{8T} + O\left(\frac{1}{T^2} + \frac{T^2}{N}\right). \quad (27)$$

On the other hand $\alpha = n/N$ has $|x| \leq k/N$, so (24) applies to n . The bounds on U give $G_T(n/N) = c(\alpha_F^*)/T + O(1/N)$, since $\log(C_F/K_F) = O(|x|)$, $T(I_F - \lambda_F) = O(T|x|^2)$ and $c(\alpha) - c(\alpha_F^*) = O(|x|)$. Hence $\log(P_N(n)/X) = c(\alpha_F^*)/T + O(T^{-2} + T^2/N)$. Comparing with (27) at $T = L$, which lies in $[T_0, N^{1/3}]$ and in $[\frac{1}{2}L, 2L]$ for large N , gives $(c(\alpha_F^*) + \frac{k-1}{8})/L = O(L^{-2})$. So $c_F = c(\alpha_F^*) = -(k-1)/8$. This value rests on the cited theorem of [3]. $\square$

C Proofs for the planar walk

This appendix has three parts. We first prove the bound on $o_N^{(4)}$ in Proposition 6.2 and Theorem 6.3. Then we prove the local limit of Theorem 6.5, and finally parts (b) and (c) of Theorem 6.9. We identify $\mathbb{R}^2$ with $\mathbb{C}$, so $e_z = i^z$ and $Y_N = \sum_{t=0}^m \ell_t i^{Z_t}$ in the notation of Lemma 6.1. Let $\text{bin}(m) = \binom{N-1}{m} (\beta h)^m p_0^{N-1-m}$ be the law of m , with mean $\bar{m} = (N-1)\beta h$. We write $\mathbb{P}_z$ and $\mathbb{E}_z$ for the skeleton walk of Lemma 6.1(c) started at z , $\mathbb{P}$ for its uniform start, and ξ for one of its increments. Put $\varkappa = s/\beta$, so $\varkappa < 1$ when $r > 0$. For a skeleton, let a, b, c, d be the numbers of runs going east, west, north and south. So $a + b + c + d = m + 1$, and the path uses all four directions exactly when $a, b, c, d \geq 1$. Throughout, $N = 2\nu$, and $[\cdot]$ is the indicator of the statement inside.

For $k \geq 1$ let $\chi_k(X) = \binom{X-1}{k-1} [X \geq 1]$, and let $\chi_0(X) = [X = 0]$. Then $\chi_k(X)$ counts the compositions of X into k positive parts.

Lemma C.1 (return counts). *Let Z be a skeleton with m switches and counts a, b, c, d . The number of compositions ℓ of N into $m+1$ parts with $\sum_t \ell_t i^{Z_t} = 0$ is*

$$\Omega_N(a, b, c, d) = \sum_{X=0}^{\nu} \chi_a(X) \chi_b(X) \chi_c(\nu - X) \chi_d(\nu - X).$$

If $a, b, c, d \geq 1$, then $m \geq 3$ and $\Omega_N(a, b, c, d) \leq \frac{1}{2} \binom{N}{m-2}$.

Proof. The sum vanishes exactly when the runs going east and west have the same total length X , and those going north and south have total length $\nu - X$. The four compositions of these totals into runs are independent, which gives the formula. Now let $a, b, c, d \geq 1$. If $m = 3$, there is one run in each direction and $\Omega_N = \nu - 1 < \frac{1}{2} \binom{N}{1}$. Let $m \geq 4$. Pick t_z with $Z_{t_z} = z$ for each z , and fix the other $m - 3$ run lengths. If they add up to $N - j$, there are $\binom{N-j-1}{m-4}$ ways to choose them. The four free lengths add up to j , and $Y_N = 0$ fixes $\ell_{t_0} - \ell_{t_2}$ and $\ell_{t_1} - \ell_{t_3}$. So ℓ_{t_2} determines the other three, and $\ell_{t_0} + \ell_{t_2}$ lies in $[2, j - 2]$ with a fixed parity. So there are at most $j/2$ solutions, and

$$\Omega_N(a, b, c, d) \leq \sum_{j=0}^{N-1} \frac{j}{2} \binom{N-1-j}{m-4} = \frac{1}{2} \binom{N}{m-2},$$

where the identity counts the subsets of $\{0, \dots, N-1\}$ of size $m-2$ by their second smallest element j . $\square$

By Lemma 6.1(b) and (d), $\varpi_N = \Omega_N(a, b, c, d) / \binom{N-1}{m}$ is the probability that $Y_N = 0$ given m and Z , so

$$o_N^{(4)} = \sum_{m \geq 3} \text{bin}(m) \mathbb{E}[\varpi_N \mathbf{1}\{a, b, c, d \geq 1\} \mid m]. \quad (28)$$

Lemma C.2 (skeleton returns). *Let $W_m = \sum_{t=0}^m i^{Z_t} = (a-b) + i(c-d)$. (a) $\mathbb{E}_z |W_m|^2 \leq m+1$ for every start z . (b) If $s > 0$ and m is odd, then $\mathbb{P}(W_m = 0) \geq \kappa/(36(m+1))$. (c) If $s = 0$ and $m \equiv 3 \pmod{4}$, then $\mathbb{P}(W_m = 0) \geq 1/(m+1)$.*

Proof. (a) Since $\mathbb{E} i^\xi = \frac{r}{\beta}(i + i^{-1}) + \frac{s}{\beta} i^2 = -\kappa$, we get $\mathbb{E}_z |W_m|^2 = \sum_{t,t'} \mathbb{E}_z i^{Z_t - Z_{t'}} = m+1 + 2 \sum_{k=1}^m (m+1-k)(-\kappa)^k$. The last sum alternates in sign, starts negative and has nonincreasing absolute terms, so it is at most 0.

(b) Write $m = 2m_1 + 1$ and split $W_m = W' + W''$, where W' sums over $t \leq m_1$ and W'' over $t > m_1$. The law of ξ is symmetric and the uniform law is stationary, so $(Z_{m_1}, \dots, Z_0)$ is again the walk from a uniform start, and $\mathbb{P}(W' = w, Z_{m_1} = z) = \frac{1}{4} \mathbb{P}_z(W_{m_1} = w)$. With probability κ , independently of $Z_0, \dots, Z_{m_1}$, the next switch is a reversal. Then given $Z_{m_1} = z$, the second half is the walk from $z+2$, and $\mathbb{P}_{z+2}(W_{m_1} = -w) = \mathbb{P}_z(W_{m_1} = w)$. So $\mathbb{P}(W_m = 0) \geq \frac{\kappa}{4} \sum_z \sum_w \mathbb{P}_z(W_{m_1} = w)^2$. Let $R^2 = m+1$. By (a) and Chebyshev's inequality, $\mathbb{P}_z(|W_{m_1}| \leq R) \geq \frac{1}{2}$. At most $9R^2$ Gaussian integers have modulus at most R , so by Cauchy-Schwarz the inner sum is at least $1/(36R^2)$. Summing over z gives (b).

(c) Now every switch is a turn, so $i^{Z_t} = i^{Z_{t-1}} \cdot i_{\epsilon_t}$ with independent fair signs ϵ_t . Hence $i^{Z_t} = i^{Z_0} i^t \sigma_t$, where $\sigma_0 = 1$ and $\sigma_t = \epsilon_1 \cdots \epsilon_t$ are independent fair signs for $t \geq 1$. Write $m = 4k - 1$. Then $i^{-Z_0} W_m$ has real part $\sum_{t \text{ even}} (-1)^{t/2} \sigma_t$, which is 1 plus a sum of $2k - 1$ fair signs, and an independent imaginary part that is a sum of $2k$ fair signs. So

$$\mathbb{P}(W_m = 0) = \binom{2k-1}{k} \binom{2k}{k} 2^{1-4k} = \prod_{j=1}^k \frac{(2j-1)^2}{(2j)^2} \geq \frac{1}{4} \prod_{j=2}^k \frac{2j-2}{2j} = \frac{1}{4k} = \frac{1}{m+1}.$$

$\square$

Lemma C.3 (balanced skeletons). *If $a = b \geq 1$, $c = d \geq 1$ and $m = 2(a+c) - 1 \leq N-1$, then $\Omega_N(a, b, c, d) / \binom{N-1}{m} \geq \sqrt{m+1}/(2N^2)$.*

Proof. Put $k = a + c = (m + 1)/2$, so $2 \leq k \leq \nu$. Only $1 \leq X \leq \nu - 1$ contribute to Ω_N , so by Cauchy–Schwarz and Vandermonde’s identity

$$\Omega_N = \sum_{X=1}^{\nu-1} \binom{X-1}{a-1}^2 \binom{\nu-X-1}{c-1}^2 \geq \frac{1}{\nu} \binom{\nu-1}{k-1}^2 = \frac{k}{\nu^2} \binom{\nu-1}{k-1} \binom{\nu}{k}.$$

Also $\binom{\nu-1}{k-1} \binom{\nu}{k} / \binom{2\nu-1}{2k-1} = \mathbb{P}(H = k - 1)$, where H is the number of red balls among $2k - 1$ balls drawn without replacement from $\nu - 1$ red and ν blue ones. If $k = \nu$, then $H = k - 1$. Otherwise the ratio $\mathbb{P}(H = j + 1)/\mathbb{P}(H = j) = (\nu - 1 - j)(2k - 1 - j)/((j + 1)(\nu - 2k + 2 + j))$ decreases in j , is $\frac{\nu-k}{\nu-k+1} < 1$ at $j = k - 1$ and exceeds 1 at $j = k - 2$. So $k - 1$ is a mode of H . Since $\text{Var } H \leq (2k - 1)/4$, Chebyshev’s inequality puts mass at least $\frac{3}{4}$ on the at most $3\sqrt{2k}$ integers within $\sqrt{2k} - 1$ of $\mathbb{E}H$, so $\mathbb{P}(H = k - 1) \geq 1/(4\sqrt{2k})$. As $\binom{N-1}{m} = \binom{2\nu-1}{2k-1}$, we get $\Omega_N / \binom{N-1}{m} \geq k/(4\nu^2\sqrt{2k}) = \sqrt{m+1}/(2N^2)$. $\square$

Proof of the bound on $o_N^{(4)}$ in Proposition 6.2. Note that $(\beta h)^m p_0^{N-1-m} = p_0^{N-1} w^m$. By (28) and Lemma C.1,

$$o_N^{(4)} \leq \frac{p_0^{N-1}}{2} \sum_{m=3}^{N-1} \binom{N}{m-2} w^m \leq \frac{p_0^{N-1} w^2}{2} ((1+w)^N - 1) = \frac{w^2}{2p_0} (1 - p_0^N),$$

since $1 + w = 1/p_0$. Dividing by c_N gives the bound on E_N . $\square$

Proof of Theorem 6.3. We first prove three bounds. Since $I_1 \leq I_0 \leq e^z$ for $z \geq 0$, we have $\psi(z) \leq 2ze^z$, and (12) gives $2S_N(u) \leq 8ue^{Nu}$. With Proposition 6.2 and $(1+w)^N \leq e^{Nw}$,

$$\frac{o_N}{c_N} \leq 8ue^{Nu} + 2w^2 e^{Nw}. \quad (29)$$

Since $o_N \geq o_N^{(0)}$, (12) with $z = Nu$ gives

$$\frac{o_N}{c_N} \geq 2S_N(u) \geq \frac{4}{N} \left(1 - \frac{z(z+2)}{2N}\right) \psi(z). \quad (30)$$

Third, we show that there are $c' > 0$ and T_0 , depending only on r and s , with

$$o_N^{(4)} \geq \frac{c'}{N^2 \sqrt{T}} \quad \text{whenever } T = Nh \geq T_0 \text{ and } \beta h \leq \frac{1}{2}. \quad (31)$$

Let $\mathcal{M}$ be the set of odd m if $s > 0$, and the set of $m \equiv 3 \pmod{4}$ if $s = 0$. Let $m \in \mathcal{M}$. The event $\{a = b \geq 1, c = d \geq 1\}$ is $\{W_m = 0\}$ minus $\{a = b = 0\}$ and $\{c = d = 0\}$. Each of these needs every switch to be a reversal, so it has probability at most $\frac{1}{2} \kappa^m$, and it is empty if $s = 0$. So Lemma C.2 gives m_2 and $c_2 > 0$ with $\mathbb{P}(a = b \geq 1, c = d \geq 1 \mid m) \geq c_2/(m+1)$ for $m \in \mathcal{M}$ with $m \geq m_2$ (take $c_2 = \kappa/72$ if $s > 0$ and $c_2 = 1$ if $s = 0$). On this event Lemma C.3 gives $\varpi_N \geq \sqrt{m+1}/(2N^2)$. Let $J = [\bar{m}/2, 2\bar{m}]$. By (28), if $\bar{m} \geq 2m_2$ then $o_N^{(4)} \geq \sum_{m \in \mathcal{M} \cap J} \binom{m}{2} c_2 / (2N^2 \sqrt{m+1}) \geq c_2 \mathbb{P}(m \in \mathcal{M} \cap J) / (2N^2 \sqrt{2\bar{m}+1})$. The Chernoff bounds for the binomial law are $\mathbb{P}(m \leq \bar{m}/2) \leq e^{-\bar{m}/8}$ and $\mathbb{P}(m \geq 2\bar{m}) \leq e^{-\bar{m}/3}$, so $\mathbb{P}(m \notin J) \leq 2e^{-\bar{m}/8}$. Put $p = \beta h \leq \frac{1}{2}$. Then $\mathbb{P}(m \text{ odd}) = \frac{1}{2}(1 - (1 - 2p)^{N-1}) \geq \frac{1}{2}(1 - e^{-2\bar{m}})$. For $s = 0$ the roots of unity filter gives

$$\mathbb{P}(m \equiv 3 \pmod{4}) = \frac{1}{4} \sum_{k=0}^3 i^{-3k} (1 - p + p i^k)^{N-1} \geq \frac{1}{4} - \frac{3}{4} e^{-\bar{m}/2},$$

since $|1 - p + p i^{\pm 1}|^2 = 1 - 2p(1 - p) \leq e^{-p}$ and $|1 - 2p| \leq e^{-2p}$. So $\mathbb{P}(m \in \mathcal{M} \cap J) \geq \frac{1}{8}$ when $\bar{m}$ is large. As $\bar{m} \geq \beta T/2$, and $2\bar{m} + 1 \leq 3\beta T$ when $\beta T \geq 1$, this proves (31) with $c' = c_2/(16\sqrt{3}\beta)$.

(a) Let $h = \tau L/N$ and $T = \tau L$. Then $u, w = O(L/N)$, $Nu = s\tau L + O(L^2/N)$ and $Nw = \beta\tau L + O(L^2/N)$. So (29) gives $o_N/c_N = O(LN^{s\tau-1} + L^2N^{\beta\tau-2})$. If $\tau < \tau^*$, then $s\tau < 1$ and $\beta\tau < 2$, and this is at most $N^{-\gamma}$ for large N whenever $0 < \gamma < \min(1 - s\tau, 2 - \beta\tau)$. Now let $\tau > \tau^*$, so $s\tau > 1$ or $\beta\tau > 2$. If $s\tau > 1$, then $z = Nu \rightarrow \infty$ and $z^2/N \rightarrow 0$. As $I_0(z) + I_1(z) \sim 2e^z/\sqrt{2\pi z}$ [38, Eq. 10.40.1], we have $\psi(z) \sim \sqrt{2z/\pi} e^z$, and (30) gives $o_N/c_N \geq N^{s\tau-1}$ for large N . If $\beta\tau > 2$, then (31) and $c_N^{-1} = 4p_0^{-(N-1)} \geq 4e^{\beta T - \beta h}$ give $o_N/c_N \geq 4c'e^{-\beta h}N^{\beta\tau-2}(\tau L)^{-1/2}$, which is at least $N^{(\beta\tau-2)/2}$ for large N .

(b) Let $\tau < \tau^* < \tau'$. By (a), for large N the ratio o_N/c_N is below 1 at $h = \tau L/N$ and above 1 at $h = \tau' L/N < 1/\beta$. It increases in h by Proposition 6.2, so $\tau L < T_N^* < \tau' L$.

(c) Put $\tau_N = T_N^*/L$, so $\tau_N \rightarrow \tau^*$ by (b), and $w^* = \beta h_N^*/(1 - \beta h_N^*)$. As in (a), $u_N^*, w^* = O(L/N)$, $Nu_N^* = s\tau_N L + o(1)$ and $Nw^* = \beta\tau_N L + o(1)$. At the crossing Proposition 6.2 gives $2S_N(u_N^*) + E_N = 1$. If $s > 2r$, then $\tau^* = 1/s$ and $\beta\tau^* < 2$. So $E_N \leq 2w^{*2}e^{Nw^*} = N^{\beta/s-2+o(1)} \rightarrow 0$, and $2S_N(u_N^*) \rightarrow 1$. The two axes carry the same mass by symmetry. If $s < 2r$, then $\tau^* = 2/\beta$ and $s\tau^* < 1$. So $2S_N(u_N^*) \leq 8u_N^*e^{Nu_N^*} = N^{2s/\beta-1+o(1)}$, which is at most $N^{-\gamma}$ for large N whenever $0 < \gamma < 1 - 2s/\beta$. $\square$

C.1 The four-direction local limit

Idea of the proof. Given the skeleton, the run lengths are close to the gaps between m uniform points in $[0, N]$. So the total lengths in the four directions are close to N times a Dirichlet vector (y_0, y_1, y_2, y_3) with parameters (a, c, b, d) . Then $N^2\varpi_N$ should be close to twice the density of $(y_0 - y_2, y_1 - y_3)$ at 0, where the factor 2 is the parity of Y_N . This is the quantity $g(a, b, c, d)$ below. Divided by m , it is close to a Gaussian function of $W_m/\sqrt{m}$, and W_m satisfies a central limit theorem. Steps 1 and 2 below remove the skeletons with too few or too many switches and those that use some direction rarely. Step 3 compares $N^2\varpi_N$ with g , Step 4 averages g over the skeleton, and Step 5 puts the steps together.

Proof of Theorem 6.5. We use the notation of the start of this appendix. Let $A = a + b$ and $B = c + d$, so $A + B = m + 1$, and put $\Gamma_4 = (a - 1)!(b - 1)!(c - 1)!(d - 1)!$ and

$$g(a, b, c, d) = \frac{m(m-1)}{2} \cdot \frac{\binom{A-2}{a-1}}{2^{A-2}} \cdot \frac{\binom{B-2}{c-1}}{2^{B-2}} = \frac{m!(A-2)!(B-2)!}{2^{m-2}(m-2)!\Gamma_4}.$$

Let $J_N = [\bar{m}/2, 2\bar{m}]$, $\mathcal{G}_m = \{a, b, c, d \geq m/8\}$ and $m_0 = 20$. Note that $\bar{m} = \beta T(1 - 1/N) \rightarrow \infty$ and $\bar{m} \leq \beta N^{1/8}$. Below, C is a constant that depends only on r and s and may change from line to line.

Step 1 (few or many switches). If $a, b, c, d \geq 1$ and $m \leq N/5$, then Lemma C.1 and $\binom{N}{m-2}/\binom{N-1}{m} = Nm(m-1)/((N-m+2)(N-m+1)(N-m))$ give $\varpi_N \leq m^2/N^2$. If $m > N/5$, then $\varpi_N \leq 1 < 25m^2/N^2$. So $N^2\varpi_N \leq 25m^2$ whenever the path uses all four directions. As in the proof of Theorem 6.3, the Chernoff bounds give $\mathbb{P}(m \notin J_N) \leq 2e^{-\bar{m}/8}$. Also $\mathbb{E}m^4 = \sum_{j=1}^4 S(4, j)(N-1)_j(\beta h)^j \leq 15\bar{m}^4$, where $S(4, j) = 1, 7, 6, 1$ are Stirling numbers of the second kind, $(N-1)_j$ is a falling factorial, and $\bar{m} \geq 1$. By the Cauchy–Schwarz inequality, the terms of (28) with $m \notin J_N$ contribute at most $25\mathbb{E}[m^2\mathbf{1}\{m \notin J_N\}] \leq 25\sqrt{30}\bar{m}^2e^{-\bar{m}/16} = o(1)$ to $N^2o_N^{(4)}$.

Step 2 (unbalanced skeletons). For $k = 1, 2, 3$ let $V_k = \sum_{t=0}^m i^{kZ_t}$. The number of t with $Z_t = z$ is $\frac{m+1}{4} + \frac{1}{4}\sum_{k=1}^3 i^{-kz}V_k$. Let $\vartheta_k = \mathbb{E}i^{k\xi}$, so $\vartheta_1 = \vartheta_3 = -\varkappa$ and $\vartheta_2 = 2\varkappa - 1$. Then

$\mathbb{E}[i^{kZ_t} \mid Z_0, \dots, Z_{t-1}] = \vartheta_k i^{kZ_{t-1}}$, so $M_m^{(k)} = \sum_{t=1}^m (i^{kZ_t} - \vartheta_k i^{kZ_{t-1}})$ is a martingale with increments of modulus at most 2, and

$$(1 - \vartheta_k)V_k = M_m^{(k)} + i^{kZ_0} - \vartheta_k i^{kZ_m}.$$

Here $1 - \vartheta_k \geq \min(1, 4r/\beta) > 0$. Burkholder's inequality [35, Theorem 2.10], applied to the real and imaginary parts, bounds $\mathbb{E}|M_m^{(k)}|^4$ by a constant times the second moment of the sum of the m squared increments. So $\mathbb{E}|M_m^{(k)}|^4 \leq Cm^2$. So $\mathbb{E}|V_k|^4 \leq Cm^2$ and $\mathbb{E}(a - \frac{m+1}{4})^4 \leq Cm^2$, and the same holds for b, c, d . Since $\frac{m+1}{4} - \frac{m}{8} \geq \frac{m}{8}$, Markov's inequality gives $\mathbb{P}(\mathcal{G}_m^c \mid m) \leq C/m^2$. Every skeleton in $\mathcal{G}_m$ uses all four directions. By Step 1, the skeletons with all four directions that are not in $\mathcal{G}_m$, with $m \in J_N$, contribute at most $\sum_m \binom{m}{4} 25m^2 C/m^2 \leq 25C$ to $N^2 o_N^{(4)}$.

Step 3 (run lengths given the skeleton). We show that $N^2 \varpi_N = g(a, b, c, d)(1 + o(1))$ uniformly over the skeletons in $\mathcal{G}_m$ with $m \in J_N$. For large N such m satisfy $m_0 \leq m \leq 2\beta N^{1/8} \leq N^{1/6}$. On $\mathcal{G}_m$ we have $A, B \geq m/4$, so $A - 1, B - 1 \geq m/8$. Put $f(X) = X^{A-2}(\nu - X)^{B-2}$. For $X \geq 1$ we have $\chi_a(X) = \frac{X^{a-1}}{(a-1)!} \prod_{1 \leq j < a} (1 - j/X)_+$, where $(y)_+ = \max(y, 0)$, and similarly for b, c, d . The terms $X = 0$ and $X = \nu$ of Ω_N vanish. So

$$\Omega_N(a, b, c, d) = \frac{1}{\Gamma_4} \sum_{X=1}^{\nu-1} f(X) \omega(X),$$

where $\omega(X) \in [0, 1]$ is the product of the four products. The function f vanishes at 0 and ν and is unimodal, and its mode $X^* = \nu(A - 2)/(m - 3)$ lies in $[\nu/6, 5\nu/6]$ since $A, B \geq m/4$ and $m \geq 18$. On each monotone piece of f every value $f(X)$ lies between the integrals of f over the two unit intervals next to X . So $|\sum_{j \leq X \leq k} f(X) - \int_j^k f| \leq 2 \max f$ for all integers $0 \leq j \leq k \leq \nu$. Moreover $f / \int_0^\nu f$ is the density of ν times a Beta variable with parameters $A - 1$ and $B - 1$, and $\int_0^\nu f = \nu^{m-2} B(A - 1, B - 1)$.

We need three estimates. First, this Beta law is the law of the $(A - 1)$ -th smallest of $m - 2$ independent uniform points in $[0, 1]$. A union bound over sets of $A - 1$ points gives, for $0 < t < 1$,

$$\mathbb{P}(\text{Beta} \leq t) \leq \binom{m-2}{A-1} t^{A-1} \leq \left(\frac{e(m-2)t}{A-1} \right)^{A-1} \leq (8et)^{A-1}.$$

We take $t = 2\nu^{-1/4}$. Since $A - 1 \geq 2$ and $16e\nu^{-1/4} < 1$ for large N , this gives $\int_0^{\nu^{3/4}+1} f \leq (16e)^2 \nu^{-1/2} \int_0^\nu f$, and the same bound holds near ν . Second, for $\nu^{3/4} \leq X \leq \nu - \nu^{3/4}$, the inequality $1 - \prod_j (1 - t_j)_+ \leq \sum_j t_j$ for $t_j \geq 0$ gives $1 - \omega(X) \leq (a^2 + b^2 + c^2 + d^2)/(2\nu^{3/4}) \leq m^2 \nu^{-3/4}$. Third, on $[\nu/12, 11\nu/12]$ we have $(\log f)'' = -(A - 2)/X^2 - (B - 2)/(\nu - X)^2 \geq -144m/\nu^2$. This interval contains every X with $|X - X^*| \leq \nu/(12\sqrt{m})$, and for these X Taylor's formula at X^* gives $f(X) \geq e^{-1/2} \max f$. So $\int_0^\nu f \geq e^{-1/2} \max f \cdot \nu/(6\sqrt{m})$, that is, $\max f \leq 6e^{1/2} \sqrt{m} \nu^{-1} \int_0^\nu f$.

Now we put the three estimates together. For the upper bound, $\sum_{X=1}^{\nu-1} f \omega \leq \sum_X f \leq \int_0^\nu f + 2 \max f$. For the lower bound we sum only over $\nu^{3/4} \leq X \leq \nu - \nu^{3/4}$, use the second estimate for ω , compare the sum of f with its integral, and use the first estimate for the two ends of the integral. This gives

$$\sum_{X=1}^{\nu-1} f \omega \geq (1 - m^2 \nu^{-3/4}) \left(\int_0^\nu f - 2(16e)^2 \nu^{-1/2} \int_0^\nu f - 2 \max f \right).$$

Since $m^2 \leq N^{1/3}$ and $\sqrt{m}/\nu \leq N^{-5/6}$, we get $\Gamma_4 \Omega_N = \nu^{m-2} B(A - 1, B - 1)(1 + o(1))$, uniformly. Also $\binom{N-1}{m} = \frac{N^m}{m!} \prod_{j=1}^m (1 - \frac{j}{N}) = \frac{N^m}{m!} (1 + O(m^2/N))$. With $\nu^{m-2}/N^{m-2} = 2^{-(m-2)}$ and $B(A - 1, B - 1) = (A - 2)!(B - 2)!/(m - 2)!$ this gives $N^2 \varpi_N = g(a, b, c, d)(1 + o(1))$.

Step 4 (average over the skeleton). We show that $\mathbb{E}[g \mathbf{1}_{\mathcal{G}_m} \mid m]/m \rightarrow c_* = \frac{2(r+s)}{\pi\beta}$ as $m \rightarrow \infty$. First we need a local limit for the fair binomial law. Let $0 \leq k \leq n$ and $y = (2k - n)/n$. If $|2k - n| \leq n^{3/5}$, Stirling's formula gives

$$\begin{aligned} \sqrt{n} \binom{n}{k} 2^{-n} &= \sqrt{\frac{2}{\pi(1-y^2)}} e^{-\frac{n}{2}[(1+y)\log(1+y) + (1-y)\log(1-y)]} (1 + O(\frac{1}{n})) \\ &= \sqrt{\frac{2}{\pi}} e^{-\frac{(2k-n)^2}{2n}} (1 + O(n^{-3/5})), \end{aligned}$$

since the bracket is $y^2 + O(y^4)$ and $ny^4 \leq n^{-3/5}$. Both sides decrease as $|2k - n|$ grows, so outside this range both are $O(e^{-n^{1/5}/3})$. Hence

$$\sup_{0 \leq k \leq n} \left| \sqrt{n} \binom{n}{k} 2^{-n} - \sqrt{\frac{2}{\pi}} e^{-(2k-n)^2/(2n)} \right| = O(n^{-1/2}). \quad (32)$$

On $\mathcal{G}_m$ we have $A-2, B-2 \geq m/8$. We apply (32) with $n = A-2$, $k = a-1$ and with $n = B-2$, $k = c-1$, and note that $2(a-1) - (A-2) = a-b$. Then, uniformly on $\mathcal{G}_m$,

$$\frac{g}{m} = \frac{m-1}{\pi \sqrt{(A-2)(B-2)}} \exp\left(-\frac{(a-b)^2}{2(A-2)} - \frac{(c-d)^2}{2(B-2)}\right) + O(m^{-1/2}).$$

Now we replace $m-1$ by m , then $A-2$ by A and $B-2$ by B , and finally B by $m(1-\theta)$, where $\theta = A/m$. The last step is needed because $B = m(1-\theta) + 1$. Each step changes the prefactor by a factor $1 + O(1/m)$ and each exponential by $O(1/m)$, since $0 \leq e^{-y/A_2} - e^{-y/A_1} \leq \frac{A_2 - A_1}{eA_1}$ for $y \geq 0$ and $0 < A_1 \leq A_2$. So, uniformly on $\mathcal{G}_m$,

$$\frac{g}{m} = \Psi\left(\frac{a-b}{\sqrt{m}}, \frac{c-d}{\sqrt{m}}, \frac{A}{m}\right) + O(m^{-1/2}), \quad \Psi(x_1, x_2, \theta) = \frac{e^{-x_1^2/(2\theta) - x_2^2/(2(1-\theta))}}{\pi \sqrt{\theta(1-\theta)}}.$$

On $\mathcal{G}_m$ we have $\theta \in [\frac{1}{4}, \frac{3}{4} + \frac{1}{m}] \subset [\frac{1}{5}, \frac{4}{5}]$, as $m \geq 20$. Let $\tilde{\Psi}$ be Ψ with θ replaced by the nearest point of $[\frac{1}{5}, \frac{4}{5}]$. It is bounded and continuous on $\mathbb{R}^3$ and equals Ψ on $\mathcal{G}_m$.

Next we prove a central limit theorem for $W_m = (a-b) + i(c-d)$. Put $\zeta_j = i\xi_j + \varkappa$, where $\xi_j = Z_j - Z_{j-1}$. The ζ_j are independent with $\mathbb{E}\zeta_j = 0$, $\mathbb{E}|\zeta_j|^2 = 1 - \varkappa^2$ and $\mathbb{E}\zeta_j^2 = \vartheta_2 - \varkappa^2 = -(1 - \varkappa)^2$. Since $i^{Z_j} = i^{Z_{j-1}}(\zeta_j - \varkappa)$,

$$(1 + \varkappa)W_m = M_m + i^{Z_0} + \varkappa i^{Z_m}, \quad M_m = \sum_{j=1}^m i^{Z_{j-1}} \zeta_j,$$

and M_m is a martingale with increments of modulus at most 2. We take the skeletons for all m to be the first steps of one infinite walk, so the σ -fields are nested. Fix $\varphi \in \mathbb{R}$. For a constant ω with $|\omega| = 1$ we have $\mathbb{E}(\text{Re } \omega \zeta_j)^2 = \frac{1}{2}(\mathbb{E}|\zeta_j|^2 + \text{Re } \omega^2 \mathbb{E}\zeta_j^2)$. So the j -th increment of $\text{Re}(e^{-i\varphi} M_m)$ has conditional variance $\frac{1}{2}[(1 - \varkappa^2) - (1 - \varkappa)^2 \cos(2\varphi)(-1)^{Z_{j-1}}]$. The average of these variances tends to $\frac{1}{2}(1 - \varkappa^2)$ in probability, because $\sum_{t=0}^m (-1)^{Z_t} = V_2$ and $\mathbb{E}|V_2/m|^4 \leq C/m^2$ by Step 2. The increments are bounded, so the martingale central limit theorem [35, Cor. 3.1] and the Cramér–Wold device give $M_m/\sqrt{m} \Rightarrow N(0, \frac{1}{2}(1 - \varkappa^2)I_2)$. Hence

$$\frac{W_m}{\sqrt{m}} \Rightarrow X \sim N(0, \sigma_*^2 I_2), \quad \sigma_*^2 = \frac{1 - \varkappa}{2(1 + \varkappa)} = \frac{r}{2(r+s)}.$$

Also $A = \frac{m+1}{2} + \frac{1}{2}V_2$, so $A/m \rightarrow \frac{1}{2}$ in probability, and $(W_m/\sqrt{m}, A/m) \Rightarrow (X, \frac{1}{2})$. By Step 2, $\mathbb{E}[\tilde{\Psi} \mathbf{1}_{\mathcal{G}_m^c} \mid m] \leq C/m^2$. So, since $\tilde{\Psi}$ is bounded and continuous,

$$\frac{\mathbb{E}[g \mathbf{1}_{\mathcal{G}_m} \mid m]}{m} = \mathbb{E} \tilde{\Psi} \left(\frac{W_m}{\sqrt{m}}, \frac{A}{m} \right) + O(m^{-1/2}) \rightarrow \mathbb{E} \Psi \left(X, \frac{1}{2} \right) = \frac{2}{\pi} \mathbb{E} e^{-|X|^2} = \frac{2}{\pi(1+2\sigma_*^2)} = c_*.$$

Step 5 (conclusion). By Steps 1 and 2, $N^2 o_N^{(4)} = \sum_{m \in J_N} \text{bin}(m) \mathbb{E}[N^2 \varpi_N \mathbf{1}_{\mathcal{G}_m} \mid m] + O(1)$. By Step 3 the expectation is $\mathbb{E}[g \mathbf{1}_{\mathcal{G}_m} \mid m](1 + o(1))$, with $o(1)$ uniform in $m \in J_N$. Since $\min J_N \rightarrow \infty$, Step 4 turns this into $N^2 o_N^{(4)} = c_*(1 + o(1)) \sum_{m \in J_N} \text{bin}(m) m + O(1)$. As in Step 1, $\mathbb{E}[m \mathbf{1}_{\{m \notin J_N\}}] \rightarrow 0$, and $\bar{m} = \beta T(1 + O(1/N))$. Hence $N^2 o_N^{(4)} = c_* \beta T(1 + o(1)) + O(1) = \frac{2(r+s)}{\pi} T(1 + o(1))$. $\square$

C.2 Turns only

We prove parts (b) and (c) of Theorem 6.9. Throughout, a word is a word of length n in the letters $\pm$, its weight is x^j if it has j switches, and we say that it switches at step t if its letters t and $t+1$ differ ($1 \leq t \leq n-1$). The words of length n have total weight $2(1+x)^{n-1}$, and those with k letters $+$ have total weight $S_{k,n-k}(x)$ if $1 \leq k \leq n-1$ and 1 if $k \in \{0, n\}$. We need one fact about Bessel functions. Let $F(y) = e^{-y}(I_0(y) + I_1(y))$ for $y > 0$, so that $\psi(y) = ye^y F(y)$. The integral forms $I_n(y) = \frac{1}{\pi} \int_0^\pi e^{y \cos \theta} \cos(n\theta) d\theta$ of I_0 and I_1 [38, Eq. 10.32.3] and the substitution $\varsigma = 1 - \cos \theta$ give

$$F(y) = \frac{1}{\pi} \int_0^\pi e^{-y(1-\cos \theta)} (1 + \cos \theta) d\theta = \frac{1}{\pi} \int_0^2 e^{-y\varsigma} \sqrt{\frac{2-\varsigma}{\varsigma}} d\varsigma.$$

So F is positive and decreasing. Replacing y by $y/2$ and ς by 2ς gives

$$F(y/2) = \frac{2}{\pi} \int_0^1 e^{-y\varsigma} \sqrt{\frac{1-\varsigma}{\varsigma}} d\varsigma.$$

This is at most $2F(y)$, because $1 - \varsigma \leq 2 - \varsigma$ and the part of the integral for $F(y)$ over $[1, 2]$ is nonnegative. Hence

$$F(y') \leq 2F(y) \quad \text{whenever } y/2 \leq y' \leq y. \quad (33)$$

Lemma C.4 (one switch of a balanced word). *Let $N \geq 4$ be even, let $x > 0$, and put $z = Nx$ and $\eta = z(z+2)/(2N)$. For $1 \leq t \leq N-1$ let $S_N^{(t)}(x)$ be the total weight of the balanced words of length N that switch at step t . If $\eta \leq \frac{1}{2}$, then $S_N^{(t)}(x) \leq 2x + 8x S_N(x)$ for every t , and $\sum_t S_N^{(t)}(x) = x S'_N(x) \leq (2z+2) S_N(x)$.*

In words, a balanced word switches at a given step with probability of order x , as a word without any condition does, and it has at most about $2z$ switches on average.

Proof. Reversing a word maps the balanced words that switch at t to those that switch at $N-t$, and it keeps the weight. So we may take $t \geq N/2$. A balanced word that switches at t is a word w_1 of length t followed by a word w_2 of length $N-t$ whose first letter differs from the last letter of w_1 , and its weight is x times the product of the two weights. If w_1 has k letters $+$, then w_2 has $N/2 - k$ of them, so $k \leq N/2 \leq t$ and $t-k \leq N/2 \leq t$.

Consider first the balanced words with $1 \leq k \leq t-1$. Fix w_2 . Then k is fixed, and the words w_1 that can precede w_2 have total weight at most $S_{k,t-k}(x)$. By [2, Proposition 4.1] the

bin polynomials grow coefficientwise as the bin moves toward the middle, that is, $S_{a,b} \leq S_{a+1,b-1}$ coefficientwise when $a < b - 1$. Since also $S_{a,b} = S_{b,a}$, this gives $S_{k,t-k}(x) \leq S_t(x)$. The words w_2 of length $N - t$ have total weight at most $2(1+x)^{N-t-1} \leq 2e^{(N-t)x}$. By (12) with $t \geq 2$ steps, $S_t(x) \leq \frac{2}{t}\psi(tx) = 2xe^{tx}F(tx)$. So these words contribute at most

$$x \cdot 2xe^{tx}F(tx) \cdot 2e^{(N-t)x} = 4x^2e^zF(tx) \leq 8x^2e^zF(z) = 4x \cdot \frac{2}{N}\psi(z) \leq \frac{4x}{1-\eta}S_N(x) \leq 8xS_N(x),$$

to $S_N^{(t)}(x)$. Here the first inequality is (33) with $z/2 \leq tx \leq z$, and the next one is the lower bound in (12). If $k = 0$ or $k = t$, then w_1 is constant and $t = N/2$, and w_2 is the constant word of the other letter. These two words have weight x each. This proves the first claim.

Each balanced word with j switches is counted j times in $\sum_t S_N^{(t)}$, so the sum is $xS'_N(x)$. The function $y \mapsto \log S_N(e^y)$ is convex, since S_N has nonnegative coefficients. So its derivative at $y = \log x$ is at most its increase over $[\log x, \log x + 1]$, that is, $xS'_N(x)/S_N(x) \leq \log(S_N(ex)/S_N(x))$. By (12), $S_N(ex) \leq \frac{2}{N}\psi(ez)$ and $S_N(x) \geq \frac{1}{2} \cdot \frac{2}{N}\psi(z)$, and $\psi(ez)/\psi(z) = e \cdot e^{(e-1)z}F(ez)/F(z) \leq e^{1+(e-1)z}$. Hence $xS'_N(x)/S_N(x) \leq 1 + \log 2 + (e-1)z \leq 2z + 2$. $\square$

Proof of Theorem 6.9(b). A pair of balanced words with a common switch has a common switch at some step t , and the pairs that both switch at t have total weight $S_N^{(t)}(x)^2$. So $\Delta_N(x) \leq \sum_t S_N^{(t)}(x)^2 \leq \max_t S_N^{(t)}(x) \cdot \sum_t S_N^{(t)}(x)$, and Lemma C.4 gives the bound, because $z(z+2) \leq N$ means $\eta \leq \frac{1}{2}$. $\square$

Proof of Theorem 6.9(c). We compare v_N^* with a value x_1 of the odds slightly above u_N , at which the bound of part (b) is already smaller than the gain in S_N^2 .

Put $z_0 = Nu_N$. Since $S_N(u) \geq 2u$, we have $u_N \leq \frac{1}{2}$ and $p_N \geq \frac{2}{3}$. With $z_0 = z_N/p_N$ and $z_N \leq L$ [2, Proposition 3.8] this gives $z_0 \leq \frac{3}{2}L$. By (12) and $S_N(u_N) = 1$, $\psi(z_0) \geq N/2$. Let $\kappa = 40(z_0 + 2)^2/N$, which tends to 0, and let $x_1 > u_N$ be the root of $S_N(x_1) = 1 + \kappa$, with $z_1 = Nx_1$ and $\eta_1 = z_1(z_1 + 2)/(2N)$.

We first show that $z_1 \leq z_0 + 1$ for large N . Since $(\log \psi)' \geq 1$, we have $\psi(z_0 + 1) \geq e\psi(z_0) \geq eN/2$. So by (12), at $x = (z_0 + 1)/N$ we get $S_N(x) \geq (1 - O(L^2/N))e$, which exceeds $1 + \kappa$ for large N . As S_N is increasing, $z_1 \leq z_0 + 1$. In particular $z_1(z_1 + 2) \leq N$ and $\kappa \leq 1$ for large N , so part (b) gives

$$\Delta_N(x_1) \leq (2x_1 + 16x_1)(2z_1 + 2) \cdot 2 = \frac{72z_1(z_1 + 1)}{N} \leq \frac{72(z_0 + 1)(z_0 + 2)}{N} \leq 2\kappa.$$

By part (a), $o_N/c_N = (1 + \kappa)^2 - \Delta_N(x_1) \geq 1$ at $v = x_1$. The ratio o_N/c_N increases with v , so $v_N^* \leq x_1$. Now $\psi(z_1) \leq \frac{N}{2}(1 + \kappa)/(1 - \eta_1)$ by (12), and $(\log \psi)' \geq 1$. Hence

$$0 \leq Nv_N^* - z_0 \leq z_1 - z_0 \leq \log \psi(z_1) - \log \psi(z_0) \leq \log \frac{1 + \kappa}{1 - \eta_1} = O\left(\frac{L^2}{N}\right),$$

where the first inequality is part (a). This is the first claim. For the second, $v = rh/(1 - 2rh)$ gives $rT_N^* = Nv_N^*/(1 + 2v_N^*)$, and $z_N = z_0/(1 + u_N)$. Both v_N^* and u_N are $O(L/N)$, so $rT_N^* = Nv_N^* + O(L^2/N)$ and $z_N = z_0 + O(L^2/N)$. $\square$

D Proofs for the table of outcomes

This appendix proves the rows P_d and Θ_ε of Table 1. In both rows Q is symmetric and the start has full support. So the admissible faces are the connected faces (Lemma 2.2), and the winners at

τ are the connected faces that minimize $\Phi_\tau(F) = \tau\lambda_F + |F| - 1$ (Theorem 3.4). We use three facts about a connected face F . First, $-Q_F$ is symmetric, λ_F is its least eigenvalue, and an eigenvector of $-Q_F$ with positive entries belongs to λ_F (the second fact at the start of Appendix A). Second, $-Q_F$ is the Laplacian of the graph on F , which is positive semidefinite, plus the diagonal matrix of the rates out of F . So λ_F is the least value of

$$\sum_{\{i,j\} \subseteq F} q_{ij}(f_i - f_j)^2 + \sum_{i \in F} q_i^{\text{out}}(F)f_i^2 \quad (34)$$

over unit vectors $f \in \mathbb{R}^F$. Third, if $F' \subsetneq F$ is nonempty, then $\lambda_{F'} > \lambda_F$ (Proposition 3.1(d)).

Each proof has two steps, namely the exit rates and then the lower-left boundary of the hull. For the second step we use the following lemma. It says that a chain of points with increasing switch values is its own lower-left boundary.

Lemma D.1 (a chain of lines). *Let $x_0 > x_1 > \dots > x_m \geq 0$ and $y_0 < y_1 < \dots < y_m$, and put $\ell_i(\tau) = \tau x_i + y_i$ and $t_i = (y_i - y_{i-1})/(x_{i-1} - x_i)$ for $1 \leq i \leq m$. Put $t_0 = 0$ and $t_{m+1} = \infty$, and suppose that $t_1 < t_2 < \dots < t_m$. Then for $1 \leq j \leq m$ and every τ*

$$\ell_j(\tau) - \ell_{j-1}(\tau) = (x_{j-1} - x_j)(t_j - \tau). \quad (35)$$

Let $0 \leq i \leq m$ and let τ be finite with $t_i \leq \tau \leq t_{i+1}$. Then $\ell_0(\tau) \geq \dots \geq \ell_i(\tau) \leq \dots \leq \ell_m(\tau)$. If also $t_i < \tau < t_{i+1}$, then all these inequalities are strict, so $\ell_i(\tau) < \ell_j(\tau)$ for every $j \neq i$.

Proof. The identity (35) is the definition of t_j . If $j \leq i$, then $t_j \leq t_i \leq \tau$, so $\ell_j(\tau) \leq \ell_{j-1}(\tau)$. If $j > i$, then $\tau \leq t_{i+1} \leq t_j$, so $\ell_j(\tau) \geq \ell_{j-1}(\tau)$. If $t_i < \tau < t_{i+1}$, these inequalities are strict. $\square$

D.1 The path

Let P_d be the path with states $1, \dots, d$ and $q_{i,i+1} = q_{i+1,i} = 1$ for $1 \leq i < d$. An *end-arc* of $k < d$ states is one of the two faces $\{1, \dots, k\}$ and $\{d - k + 1, \dots, d\}$. For $k \geq 1$ put

$$\lambda'_k = 2 - 2 \cos \frac{\pi}{2k+1}, \quad \tau'_k = \frac{1}{\lambda'_{k-1} - \lambda'_k} \quad (k \geq 2), \quad D'_k = k + \lambda'_k \tau'_k \quad (k \geq 2).$$

Lemma D.2 (the numbers D'_k). *(a) The sequence λ'_k is positive, strictly decreasing and strictly convex in $k \geq 1$. Hence $0 < \tau'_2 < \tau'_3 < \dots$. Also $\tau'_2 = (1 + \sqrt{5})/2$.*

(b) $\frac{3}{2}k - \frac{1}{2} < D'_k < \frac{3}{2}k$ for every $k \geq 2$.

Proof. (a) Let $g(x) = 2 - 2 \cos(\pi/x)$, so that $\lambda'_k = g(2k+1)$. For $x \geq 2$,

$$g'(x) = -\frac{2\pi \sin(\pi/x)}{x^2} < 0, \quad g''(x) = 2\pi \left(\frac{\pi \cos(\pi/x)}{x^4} + \frac{2 \sin(\pi/x)}{x^3} \right) > 0.$$

So $\lambda'_{k-1} - \lambda'_k > \lambda'_k - \lambda'_{k+1} > 0$ for $k \geq 2$, and the τ'_k increase. Since $\lambda'_1 = 1$ and $\lambda'_2 = 2 - 2 \cos \frac{\pi}{5} = (3 - \sqrt{5})/2$, we get $\tau'_2 = 2/(\sqrt{5} - 1) = (1 + \sqrt{5})/2$.

(b) Put $a = \pi/(2(2k-1))$ and $b = \pi/(2(2k+1))$, so that $0 < b < a \leq \pi/6$. Since $2 - 2 \cos 2x = 4 \sin^2 x$, the ratio $\rho = \lambda'_k/\lambda'_{k-1}$ equals $(\sin b/\sin a)^2$. Also $D'_k - k = \rho/(1 - \rho)$, which increases with ρ . So the claim is

$$\frac{k-1}{k+1} < \rho < \frac{k}{k+2}.$$

For the lower bound, $\sin x/x$ decreases on $(0, \pi]$, so $\sin b/\sin a > b/a = (2k-1)/(2k+1)$. Then $\rho > (2k-1)^2/(2k+1)^2 > (k-1)/(k+1)$, because $(2k-1)^2(k+1) - (k-1)(2k+1)^2 = 2$.

For the upper bound write $s(x) = \sin x/x$ and $y = (8k-2)/((k+2)(2k-1)^2)$. Then

$$\rho = \left(\frac{2k-1}{2k+1}\right)^2 \left(\frac{s(b)}{s(a)}\right)^2 \quad \text{and} \quad \frac{k}{k+2} = \left(\frac{2k-1}{2k+1}\right)^2 (1+y),$$

the second because $k(2k+1)^2 - (k+2)(2k-1)^2 = 8k-2$. So we must show that $s(b)/s(a) < \sqrt{1+y}$. Now $0 < y < 1$, because $(k+2)(2k-1)^2 \geq 12(2k-1) > 8k-2$, and $\sqrt{1+y} \geq 1 + \frac{y}{3}$ for $0 \leq y \leq 3$. So it suffices that $s(b) - s(a) < \frac{y}{3}s(a)$. We use $1 - \frac{x^2}{6} \leq s(x) \leq 1 - \frac{x^2}{6} + \frac{x^4}{120}$ for $x > 0$. This gives $s(a) \geq 1 - \frac{\pi^2}{216} > 0.954$ and

$$s(b) - s(a) \leq \frac{a^2 - b^2}{6} + \frac{b^4}{120} = \frac{\pi^2 k}{3(4k^2 - 1)^2} + \frac{\pi^4}{1920(2k+1)^4}.$$

So it suffices that this is less than $0.318y$. Multiply by $(2k-1)^2$ and use $k/(2k+1)^2 < 1/(4k)$ and $(2k-1)^2/(2k+1)^4 < 1/(4k^2)$. It then suffices that

$$\frac{\pi^2}{12k} + \frac{\pi^4}{7680k^2} < 0.318 \frac{8k-2}{k+2}.$$

The left side decreases in k and the right side, which is $0.318(8 - \frac{18}{k+2})$, increases. At $k=2$ they are less than 0.415 and more than 1.11. $\square$

Proposition D.3 (the path). *Let $d \geq 3$, let Q be the path P_d with unit rates, and let the start have full support. Put $K = \lfloor 2d/3 \rfloor$ and $\tau_{\text{full}} = \lceil d/3 \rceil / \lambda'_K$.*

- (a) *The connected faces are the sets of consecutive states. An end-arc of $k < d$ states has exit rate λ'_k . A set of k consecutive states that contains neither 1 nor d has exit rate $2 - 2\cos \frac{\pi}{k+1} > \lambda'_k$.*
- (b) *We have $2 \leq K \leq d-1$ and $\tau'_2 < \dots < \tau'_K < \tau_{\text{full}}$.*
- (c) *The winners are the states 1 and d for $0 < \tau < \tau'_2$, the two end-arcs of k states for $\tau'_k < \tau < \tau'_{k+1}$ and $2 \leq k \leq K-1$, the two end-arcs of K states for $\tau'_K < \tau < \tau_{\text{full}}$, and P_d alone for $\tau > \tau_{\text{full}}$. No other face wins for any $\tau > 0$.*

In words, the mode grows from an end of the path, one state at a time, until it covers two thirds of the path, and then it jumps to the whole path. The first step is at the golden ratio.

Idea of the proof. Growing an end-arc by one state lowers its exit rate by less and less, since λ'_k is convex. So the end-arcs alone win in the order of their size, and the arc of k states starts to win at τ'_k . The whole path enters when its exponent $d-1$ falls below the least exponent of the arcs. At τ'_k that least exponent is $D'_k - 1$, so the arcs of k states win before the whole path exactly when $D'_k < d$. Lemma D.2(b) puts D'_k just below $\frac{3}{2}k$, and this gives $k \leq 2d/3$.

Proof. (a) A connected face of a path is a set of consecutive states. Let F have $k < d$ states. If $F = \{1, \dots, k\}$, then $-Q_F$ is tridiagonal with off-diagonal entries -1 and diagonal $(1, 2, \dots, 2)$. Put $\vartheta = \pi/(2k+1)$ and $r_j = \cos((2j-1)\vartheta/2)$ for $1 \leq j \leq k$. Then $r_j > 0$, since $(2j-1)\vartheta/2 < \pi/2$. With $r_0 = r_1$ and $r_{k+1} = \cos(\pi/2) = 0$ we have $r_{j-1} + r_{j+1} = 2\cos\vartheta r_j$ for $1 \leq j \leq k$, and this says that $-Q_F r = (2 - 2\cos\vartheta)r$. So $\lambda_F = \lambda'_k$, and the same holds for $\{d-k+1, \dots, d\}$ by the reflection $i \mapsto d+1-i$. If F contains neither 1 nor d , then $-Q_F = 2I - A$ with A the adjacency

matrix of a path on k states. Now $r_j = \sin(j\pi/(k+1))$ is a positive eigenvector with eigenvalue $2 - 2\cos \frac{\pi}{k+1}$, and this is larger than λ'_k because $\frac{\pi}{k+1} > \frac{\pi}{2k+1}$.

(b) Since $d \geq 3$ we have $2 \leq K \leq d-1$. The τ'_k increase by Lemma D.2(a). Note that $d-K = \lceil d/3 \rceil$, so $\tau_{\text{full}} = (d-K)/\lambda'_K$. By Lemma D.2(b), $D'_K < \frac{3}{2}K \leq d$, that is $\lambda'_K \tau'_K < d-K$. So $\tau'_K < \tau_{\text{full}}$.

(c) Write $\phi_k(\tau) = \tau\lambda'_k + k - 1$ for the exponent of an end-arc of k states, $1 \leq k \leq d-1$. First we apply Lemma D.1 to the points $(\lambda'_k, k-1)$ for $1 \leq k \leq K$ followed by $(0, d-1)$, the point of P_d . Its switch values are $\tau'_2, \dots, \tau'_K$ and $(d-K)/\lambda'_K = \tau_{\text{full}}$, which increase by (b). So on each of the intervals in (c) the face named there has a smaller exponent than the other end-arcs of at most K states and than P_d , and by (a) also than every set of at most K consecutive states that is not an end-arc.

It remains to show that a face F of k consecutive states with $K < k < d$ never wins. By (a) its exponent is at least $\phi_k(\tau)$. We show that $\phi_k(\tau) > \min(\phi_K(\tau), d-1)$ for every $\tau > 0$. Now we apply Lemma D.1 to the points $(\lambda'_j, j-1)$ for $1 \leq j \leq d-1$, with switch values $\tau'_2 < \dots < \tau'_{d-1}$. If $\tau < \tau'_{K+1}$, then $\tau < \tau'_j$ for $K < j \leq k$, and (35) gives $\phi_k(\tau) > \phi_{k-1}(\tau) > \dots > \phi_K(\tau)$. If $\tau \geq \tau'_{K+1}$, take j with $K+1 \leq j \leq d-1$ and $\tau'_j \leq \tau$, and with $\tau \leq \tau'_{j+1}$ if $j < d-1$. Then

$$\phi_k(\tau) \geq \phi_j(\tau) \geq \phi_j(\tau'_j) = D'_j - 1 > \frac{3}{2}j - \frac{3}{2} \geq \frac{3}{2}K \geq d-1.$$

The first step is Lemma D.1, the second holds since ϕ_j increases, the third is the definition of D'_j , the fourth is Lemma D.2(b), the fifth is $j \geq K+1$, and the last holds because $3K \geq 2d-2$. So F never wins. The winners at τ are the admissible faces with the least exponent, and this gives (c). $\square$

At a switch value the faces on both sides tie. On P_4 we have $K=2$ and $D'_3 = 4.077 > 4$, so the end-arc of three states never wins. Its point lies only $\lambda'_3 - \frac{1}{2}\lambda'_2 = 0.0071$ to the right of the edge from the end-arcs of two states to P_4 . The distance is small because the lower bound in Lemma D.2(b) is nearly attained.

D.2 Two triangles

Let $\varepsilon > 0$. The graph Θ_ε has states $1, \dots, 6$, unit rates on the triangles $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$, and one more link between 3 and 4 of rate ε . We call $1, 2, 5, 6$ the inner states and $\{1, 2\}, \{5, 6\}$ the inner edges. The graph has the symmetries $1 \leftrightarrow 2$, $5 \leftrightarrow 6$ and the mirror map $i \mapsto 7-i$. Put

$$\begin{aligned} p_A(x) &= x^2 - (3 + \varepsilon)x + \varepsilon, \\ p_4(x) &= x^3 - (5 + 2\varepsilon)x^2 + (6 + 6\varepsilon)x - 2\varepsilon, \\ p_5(x) &= x^3 - (4 + 2\varepsilon)x^2 + (3 + 4\varepsilon)x - \varepsilon, \end{aligned}$$

and $\varepsilon_0 = (64 - \sqrt{46})/60 = 0.95363$, $\varepsilon_1 = (64 + \sqrt{46})/60 = 1.17971$, the roots of $120\varepsilon^2 - 256\varepsilon + 135$.

Proposition D.4 (two triangles joined by a link). *Let the start have full support.*

- (a) $\lambda_A = \lambda_B = \frac{1}{2}(3 + \varepsilon - \sqrt{\varepsilon^2 + 2\varepsilon + 9})$ is the smaller root of p_A , and $0 < \lambda_A < 1$.
- (b) Among the connected faces of one, two and three states the least exit rates are 2, 1 and λ_A . They are attained exactly by the inner states, by the inner edges and by A and B . Put $\lambda_4 = \lambda_{\{1,2,3,4\}}$ and $\lambda_5 = \lambda_{\{2,\dots,6\}}$. Every connected face of four states has exit rate at least λ_4 , and every connected face of five states has exit rate λ_5 . Here λ_4 and λ_5 are the least roots of p_4 and p_5 , and $\lambda_5 < \lambda_4 < \lambda_A$.

(c) $\lambda_5 > \frac{1}{2}\lambda_4$ for every $\varepsilon > 0$, and $\lambda_4 > \frac{2}{3}\lambda_A$ if and only if $\varepsilon < \varepsilon_1$.

(d) Let $\varepsilon < \varepsilon_1$, and put

$$\tau_A = \frac{1}{1 - \lambda_A} = \frac{1 + \varepsilon + \sqrt{\varepsilon^2 + 2\varepsilon + 9}}{4}, \quad \tau^* = \frac{3}{\lambda_A} = \frac{3(3 + \varepsilon + \sqrt{\varepsilon^2 + 2\varepsilon + 9})}{2\varepsilon}.$$

Then $1 < \tau_A < \tau^*$. The winners are the inner states for $0 < \tau < 1$, the inner edges for $1 < \tau < \tau_A$, the triangles A and B for $\tau_A < \tau < \tau^*$, and $[6]$ alone for $\tau > \tau^*$. No other face wins for any $\tau > 0$. As $\varepsilon \rightarrow 0$, $\tau_A = 1 + \frac{\varepsilon}{3} + O(\varepsilon^2)$ and $\tau^* = \frac{9}{\varepsilon} + 2 + O(\varepsilon)$.

(e) Let $\varepsilon > \varepsilon_1$. Then at $\tau = 3/\lambda_A$, where the triangles and $[6]$ have the same exponent, the face $\{1, 2, 3, 4\}$ has a smaller exponent. So the list in (d) fails.

In words, for a weak link each triangle is a community that the chain rarely leaves. The mode sits on a triangle from $\tau \approx 1$ until $\tau \approx 9/\varepsilon$, and then it spreads over all six states at once. Once the link is stronger than ε_1 this list fails, because a triangle together with the far end of the link beats both the triangles and $[6]$ at the value of τ where they tie.

Idea of the proof. The exit rates of small faces are explicit, and those of the faces with four and five states are roots of cubics. The points of the inner states, the inner edges, the triangles and $[6]$ form a chain as in Lemma D.1. The points of the faces with four and five states must lie strictly to the right of the edge from a triangle to $[6]$, and this is what (c) says. We prove (c) by continuity in ε . A change of sign needs a common root of two polynomials, hence a zero of a resultant, and the resultants are explicit polynomials in ε .

Proof. (a) On the vectors (a, a, b) the matrix $-Q_A$ acts as $(a, b) \mapsto (a - b, -2a + (2 + \varepsilon)b)$, with characteristic polynomial p_A , and $(1, -1, 0)$ is an eigenvector with eigenvalue 3. As $p_A(0) = \varepsilon > 0$ and $p_A(1) = -2 < 0$, the smaller root of p_A lies in $(0, 1)$ and the other root is larger than 1. So the smaller root is λ_A . The mirror map gives $\lambda_B = \lambda_A$.

(b) By (34), $\lambda_F \geq 1$ whenever every state of F has rate at least 1 out of F . We go through the connected faces by size. The states 3 and 4 have exit rate $2 + \varepsilon$, and the inner states have 2. The connected pairs are the inner edges, with $\lambda = 1$, the pair $\{3, 4\}$, with $\lambda = 2$, and $\{1, 3\}$, $\{2, 3\}$ and their mirror images, with $\lambda = \frac{1}{2}(4 + \varepsilon - \sqrt{\varepsilon^2 + 4}) > 1$. The connected faces of three states are A , B and the sets $\{a, 3, 4\}$ and $\{3, 4, b\}$ with $a \in \{1, 2\}$ and $b \in \{5, 6\}$, because a connected face that meets both triangles contains 3 and 4. In each of these four sets every state has rate at least 1 out of the set, so $\lambda \geq 1 > \lambda_A$. A connected face of four states meets both triangles, so it contains 3 and 4. So it is $\{1, 2, 3, 4\}$, its mirror image $\{3, 4, 5, 6\}$, or a set $\{a, 3, 4, b\}$, in which every state has rate 1 out of the set, so that $\lambda \geq 1$. A connected face of five states is the complement of an inner state (the complements of 3 and of 4 are disconnected), and these four faces are images of $\{2, \dots, 6\}$ under the symmetries. By the third fact above, $\lambda_5 < \lambda_{\{3, 4, 5, 6\}} = \lambda_4 < \lambda_A < 1$, since $\{2, \dots, 6\} \supsetneq \{3, 4, 5, 6\}$ and $\{1, 2, 3, 4\} \supsetneq A$.

For the polynomials, note that $(1, -1, 0, 0)$ is an eigenvector of $-Q_F$ for $F = \{1, 2, 3, 4\}$ with eigenvalue 3. On the vectors (a, a, b, c) the matrix acts as $(a, b, c) \mapsto (a - b, -2a + (2 + \varepsilon)b - \varepsilon c, -\varepsilon b + (2 + \varepsilon)c)$, with characteristic polynomial $(x - 1)(x - 2)(x - 2 - 2\varepsilon) - 2(x - 2 - \varepsilon) = p_4(x)$. So the eigenvalues of $-Q_F$ are 3 and the roots of p_4 . As $\lambda_4 < 1$, it is the least root of p_4 . In the same way, for $F = \{2, \dots, 6\}$ the vector that is 1 at 5, -1 at 6 and 0 elsewhere has eigenvalue 3, and on the vectors that agree at 5 and 6 the characteristic polynomial is $(x - 3)p_5(x)$. So λ_5 is the least root of p_5 . (Both characteristic polynomials are also checked by computer algebra, see Section D.3.)

(c) Put $D_1 = \lambda_5 - \frac{1}{2}\lambda_4$ and $D_2 = \lambda_4 - \frac{2}{3}\lambda_A$. Both are continuous in $\varepsilon > 0$, because the least eigenvalue of a symmetric matrix is continuous in its entries. If $D_1 = 0$, then $x = \lambda_5$ is a common root of $p_5(x)$ and $p_4(2x)$. If $D_2 = 0$, then $x = \lambda_4$ is a common root of $p_4(x)$ and $p_A(\frac{3}{2}x)$. A common root makes the resultant vanish, and an exact computation gives

$$\begin{aligned}\text{Res}_x(p_5(x), p_4(2x)) &= -8\varepsilon^3(16\varepsilon^2 + 12\varepsilon + 9), \\ \text{Res}_x(p_A(\frac{3}{2}x), p_4(x)) &= -\frac{1}{16}\varepsilon^2(120\varepsilon^2 - 256\varepsilon + 135).\end{aligned}$$

So D_1 has no zero in $\varepsilon > 0$, and D_2 can vanish only at ε_0 and at ε_1 .

We find the signs for small ε . By (a) and (b), $0 < \lambda_5 < \lambda_4 < \lambda_A < \varepsilon$. Here $\lambda_5 > 0$ because $\lambda_5 \geq 0$ by (34) and $p_5(0) = -\varepsilon \neq 0$, and $\lambda_A < \varepsilon$ because $p_A(\varepsilon) = -2\varepsilon < 0$. At $\varepsilon = 0$ the cubics are $x(x-2)(x-3)$ and $x(x-1)(x-3)$, with the simple root 0. So the implicit function theorem gives one root $\frac{\varepsilon}{3} + O(\varepsilon^2)$ of each cubic near 0, and no other root near 0. So $\lambda_4 = \frac{\varepsilon}{3} + O(\varepsilon^2)$ and $\lambda_5 = \frac{\varepsilon}{3} + O(\varepsilon^2)$, and (a) gives $\lambda_A = \frac{\varepsilon}{6} + O(\varepsilon^2)$. Hence $D_1 = \frac{\varepsilon}{6} + O(\varepsilon^2)$ and $D_2 = \frac{\varepsilon}{9} + O(\varepsilon^2)$ are positive for small ε . So $D_1 > 0$ for every $\varepsilon > 0$, and $D_2 > 0$ on $(0, \varepsilon_0)$.

Next, $D_2 < 0$ for large ε . Indeed $\lambda_A \rightarrow 1$ as $\varepsilon \rightarrow \infty$ by (a). On the other hand take $f = (a, a, c, c)$ in (34) for $\{1, 2, 3, 4\}$. The value is $2(a-c)^2 + 2c^2$ and $|f|^2 = 2a^2 + 2c^2$, so $\lambda_4 \leq \min_{a,c}((a-c)^2 + c^2)/(a^2 + c^2) = (3 - \sqrt{5})/2 < \frac{1}{2}$ for every ε .

Now we show that $D_2 > 0$ at ε_0 . Put $x_0 = \frac{7}{40}$. Then

$$p_4(x_0) = \frac{57743}{64000} - \frac{809}{800}\varepsilon, \quad p_A(\frac{3}{2}x_0) = \frac{59}{80}\varepsilon - \frac{4599}{6400}$$

are both negative for $0.8923 < \varepsilon < 0.9743$, and this interval contains ε_0 . For $0 \leq x \leq \frac{1}{2}$ we have $p'_4(x) = 3x^2 - 2(5+2\varepsilon)x + 6+6\varepsilon \geq 1+4\varepsilon > 0$. So p_4 increases on $[0, x_0]$ and is negative there, and its positive root λ_4 is larger than x_0 . Also $p_A(\frac{3}{2}x_0) < 0$ puts $\frac{3}{2}x_0$ between the roots of p_A , so $\lambda_A < \frac{3}{2}x_0$. Hence $\lambda_4 > x_0 > \frac{2}{3}\lambda_A$ at ε_0 .

So $D_2 > 0$ on $(0, \varepsilon_0]$. It has no zero in $(\varepsilon_0, \varepsilon_1)$, so $D_2 > 0$ on $(0, \varepsilon_1)$. It has no zero in (ε_1, ∞) and is negative for large ε , so $D_2 < 0$ on (ε_1, ∞) . By continuity $D_2 = 0$ at ε_1 . This proves (c).

(d) As $p_A(\frac{3}{4}) = \frac{\varepsilon}{4} - \frac{27}{16} < 0$ for $\varepsilon < \frac{27}{4}$, we have $\lambda_A < \frac{3}{4}$. So $1 - \lambda_A > 0$ and $\lambda_A < 3(1 - \lambda_A)$, that is $1 < \tau_A < \tau^*$. Apply Lemma D.1 to the points $(2, 0)$, $(1, 1)$, $(\lambda_A, 2)$ and $(0, 5)$ of the inner states, the inner edges, the triangles and [6]. The switch values are 1, τ_A and τ^* . So on each interval in (d) the faces named there have a smaller exponent than the other three kinds, and by (b) than every other connected face of at most three states. Let F be a connected face of four or five states. By (b) and (c), for every $\tau > 0$

$$\begin{aligned}\Phi_\tau(F) &\geq \tau\lambda_4 + 3 > \frac{2}{3}(\tau\lambda_A + 2) + \frac{1}{3} \cdot 5 & \text{if } |F| = 4, \\ \Phi_\tau(F) &= \tau\lambda_5 + 4 > \frac{1}{3}(\tau\lambda_A + 2) + \frac{2}{3} \cdot 5 & \text{if } |F| = 5,\end{aligned}$$

where the second line uses $\lambda_5 > \frac{1}{2}\lambda_4 > \frac{1}{3}\lambda_A$. Each right side is at least the smaller of the exponents $\tau\lambda_A + 2$ of a triangle and 5 of [6]. So F never wins, and this gives the winners. The closed forms follow from (a) by rationalizing, since $(\varepsilon^2 + 2\varepsilon + 9) - (1 + \varepsilon)^2 = 8$ and $(3 + \varepsilon)^2 - (\varepsilon^2 + 2\varepsilon + 9) = 4\varepsilon$. The expansions follow from $\sqrt{\varepsilon^2 + 2\varepsilon + 9} = 3 + \frac{\varepsilon}{3} + O(\varepsilon^2)$.

(e) At $\tau = 3/\lambda_A$ the triangles and [6] have the exponent 5, and $\{1, 2, 3, 4\}$ has $3\lambda_4/\lambda_A + 3 < 5$ by (c). $\square$

At $\varepsilon = 1$, for example, $\tau_A = (1 + \sqrt{3})/2$ and $\tau^* = 6 + 3\sqrt{3}$.

D.3 What was checked by computation

The proofs above use computer algebra in one place only, namely the two resultants in the proof of Proposition D.4(c). The script `verify_examples.py` computes them in sympy. It also expands the characteristic polynomials of $-Q_F$ for $F = A, \{1, 2, 3, 4\}$ and $\{2, \dots, 6\}$ and finds $(x-3)p_A$, $(x-3)p_4$ and $(x-3)^2p_5$, and it recomputes $p_4(\frac{7}{40})$ and $p_A(\frac{21}{80})$ in rational arithmetic. The rest of its checks for these two rows are in floating point, and no proof rests on them. For $3 \leq d \leq 9$ it computes the exit rates of all connected faces of P_d , compares those of the end-arcs with λ'_k , and finds the sizes of the winners as in Proposition D.3(c). With 50 digits it confirms $\frac{3}{2}k - \frac{1}{2} < D'_k < \frac{3}{2}k$ for $2 \leq k \leq 2000$. For $\varepsilon = 0.3, 1$ and 1.1 it computes the exit rates of all 63 faces of Θ_ε and finds the least exit rates, the winners and the switch values of Proposition D.4. For $\varepsilon = 1.5, 3, 4.4, 4.6$ and 8 it finds that the faces $\{1, 2, 3, 4\}$ and $\{3, 4, 5, 6\}$ win on an interval of τ , which agrees with Proposition D.4(e).

E The large-concentration limit of the direct jump

We prove Proposition 7.4(b), with B_1, B_2, B_3 independent Beta(a, a) as in part (a) and $a = \omega/3 \geq 1$.

Idea of the proof. We pass to the logits ℓ_j of the B_j . For large a their law concentrates near 0 on the scale $a^{-1/2}$, and D is a thin tube around the line of cyclic kernels. Let t be the coordinate along this line. The section of D at t is close to a triangle of area $\frac{9}{8}t^4$. So we expect $\mathbb{P}_\omega(D) \approx a^{3/2} \int t^4 e^{-cat^2} dt \approx 1/a$, and a Gaussian integral gives the constant.

Put $\ell_j = \log(B_j/(1-B_j))$, $g(y) = 2 \log \cosh(y/2)$ and $g_j = g(\ell_j)$. Then g is even and convex, $g \geq 0 = g(0) = g'(0)$, $B_j(1-B_j) = \frac{1}{4}e^{-g_j}$, $\log(2B_j) = \frac{1}{2}(\ell_j - g_j)$ and $\log(2(1-B_j)) = -\frac{1}{2}(\ell_j + g_j)$. As $dB_j/d\ell_j = B_j(1-B_j)$, ℓ_j has density $e^{-ag(\ell_j)}/Z$ with $Z = 4^a \Gamma(a)^2 / \Gamma(2a) = 2\sqrt{\pi} \Gamma(a) / \Gamma(a + \frac{1}{2})$. Taking logarithms in the three conditions $B_1 B_2 \leq \frac{1}{4}$, $(1-B_1)B_3 \leq \frac{1}{4}$ and $(1-B_2)(1-B_3) \leq \frac{1}{4}$ of Proposition 7.4, we see that D is the set where

$$M_1 \leq g_1 + g_2, \quad M_2 \leq g_1 + g_3, \quad M_3 \leq g_2 + g_3, \quad (36)$$

with $M_1 = \ell_1 + \ell_2$, $M_2 = \ell_3 - \ell_1$, $M_3 = -\ell_2 - \ell_3 = -M_1 - M_2$. With $\psi = g_1 + g_2 + g_3$ (in this appendix ψ is not the Bessel function of Section 6),

$$\mathbb{P}_\omega(D) = Z^{-3} \int_D e^{-a\psi(\ell)} d\ell. \quad (37)$$

For real y ,

$$\frac{y^2}{4} - \frac{y^4}{96} \leq g(y) \leq \frac{y^2}{4}, \quad (g1)$$

$$g(y) \geq \frac{1}{10} \min(y^2, |y|). \quad (g2)$$

With $u = y/2$, the upper bound is $\cosh u \leq e^{u^2/2}$ (termwise), and the lower bound holds as $\chi = \log \cosh u - \frac{u^2}{2} + \frac{u^4}{12}$ has $\chi(0) = \chi'(0) = 0$ and $\chi'' = u^2 - \tanh^2 u \geq 0$. For (g2), $g'' = \frac{1}{2} \operatorname{sech}^2(y/2) > 0.2$ on $|y| \leq 2$, and for $|y| \geq 2$ convexity gives $g(y) \geq \frac{1}{2}|y|g(2) > 0.43|y|$.

We now choose coordinates along and across the cyclic kernels. A cyclic kernel has $B_1 = 1 - B_2 = B_3$, so its logits are $\ell = tv$ with $v = (1, -1, 1)$ and $t \in \mathbb{R}$. For general ℓ let $t = \langle \ell, v \rangle / 3$. The map $\ell \mapsto (t, M_1, M_2)$ has determinant 1 and inverse $\ell = tv + w$ with $w = \frac{1}{3}(M_1 - M_2, 2M_1 + M_2, M_1 + 2M_2) \perp v$. So $|\ell|^2 = 3t^2 + |w|^2$ with $|w|^2 = \frac{1}{3}(M_1^2 + M_2^2 + M_3^2)$, and with $|M|$, $|M|_\infty$ the 2-norm and max norm of (M_1, M_2) ,

$$\frac{1}{3}|M|^2 \leq |w|^2 \leq |M|^2 \leq 2|M|_\infty^2. \quad (38)$$

Proof of Proposition 7.4(b). Step 1: an a priori bound. On D , (36) gives $M_1 \leq \psi$ and $M_1 = -M_2 - M_3 \geq -2\psi$, and the same for M_2 . So by (g1)

$$|M|_\infty \leq 2\psi(\ell) \leq \frac{1}{2}|\ell|^2 = \frac{1}{2}(3t^2 + |w|^2). \quad (39)$$

Step 2: localization. Put $\varepsilon = 1/12$ and let U be the set where $|t| \leq \varepsilon$ and $|M|_\infty \leq \varepsilon$. Off U , $|\ell| \geq \max(\sqrt{3}|t|, |M|/\sqrt{3}) \geq \varepsilon/\sqrt{3}$, and some $|\ell_j| \geq |\ell|/\sqrt{3}$. So (g2) gives $\psi \geq \gamma := 0.1 \min(|\ell|^2/3, |\ell|/\sqrt{3}) \geq \gamma_0 = \varepsilon^2/90$. Hence $e^{-a\psi} \leq e^{-(a-1)\gamma_0} e^{-\gamma}$ off U , and $\int_{D \setminus U} e^{-a\psi} = O(e^{-\gamma_0 a})$.

Step 3: the section of D at t . On $D \cap U$, (39) and (38) give $|M|_\infty \leq \frac{3}{2}t^2 + \varepsilon|M|_\infty$, so $|M|_\infty \leq 2t^2$. On $E = \{|t| \leq \varepsilon, |M|_\infty \leq 2t^2\}$ we have $|w|^2 \leq 8t^4$, so $\ell_j = \pm t + w_j$ gives $|\ell_j| \leq 2|t|$ and $|\ell_j^2 - t^2| \leq 7|t|^3$. Then (g1) gives $|g_j - \frac{1}{4}t^2| \leq 2|t|^3$, and with $|\ell|^2 = 3t^2 + |w|^2$ it bounds ψ . Hence

$$|g_i + g_j - \frac{1}{2}t^2| \leq 4|t|^3 \quad (i \neq j), \quad |\psi - \frac{3}{4}t^2| \leq 2t^4. \quad (40)$$

Let $\mathcal{T}(h) = \{M : M_1, M_2 \leq h, M_1 + M_2 \geq -h\}$, a triangle of area $\frac{9}{2}h^2$ in $\{|M|_\infty \leq 2h\}$. Let D_t be the section of $D \cap U$ at t and $h_\pm = \frac{1}{2}t^2(1 \pm 8|t|) \geq 0$. If $M \in D_t$, then $(t, M) \in E$, and (36) and (40) give $M_1, M_2, M_3 \leq h_+$. If $M \in \mathcal{T}(h_-)$, then $|M|_\infty \leq 2h_- \leq \min(t^2, \varepsilon)$, so $(t, M) \in E \cap U$, and (40) gives (36). So $\mathcal{T}(h_-) \subseteq D_t \subseteq \mathcal{T}(h_+)$, and

$$\frac{9}{8}t^4(1 - 8|t|)^2 \leq |D_t| \leq \frac{9}{8}t^4(1 + 8|t|)^2. \quad (41)$$

Step 4: integration. By Fubini (Jacobian 1), (40) and (41), $J = \int_{D \cap U} e^{-a\psi}$ lies between the integrals over $|t| \leq \varepsilon$ of $\frac{9}{8}t^4(1 \mp 8|t|)^2 e^{-a(\frac{3}{4}t^2 \pm 2t^4)}$. With $t = s/\sqrt{a}$ these are $a^{-5/2}$ times integrals whose integrands tend to $\frac{9}{8}s^4 e^{-3s^2/4}$ and are at most $\frac{9}{2}s^4 e^{-s^2/2}$, since $8|t| < 1$ and $2s^4/a \leq 2\varepsilon^2 s^2$. By dominated convergence and $\int s^4 e^{-3s^2/4} ds = \frac{3}{4}\sqrt{\pi}(\frac{4}{3})^{5/2}$, $J = a^{-5/2}(\sqrt{3\pi} + o(1))$, and the same holds over D . As Γ is log-convex, $\Gamma(a + \frac{1}{2})^2 \leq a\Gamma(a)^2$ and $\Gamma(a + 1)^2 \leq (a + \frac{1}{2})\Gamma(a + \frac{1}{2})^2$, so $Z = 2\sqrt{\pi/a}(1 + O(1/a))$. Then (37) gives $\mathbb{P}_\omega(D) = \frac{\sqrt{3}}{8\pi a}(1 + o(1)) = \frac{3\sqrt{3}}{8\pi\omega}(1 + o(1))$. $\square$

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